

Cargo Securing Manual

PASHA HAWAII C9 Class Vessel

APPROVED

by ABS on behalf of the
USCG for compliance with
applicable USCG
Regulations subject to
any amendments or
comments in ABS letter



USCG approval applies
only to specified vessel(s)

GEORGE II
ABS ID No. 8012487

Latest Revision No. 15, December 2023
See Revision page for details

TABLE OF CONTENTS

INTRODUCTION

CHAPTER 1 - GENERAL INFORMATION, DEFINITIONS AND VESSEL OVERVIEW

- 1.1 General Information and Definitions
- 1.2 Principle Characteristics
- 1.3 Stowage Profile
- 1.4 Hatch Cover Arrangements, Weight and Typical Fitting Layout
- 1.5 Visibility Restrictions
- 1.6 Hazardous Material Information

CHAPTER 2 - SECURING DEVICES

- 2.1 Handling and Safety Instructions
- 2.2 Specifications of Fixed Securing Devices
(Fixed sockets, lashing plates, D rings cell guides, etc.)
- 2.3 Specifications for Portable Securing Devices
(Twistlocks, lashing rods, stackers, etc...)
- 2.4 Table of required quantity of Portable Securing Devices and Test Certificates
- 2.5 Inspection and Maintenance Procedures

CHAPTER 3 - SECURING ARRANGEMENTS

- 3.1 Allowable Stowage Patterns
- 3.2 Stowage and Securing Instructions
- 3.3 Stowage of Refrigerated Containers
- 3.4 Allowable Stack Weights
- 3.5 Allowable Stack Weights 20' in 40' Cell Guides
- 3.6 Evaluation of Forces Acting on Containers

CHAPTER 4 - SECURING ARRANGEMENTS FOR NON-STANDARD CONTAINERS

- 4.1 Handling Arrangements of non-standard cargo
- 4.2 Method to assess securing arrangements for non-standardized cargo
- 4.3 Allowable Stowage Patterns
(Flatracks, oversizes containers, etc...)

APPENDIX

- A. Inspection and Maintenance Forms
- B. USCG NVIC 10-97
- C. Equipment (container) List
- D. Resolution A748(18)

Revision History

Revision No.	Revision	Date
1	Revised to satisfy ABS approval letter dated 10 November 1997 (Sea-Land Service, Inc. document), vessel name changed to <i>HORIZION Reliance</i> ; revised format of Section 3.4; added Section 1.7, Section 3.5 (formally 4.2), added new Section 4.2, Appendix D and E.	May 2003
1	Revised according to ABS Letter Dated 13 June 2003; added Section 3.5 and re-numbered Section 3.5 to 3.6 to be consistent with other manuals in Horizon Fleet; added calculation sheets to Section 4.2; added information to Section 2.4 .	June 2003
2	Revised error in color drawing in Section 3.1 .	April 2004
3	Revised Sections 2.1 and 2.5 added to Section 3.4 - 4 and 5, high tiers with twistlocks and no lashings.	April 2004
3	Revised according to ABS letters of May 14 and 19, 2004.	June 2004
4	Revised Section 3.4 to add stack weights for the paralash lashing system, according to ABS letter of August 26, 2004.	August 2004
5	Revised reefer container count in Sections 1.3, 1.7 and 3.3; revised container loose fitting count in Section 2.1 .	November 2004
6	Corrected typos in Introduction and Sections 1.1, 1.2, 1.3, 1.6, 1.7, 2.2, 2.5, 3.1, 3.2 and 3.4; corrected container fitting VCG's in Section 3.1 drawings.	April 2005
7	Corrected reefer outlet number and location in Sections 1.3, 1.7 and 3.3; added load case for five high paralash with extra outboard lash, Section 3.4 and Appendix; added twist lock C5AM – HC to Section 2.3 .	June 2005
8	Added 6/7 tier empty stack weights, ABS 12/2006, Section 3.4.	Dec. 2006
9	Revised typos in Sections 1.7, 3.1, 3.2 and 3.4; Summer Loadline displacement and deadweight corrected to reflect as-built drawings, Section 1.2; corrected Hatch 1 container quantities, Section 1.3; Hazardous Material Table modified to show mechanical ventilation below deck all holds, Section 1.7, also note added to the Hazardous Material Table on the exemption of limited quantities from segregation requirements; added test certificates of container lashing gear to Section 2.5; added allowable stack weights for GM = 15' in tables of Section 3.4, modified sketches for two high and three high container loading with twistlocks only to show no lashes on the outboard stack; Allowable stowage patterns modified to show 45' stowage above the 1 st tier on Hatches 2, 3, 7 and 12 in Section 3.1; reorganized Appendix C.	Jan. 2011
10	Reefer receptacles added to the coamings on deck aft at Hatch 5 and fwd at Hatch 6; updated Sections 1.7 and 3.3 .	June 2013
11	Revised Section 2.3, to include BD-J1 semi automatic twistlock information, Section 2.3. Revised Section 3.1, with revised bay plans. Revised Section 1.3 Stowage Profile & Summary Table.	August 2016

Revision History

Revision No.	Revision	Date
12	Revised to reflect changes after repowering and modification. Vessel name changed to <i>GEORGE II</i> .	Nov. 2022
13	1. Removed 6 & 7 tier high stowage arrangement on hatch covers-Section 3.4. 2. Added long lashing rods and certificates for outboard pedestals at hatch 9 & 10-Section 2.4. 3. Added and corrected dimensions for stowage section above deck at hatch 1, 9 and 10.	Aug. 2023
14	1. Hatch 2 above deck stowage updated. Section 1.3, 1.4, 3.1 & 3.2 revised. 2. Main particulars updated per approved Lightship. 3. Hatch 3 & 4 above deck stowage updated. Section 3.1&3.2 revised.	Nov. 2023 Dec. 2023
15	1. 5-high stowage on deck updated for hatch 1-6, 9 & 10. Section 1.3, 3.1, 3.2 & 3.4 revised 2. Main particulars (Sec 1.2) updated per approved Trim & Stability Booklet.	Dec. 2023

Introduction

- 1 The information contained in this Cargo Securing Manual, required in accordance with the 1994 amendments to the International Convention for the Safety of Life at Sea, 1974 (SOLAS) VI/5 and VII/5, is in an approved form in accordance with the Guidelines for the Preparation of the Cargo Securing Manual, MSC/Circ.745.
- 2 The purpose of this manual is to provide guidance to the Master and crew on board the vessel with respect to the proper stowage and securing of cargo units.
- 3 It is the Masters responsibility to ensure that cargo units (as defined in MSC/Circ.745) are at all times stowed and secured in an efficient manner, taking into account the prevailing conditions and the general principals of safe stowage set out in this Manual, and that the securing equipment and timber used are adequate for the loadings calculated in accordance with this Manual.
- 4 This Manual should be kept on board the vessel for inspection by Port/Flag State inspectors, Classification Society Surveyors and other interested parties.
- 5 This manual has been approved by American Bureau of Shipping on behalf of the Government of the United States of America. It is to be retained on board the vessel for examination and/or reference by American Bureau of Shipping Surveyor, Port/Flag State inspector, and those involved with the safe stowage and securing of cargoes carried.
- 6 The information contained in or appended to this manual should be regularly reviewed and updated. With the exception of the lists of portable cargo securing devices where equipment is replaced with new equipment of an identical type, amendments should not be made to this Manual without the prior approval of the American Bureau of Shipping.

1.1 GENERAL INFORMATION & DEFINITIONS

1.1.1 INFORMATION & DEFINITIONS

- 1 The guidance given herein should by no means rule out the principles of good seamanship, neither can they replace experience in stowage and securing practice.
- 2 The information and requirements set forth in this Manual are consistent with the requirements of the C9 CLASS vessel's trim and stability booklet, hull strength, International Load Line Certificate (1966) and with the requirements of the International Maritime Dangerous Goods (IMDG) Code. Verification of trim, stability and longitudinal strength is accomplished by using the PC based CargoMax program, the loading instrument.
- 3 This Cargo Securing Manual specifies arrangements and cargo securing devices provided on board the ship for the correct application to and the securing of cargo units, containers, vehicles and other entities, based on transverse, longitudinal and vertical forces which may arise during adverse weather and sea conditions.
- 4 It is imperative to the safety of the ship and the protection of the cargo and personnel that the securing of the cargo is carried out properly and that only appropriate securing points or fittings should be used for cargo securing.
- 5 The cargo securing devices mentioned in this manual should be applied so as to be suitable and adapted to the quantity, type of packaging, and physical properties of the cargo to be carried. When new or alternative types of cargo securing devices are introduced, the Cargo Securing Manual should be revised accordingly. Alternative cargo securing devices introduced should not have less strength than the equipment which they replace.
- 6 There should be a sufficient quantity of reserve cargo securing devices on board the ship.
- 7 Information on the strength and instructions for the use and maintenance of each specific type of cargo securing device, where applicable, is provided in this manual. The cargo securing devices should be maintained in a satisfactory condition. Items worn or damaged to such an extent that their quality is impaired should be replaced.
- 8 Any equipment supplied by stevedoring companies for the securing of specific cargoes shall be appropriately certified in accordance with national or international standards.
- 9 "Cargo Securing Devices" are all the fixed and portable devices used to secure and support Cargo Units.

- 10 “Maximum Securing Load” (MSL) is a term used to define the allowable load capacity for a device used to secure cargo to a ship. “Safe Working Load” (SWL) may be substituted for MSL for securing purposes, provided this is equal to or exceeds the strength defined by MSL.
- 11 “Standardized Cargo” means cargo for which the ship is provided with an approved securing system based upon cargo units of specific types, such as ISO 45’, 40’, 40’ high-cube and 20’ containers.
- 12 *Semi-standard Cargo* means cargo for which the ship is provided with a securing system capable of accommodating a limited variety of cargo units, such as flat racks, vehicles, trailers, etc.
- 13 “Non-standardized Cargo” means cargo which requires individual stowage and securing arrangements.
- 14 Cargo *Unit* means vehicle, container, flat, pallet, portable tank, packaged unit, or any other entity, etc. or loading equipment, or any part thereof, which belongs to the ship but is not fixed to the ship.

1.1.2 Cargo Information

Prior to shipment the shipper should provide all necessary information about the cargo to enable the shipowner or ship operator to ensure that:

- the different commodities to be carried are compatible with each other or suitably separated;
- the cargo is suitable for the ship;
- the ship is suitable for the cargo; and
- the cargo can be safely stowed and secured on board the ship and transported under all expected conditions during the intended voyage.

The master should be provided with adequate information regarding cargo to be carried so that its stowage may be properly planned for handling and transport.

1.1.3 Stowage and Securing

1. General principles of cargo securing:

- All cargoes should be stowed and secured in such a way that the ship and persons on board are not put at risk.
- The safe stowage and securing of cargoes depend on proper planning, execution and supervision.
- Personnel commissioned to tasks of cargo stowage and securing should be properly qualified and experienced.
- Personnel planning and supervising the stowage and securing of cargo should have a sound practical knowledge of the application and content of this Cargo Securing Manual.
- In all cases, improper stowage and securing of cargo will be potentially hazardous to the securing of other cargoes and to the ship itself.
- Decisions taken for measures of stowage and securing cargo should be based on the most severe weather conditions which may be expected by experience for the intended voyage.
- Ship-handling decisions taken by the master, especially in bad weather conditions, should take into account the type and stowage position of the cargo and the securing arrangements.

2. Cargo carried in a cargo unit shall be packed and secured within the unit so as to prevent, throughout the voyage, damage or hazard to the ship, to the persons on board and to the marine environment.

3. Appropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with abnormal physical dimensions to ensure that no structural damage to the ship occurs and to maintain adequate stability throughout the voyage.

4. Cargo should be stowed such that the design loads for main deck, hatch covers and inner bottom, indicated in this manual, are not exceeded. Allowable weights are found in Section 1.4 for hatch covers and Section 3.2 for inner bottom.

5. Cargo should be stowed such that the design loads / safe working loads for the containers and lashing system are not exceeded. This can be accomplished by following allowable stack weights given in Section 3.4 of this manual.

C9 CLASS

GEORGE II

Principal Characteristics

Length Over All	272.29 M	893.33 Ft.
Length Between Perpendiculars	243.03 M	797.33 Ft.
Breadth (molded)	30.48 M	100.00 Ft.
Depth (molded)	18.29 M	60.00 Ft.
Draft (summer load line)	12.44 M	40.81 Ft.
Draft (design)	8.53 M	28.00 Ft.
Displacement @ Summer Load Line	63,161 MT	62,163 LT
Deadweight @ Summer Load Line	45,153 MT	44,439 LT
Displacement @ Design Draft	38,621 MT	38,011 LT
Deadweight @ Design Draft	20,613 MT	20,287 LT
Lightship Weight	18,008 MT	17,724 LT
Permanent Ballast	0 MT	0 LT

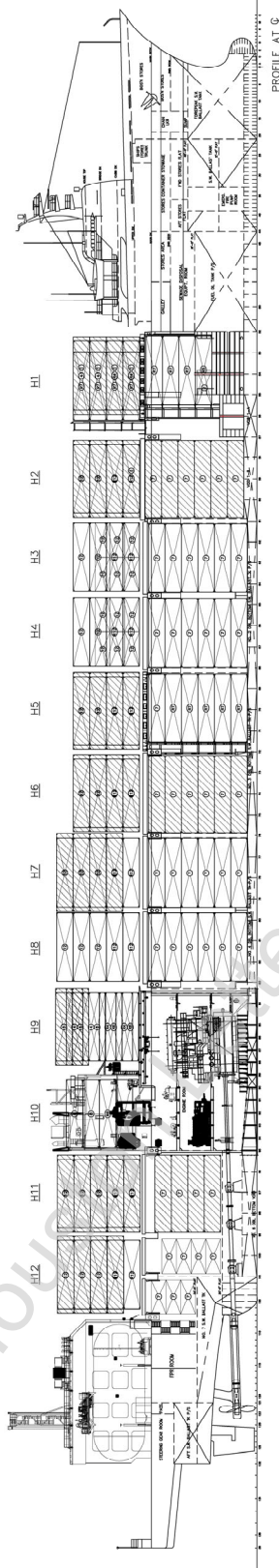
Maximum TEU Capacity 2,089

Port of Registry	Honolulu, HI
IMO Number	7729461
Official Number	625873
ABS ID No.	8012487
Call Letters	WFLH

Main Engine	MAN B&W 8G60ME-C10.5-GI-EGRBP
Rating (MCR)	22,720KW @ 103 rpm
Rating (NCR)	20,198KW @ 99 rpm
Service Speed	22.0 Knots @ NCR

Generators	Hyundai 9H27DF
Output	3x2,565KW @ 900 rp

C9 CLASS
 MV GEORGE II

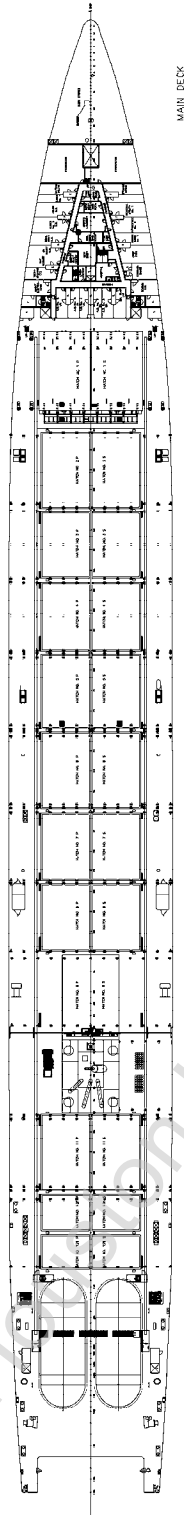


CONTAINER DATA

HATCH NUMBER	12	11	10	9	8	7	6	5	4	3	2	1	TOTALS
MAX. 40' STD. (45') CONT. ON COVER	60 (48)	60 (60)	32	56 (20)	60	60 (48)	48 (48)	48 (48)	48	48	48 (42)	44 (44)	612 (358)
MAX. 40' HI-C (45') CONT. ON COVER	60 (48)	60 (60)	32	56 (20)	60	60 (48)	48 (48)	48 (48)	48	48	48 (42)	44 (44)	612 (358)
MAX. 20' CONT. ON COVER									72	72			144
REEFER OUTLETS ON COVER	24	24	16	24	24	24	24	24	24	24	24	36	292
MAX. 40' STD. CONT. IN CELLS					42	42		42	42	42		26	236
MAX. 40' HI-C CONT. IN CELLS					42	42		42	42	42		26	236
MAX. 45' CONT. IN CELLS		28					42				42		112
MAX. 20' CONT. IN CELLS	49											2	51
REEFER PLUGS BLW. IN CELLS								35				26	61
TOTAL MAX. 40' STD. (45') CONTAINERS	60 (48)	60 (68)	32	56 (20)	102	102 (48)	48 (90)	90 (48)	90	90	48 (84)	70 (44)	848 (470)
TOTAL MAX. 40' STD. (45') CONTAINERS	60 (48)	60 (68)	32	56 (20)	102	102 (48)	48 (90)	90 (48)	90	90	48 (84)	70 (44)	848 (470)
TOTAL MAX. 20' CONTAINERS	49								72	72		2	195
TOTAL REEFER OUTLETS	24	24	16	24	24	24	24	59	24	24	24	62	353

December 2023

C9 CLASS
 MV GEORGE II



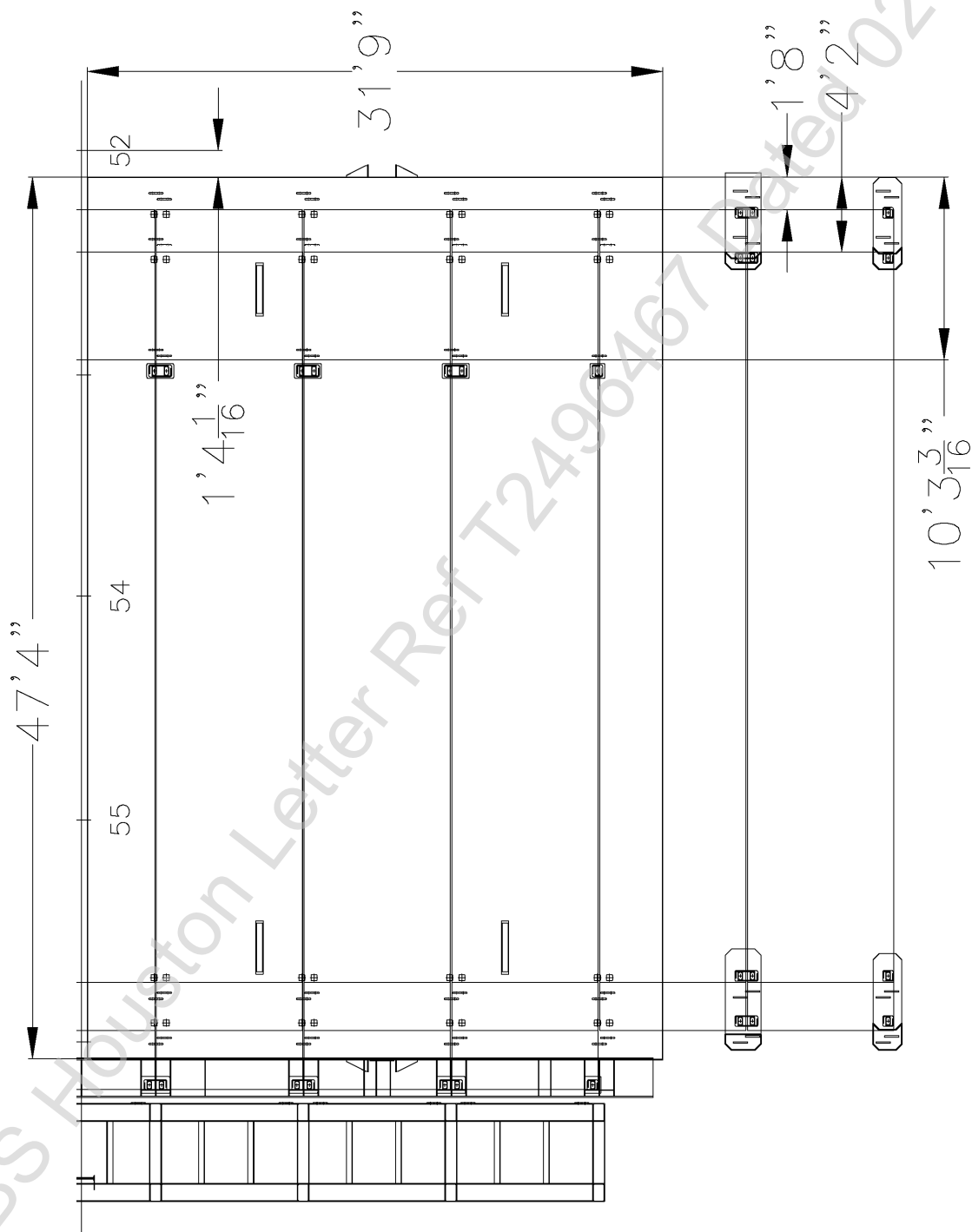
HATCH COVER DIMENSIONS

HATCH	HOLD	SIZE: LENGTH x WIDTH	WT. (LT)	QTY.
1 P/S	NO. 1 FWD	47'4" x 31'9"	28.2	2
2 P/S	NO. 1 AFT	46'5" x 31'9"	26.3	2
3 P/S	NO. 2 FWD	41'0" x 31'9"	24.1	2
4 P/S	NO. 2 AFT	41'0" x 31'9"	24.1	2
5 P/S	NO. 3 FWD	44'11" x 31'9"	25.7	2
6 P/S	NO. 3 AFT	46'5" x 31'9"	25.7	2
7 P/S	NO. 4 FWD	40'10" x 31'9"	24.1	2
8 P/S	NO. 4 AFT	40'10" x 31'9"	24.1	2
11 P/S	NO. 6 FWD	46'5" x 31'9"	26.0	2
12A P/S	NO. 6 AFT	21'11" x 31'9"	26.0	2
12B P/S	NO. 7	21'11" x 31'9"	26.4	2

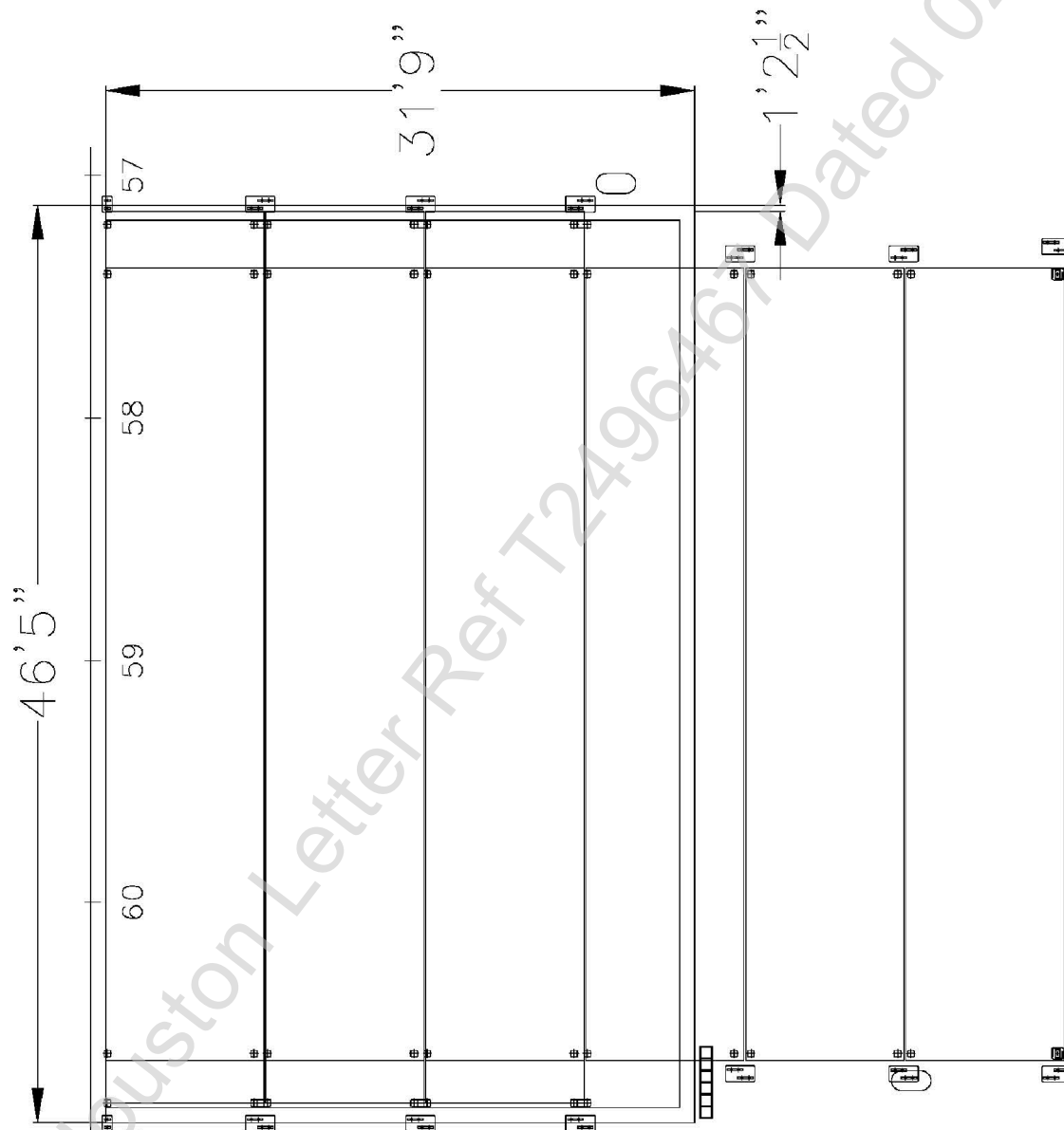
STACKING WEIGHT ON DECK

	ALL CARGO HOLD
20 FT	24 LT
40/45 FT	90 LT

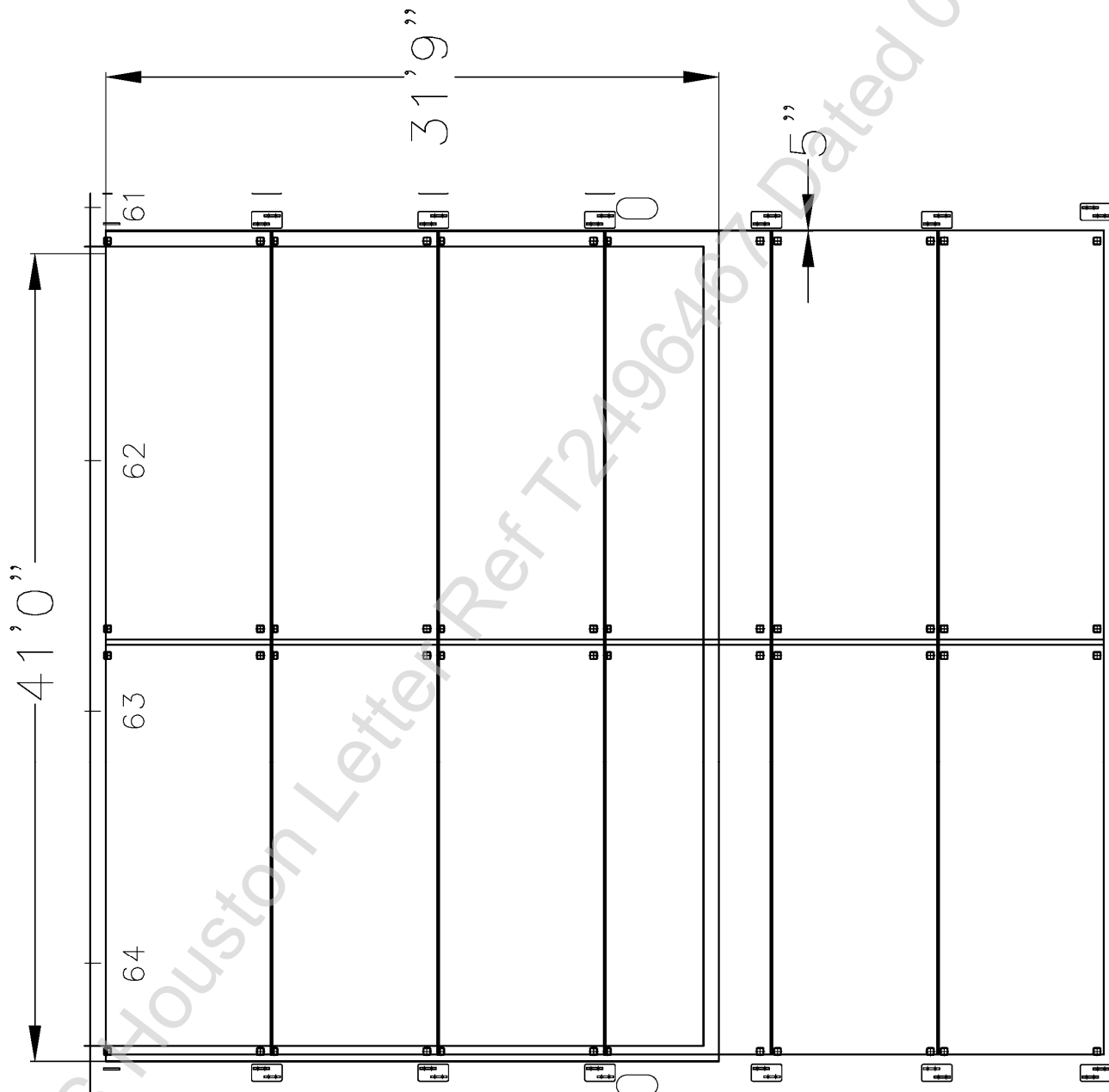
HATCH 1



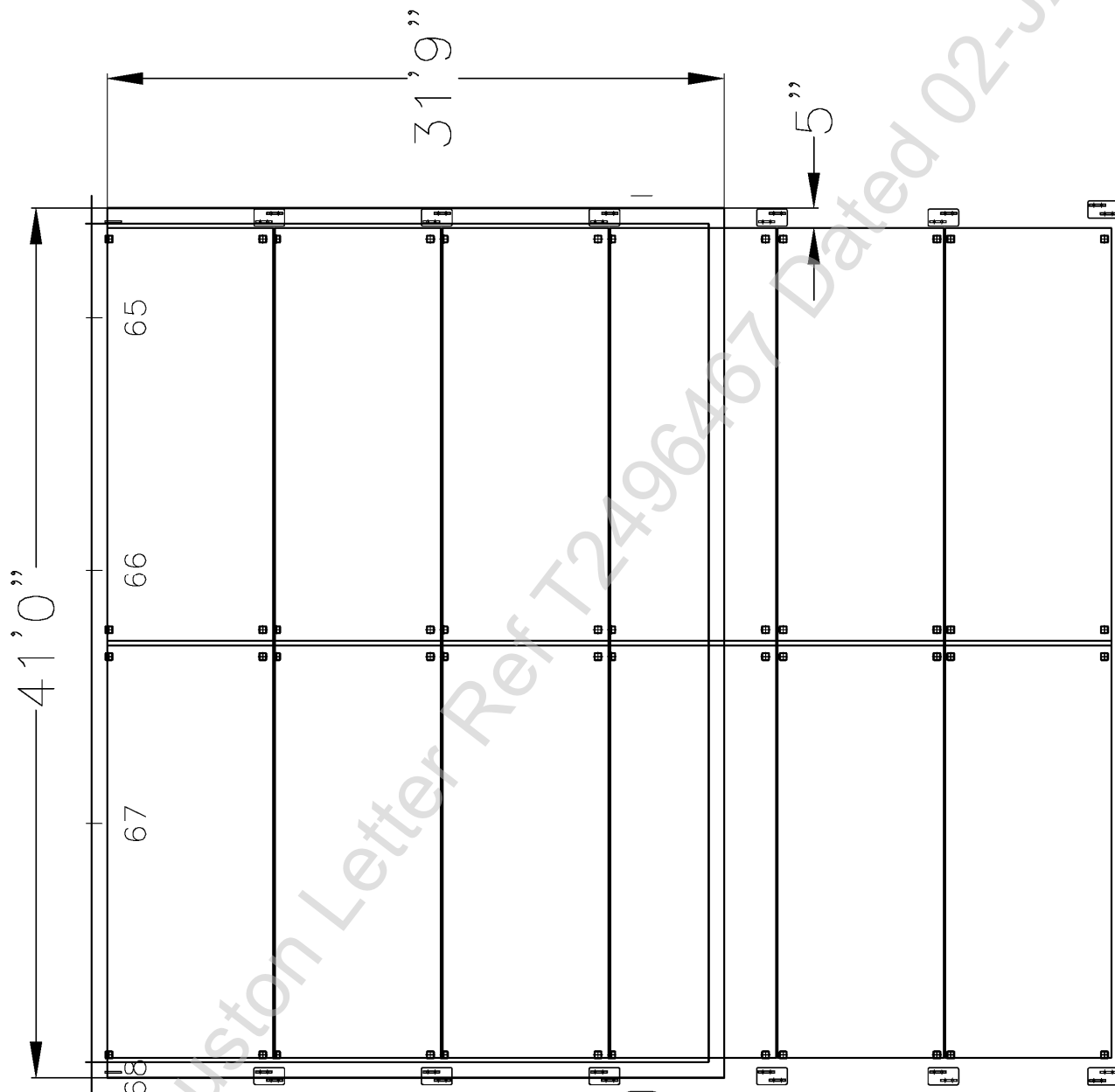
HATCH 2



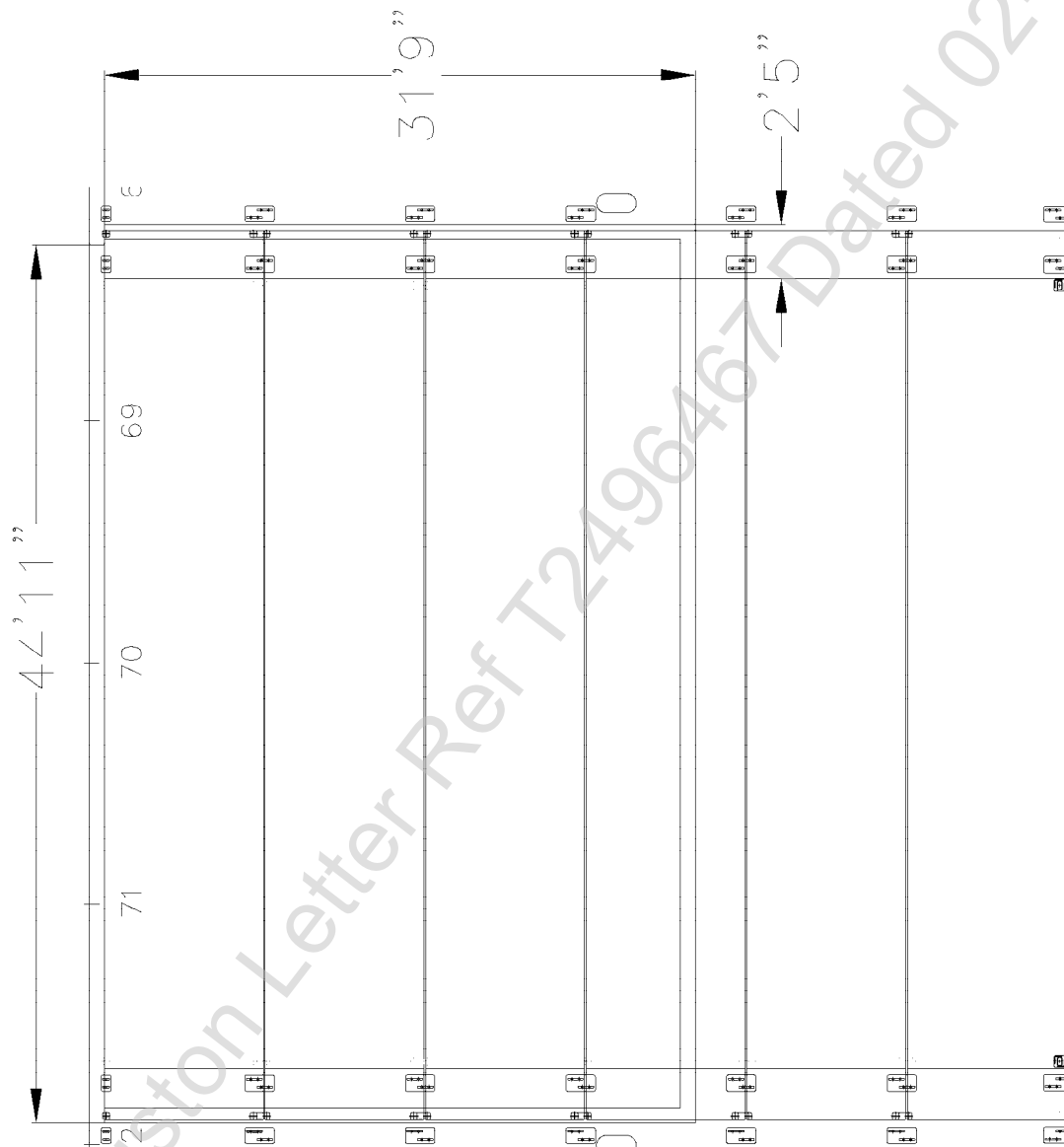
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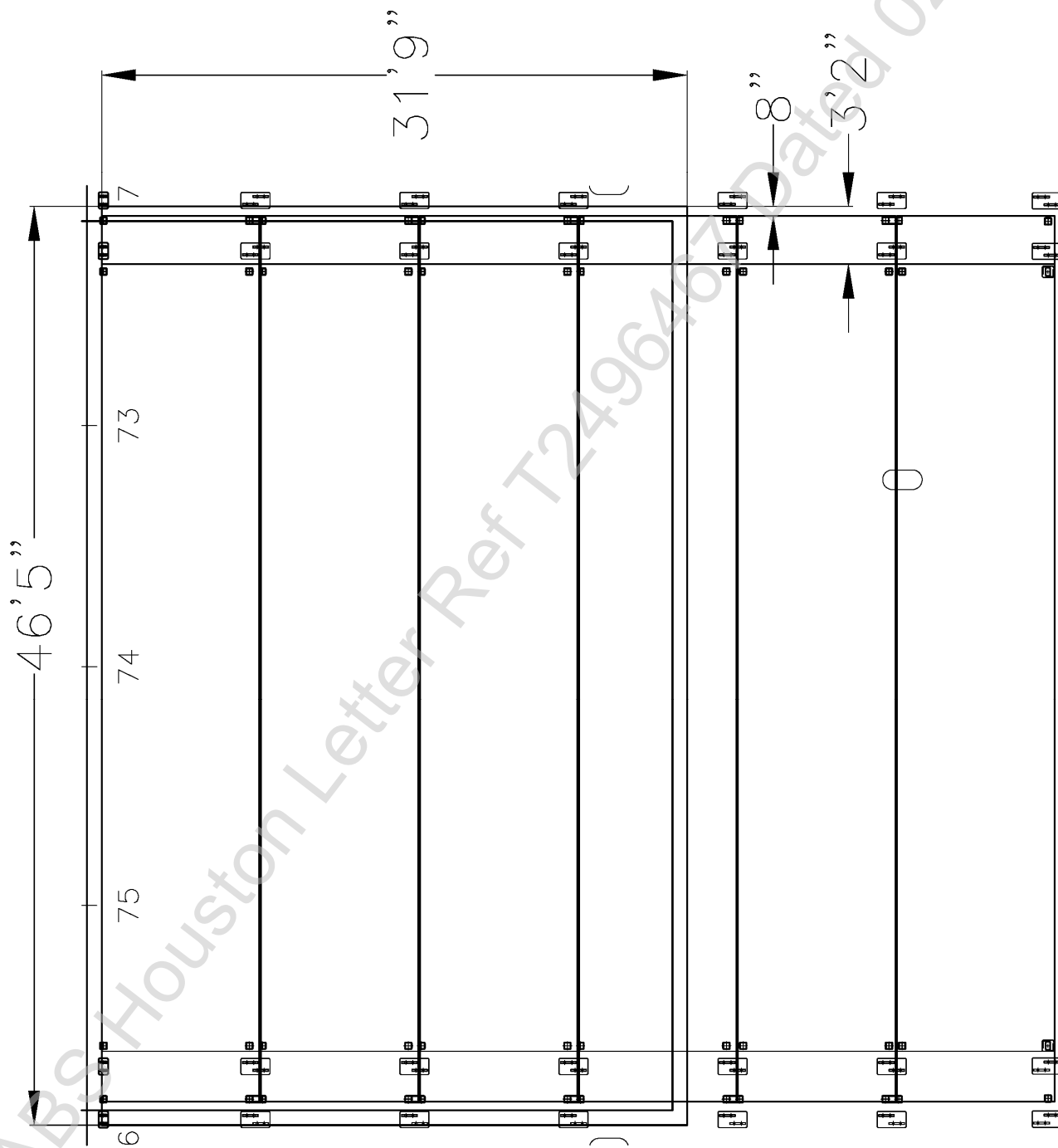
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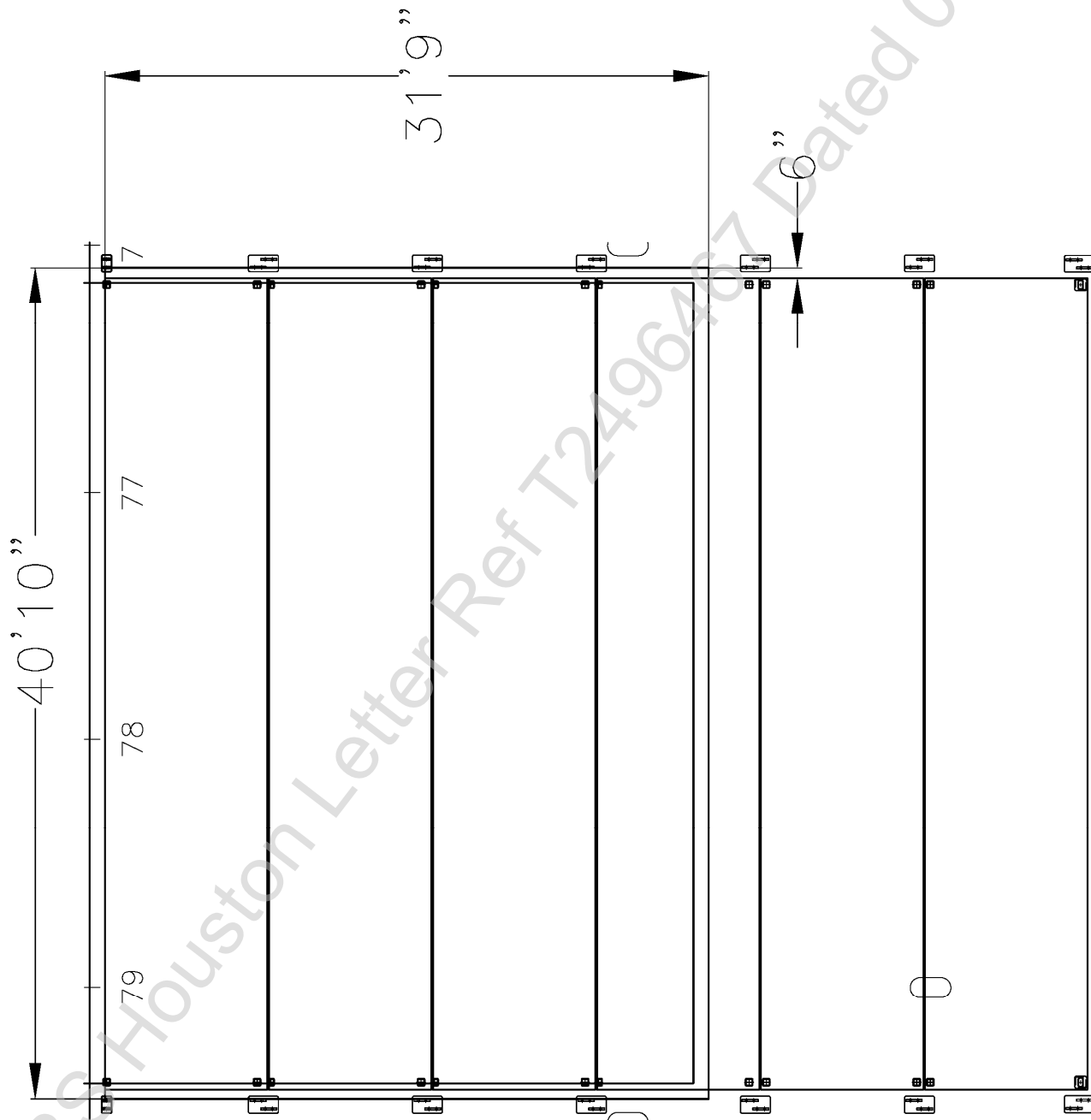
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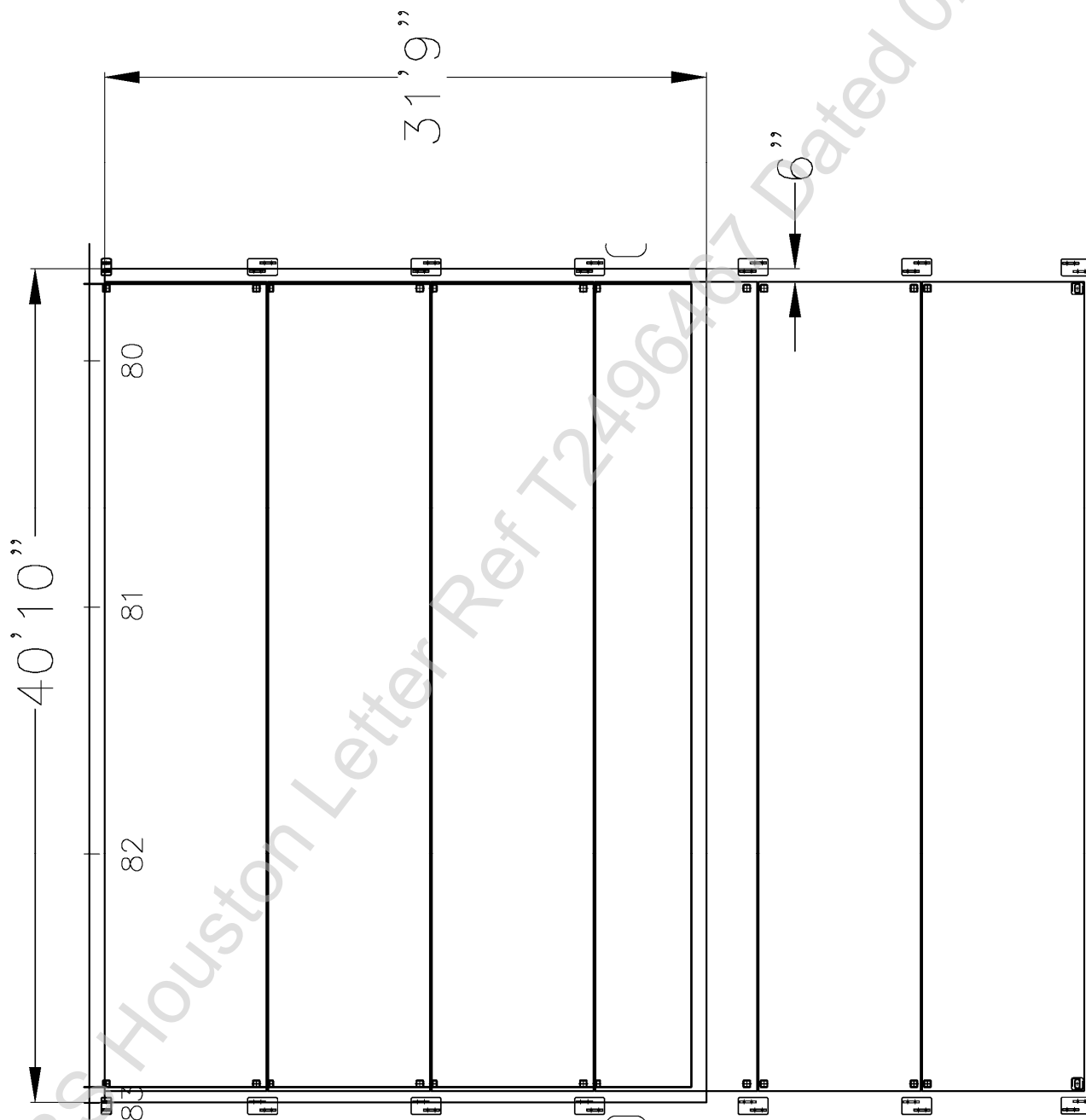
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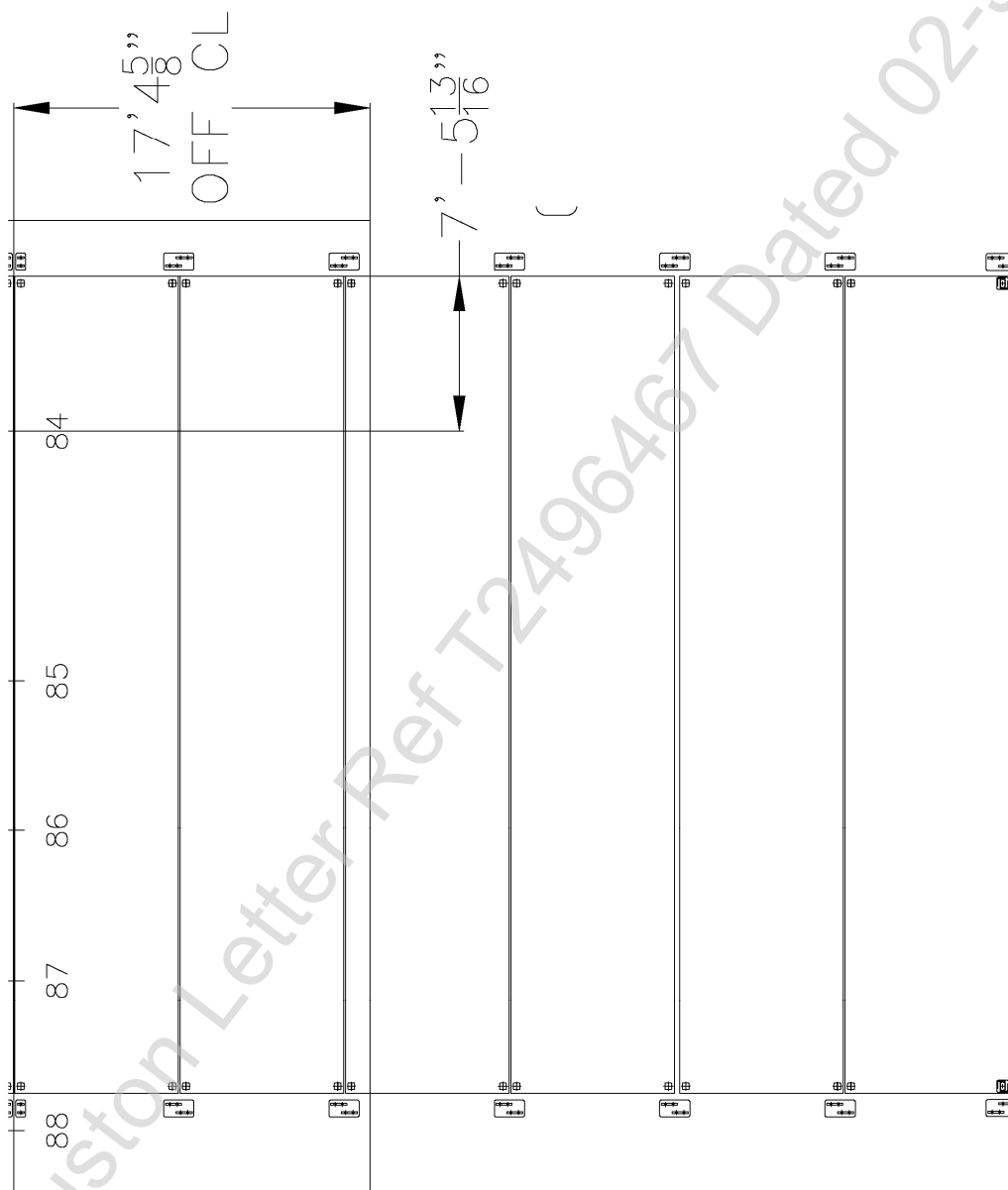
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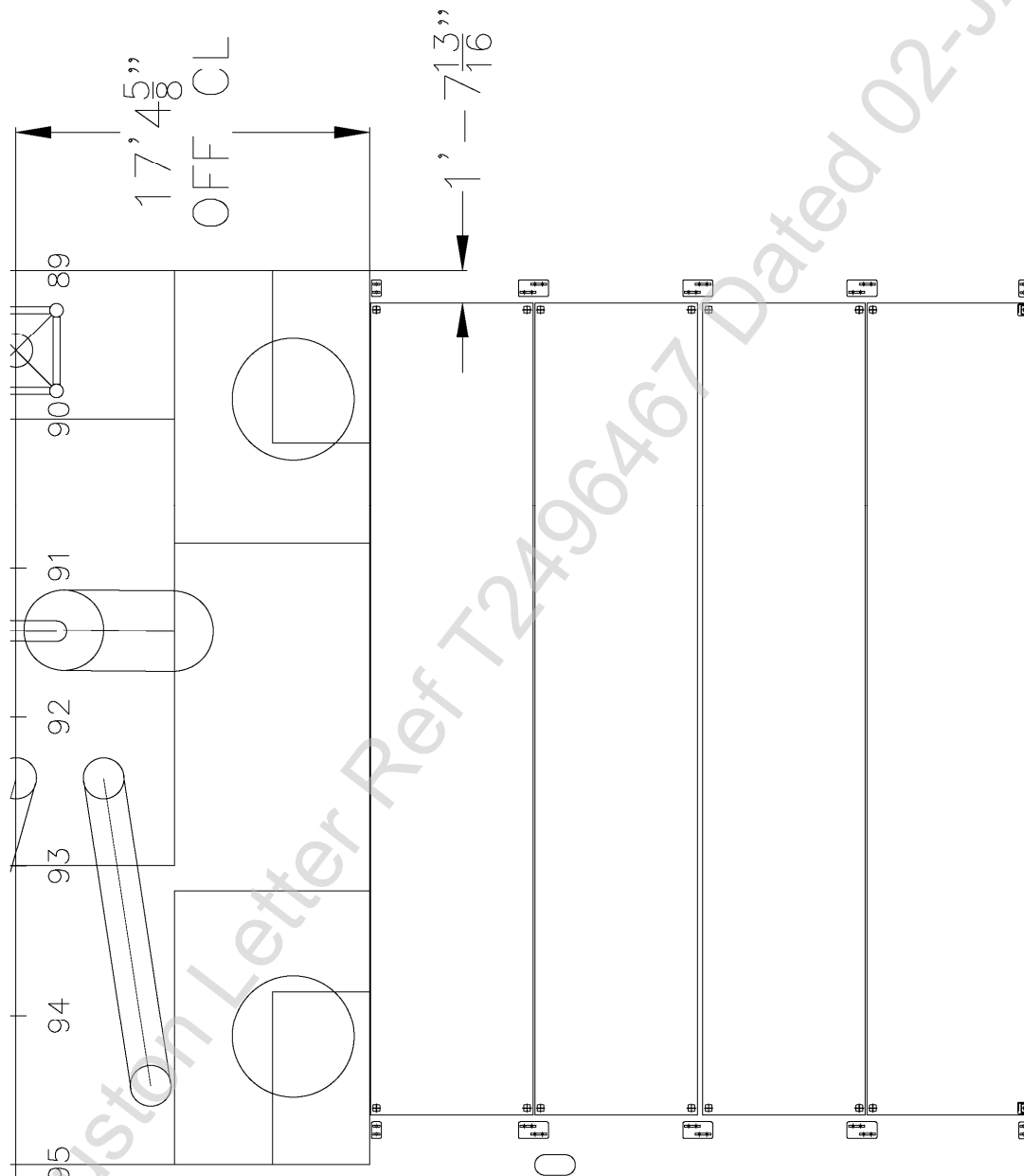
HATCH 8



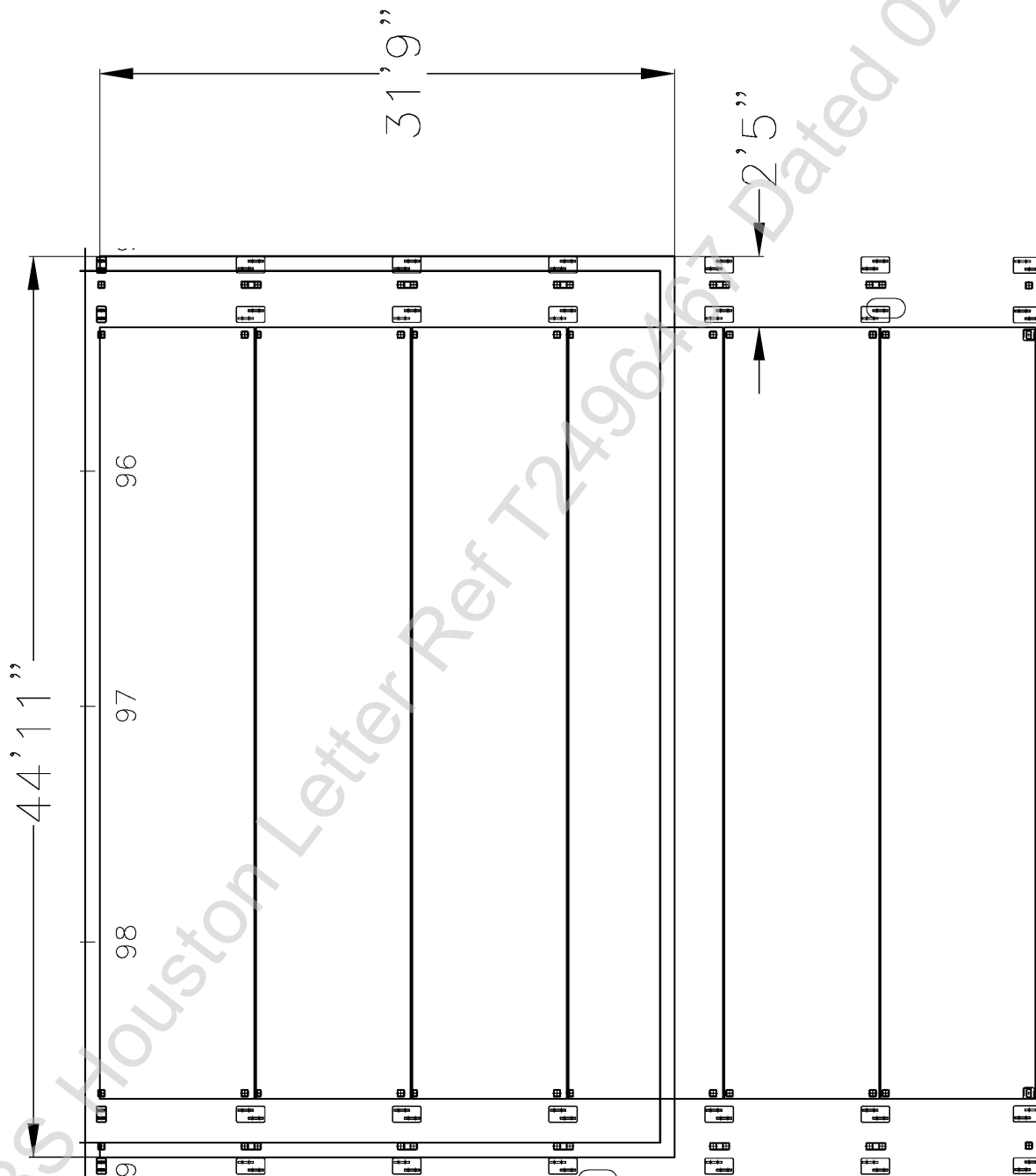
HATCH 9 (OVER ER)



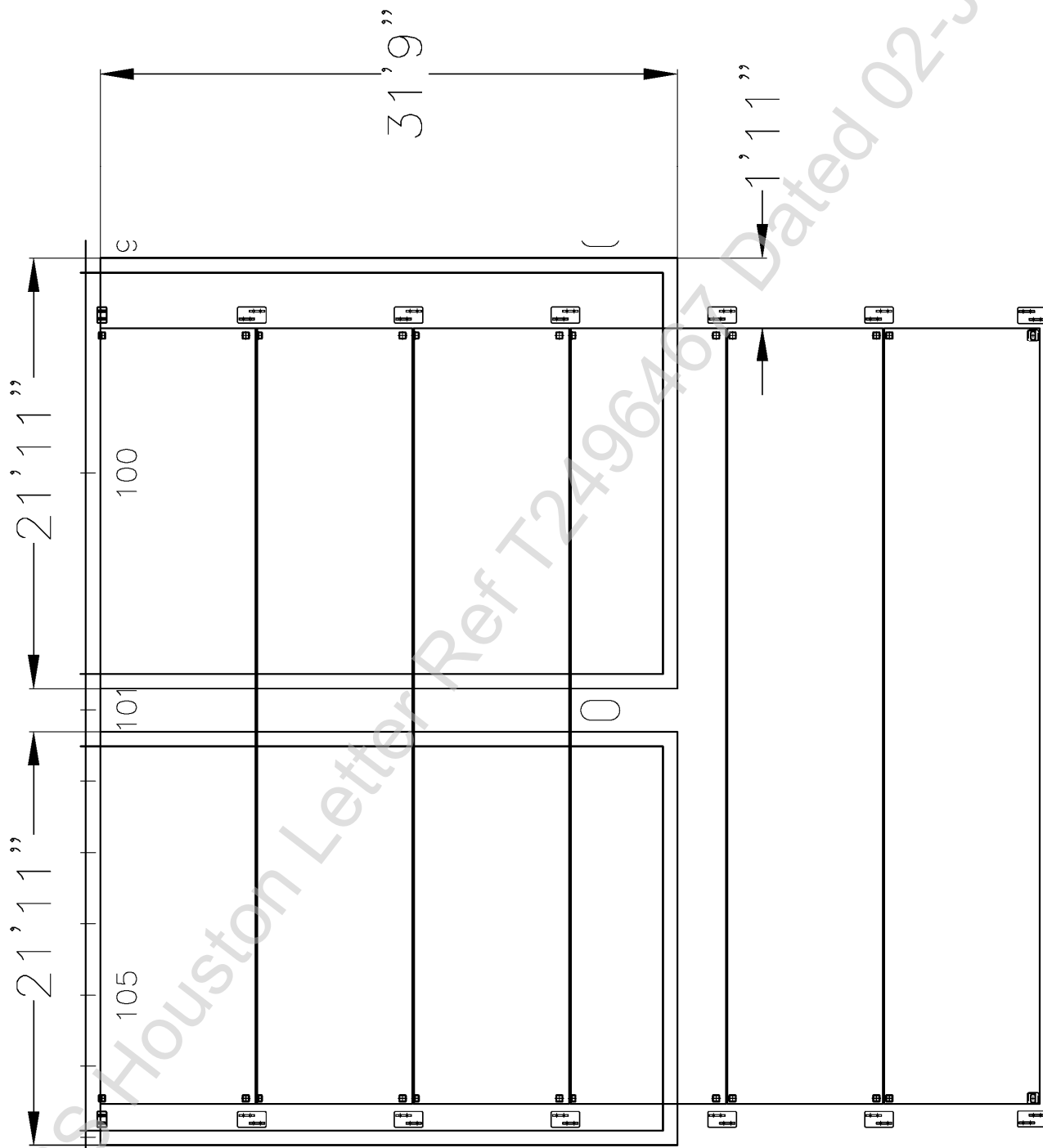
HATCH 10 (OVER ER)



HATCH 11

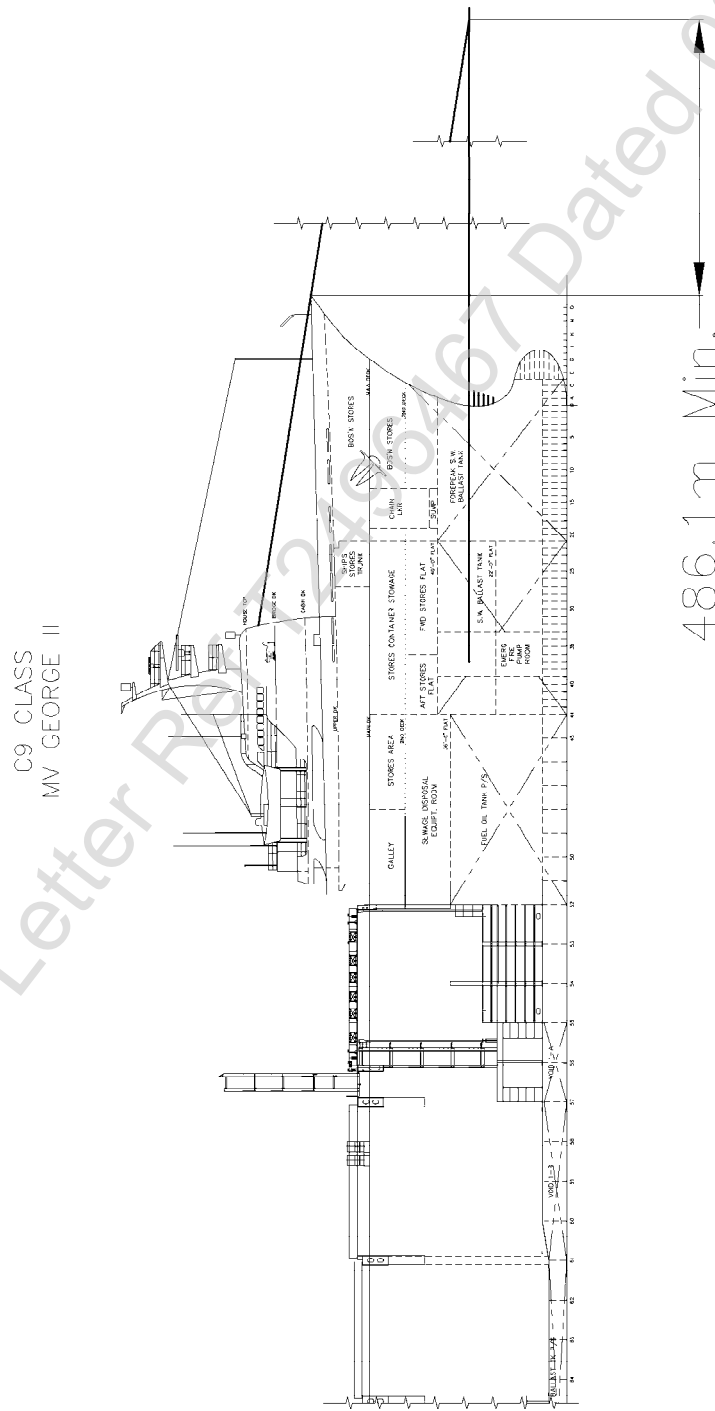


HATCH 12



1.5 Visibility Restrictions

MV George II C9 class vessel has no visibility restrictions. All containers are stowed aft of the conning position.



IMDG CARGO SHALL BE LOADED AS FOLLOWS:

CLASSES	Loading Area	
	CARGO HOLD NO.2 (HATCH 3 & 4)	On Hatch Covers, except HATCH 8, 9, 10 & 12
1.1 to 1.6	-	OK
1.4S	OK	OK
2.1	OK	OK
2.2	OK	OK
2.3 flammable ²⁰	OK	OK
2.3 non-flammable	OK	OK
3 F.P ¹⁵ < 23° C	OK	OK
3 F.P ¹⁵ ≥ 23°C to ≤ 60°C	OK	OK
4.1	-	OK
4.2	-	OK
4.3 liquids ²¹	-	OK
4.3 solids	-	OK
5.1	-	OK
5.2 ¹⁶	-	OK
6.1 liquids F.P ¹⁵ < 23° C	-	OK
6.1 liquids F.P ¹⁵ ≥ 23°C to ≤ 60°C	-	OK
6.1 liquids	-	OK
6.1 solids	-	OK
8 liquids F.P ¹⁵ < 23° C	OK	OK
8 liquids F.P ¹⁵ ≥ 23°C to ≤ 60°C	OK	OK
8 liquids	OK	OK
8 solids	OK	OK
9	OK	OK

In addition to complying with this table, the individual stowage & segregation restrictions for each commodity must be checked.

Notes

¹¹ When "mechanically-ventilated spaces" are required by the IMDG Code.

¹² Stow 3m horizontally away from the machinery space boundaries in all cases.

¹³ Refer to the IMDG Code.

¹⁴ As appropriate to the goods to be carried.

¹⁵ FP means flashpoint

¹⁶ Under the provisions of the IMDG Code, stowage of class 5.2 dangerous goods under deck or in enclosed Ro-Ro spaces is prohibited.

¹⁷ Only applicable to dangerous goods evolving flammable vapour listed in the IMDG Code.

¹⁸ Only applicable to dangerous goods having a flashpoint less than 23° C listed in the IMDG Code.

¹⁹ Only applicable to dangerous goods having a subsidiary risk class 6.1.

²⁰ Under the provisions of the IMDG Code, stowage of class 2.3 having subsidiary risk class 2.1 under deck or in enclosed Ro-Ro spaces is prohibited.

²¹ Under the provisions of the IMDG Code, stowage of class 4.3 liquids having a flashpoint less than 23° C under deck or in enclosed Ro-Ro spaces is prohibited.

TABLE 176.83(B) - General Segregation Requirements for Hazardous Materials
(Segregation must also take account of a single secondary hazard label, as required by 49CFR 176.83(a)(6))

Class	1.1 1.2 1.5	1.3	1.4 1.6	2.1	2.2	2.3	3	4.1	4.2	4.3	5.1	5.2	6.1	6.2	7	8	9
Explosives, 1.1, 1.2, 1.5	(*)	(*)	(*)	4	2	2	4	4	4	4	4	4	2	4	2	4	X
Explosives, 1.3	(*)	(*)	(*)	4	2	2	4	3	3	4	4	4	2	4	2	2	X
Explosives, 1.4., 1.6	(*)	(*)	(*)	2	1	1	2	2	2	2	2	2	X	4	2	2	X
Flammable gases 2.1	4	4	2	X	X	X	2	1	2	X	2	2	X	4	2	1	X
Non-toxic, non-flammable gases 2.2	2	2	1	X	X	X	1	X	1	X	X	1	X	2	1	X	X
Poisonous gases 2.3	2	2	1	X	X	X	2	X	2	X	X	2	X	2	1	X	X
Flammable liquids 3	4	4	2	2	1	2	X	X	2	1	2	2	X	3	2	X	X
Flammable solids 4.1	4	3	2	1	X	X	X	X	1	X	1	2	X	3	2	1	X
Spontaneously combus- tible substances 4.2	4	3	2	2	1	2	2	1	X	1	2	2	1	3	2	1	X
Substances which are dangerous when wet 4.3	4	4	2	X	X	X	1	X	1	X	2	2	X	2	2	1	X
Oxidizing substances 5.1	4	4	2	2	X	X	2	1	2	2	X	2	1	3	1	2	X
Organic peroxides 5.2	4	4	2	2	1	2	2	2	2	2	2	X	1	3	2	2	X
Poisons 6.1	2	2	X	X	X	X	X	X	1	X	1	1	X	1	X	X	X
Infectious substances 6.2	4	4	4	4	2	2	3	3	3	2	3	3	1	X	3	3	X
Radioactive materials 7	2	2	2	2	1	1	2	2	2	2	1	2	X	3	X	2	X
Corrosives 8	4	2	2	1	X	X	X	1	1	1	2	2	X	3	2	X	X
Miscellaneous dangerous substances 9	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Numbers and symbols relate to the following terms as defined in this section:

- 1 - "Away From".
- 2 - "Separated From."
- 3 - "Separated by a complete compartment or hold from."
- 4 - "Separated longitudinally by an intervening complete compartment or hold from."
- X - The segregation, if any, is shown 49CFR 172.101 (table).
- * - See 49CFR 176.144 for segregation within Class 1.

Table 176.83(F) - Segregation of Containers on Board Container Ships

	Vertical			Horizontal						
Segregation Requirement	Closed versus Closed	Closed versus Open	Open Versus Open		Closed Versus Closed		Closed Versus Open		Open versus Open	
					On Deck	Under Deck	On Deck	Under Deck	On Deck	Under Deck
1. "Away From".....	One on top of the other permitted.	Open on top of closed permitted. Otherwise as for open versus open.	Not in the same vertical line unless segregated by a deck.	Fore and aft. Athwartships	No restriction No restriction	No restriction No restriction	No restriction No restriction	No restriction No restriction	One container space. One container space.	One container space or one bulk-head. One container space.
2. "Separated from".	Not in the same vertical line unless segregated by a deck.	As for open versus open.	Not in the same vertical line unless segregated by a deck.	Fore and aft. Athwartships	One container space. One container space.	One container space or one bulk-head. One container space. One bulkhead	One container space. One container space. One bulkhead	One container space or one bulk-head. Two container spaces. One bulkhead	One container space. Two container spaces.	One bulkhead. One bulkhead. Two bulk-heads
3. "Separated by a complete compartment or hold from".	Not in the same vertical line unless segregated by a deck.	As for open versus open.	Not in the same vertical line unless segregated by a deck.	Fore and aft. Athwartships	One container space. Two container spaces.	 One bulkhead	One container space. Two container spaces.	 One bulkhead	Two container spaces. Three container spaces.	Two bulk-heads Two bulk-heads
4. "Separated longitudinally by an intervening complete compartment or hold from".	Prohibited.....			Fore and aft. Athwartships	Four container spaces. Prohibited.....	One bulkhead and four container spaces*. Prohibited.....	Four container spaces. Prohibited.....	Two bulk-heads. Prohibited.....	Four container spaces. Prohibited.....	Two bulk-heads Prohibited.

*Containers not less than 6 meters (20 feet) from intervening bulkhead.

Note: All bulkheads and decks must be resistant to fire and liquid.

2.1 Handling and Safety Instructions

This sub-chapter contains instructions on the proper handling and use of the securing devices found on board. It also contains the instructions for the proper method of securing and un- securing containers or other standardized cargo by ship or shore personnel.

Detailed instructions, supplied by lashing gear manufacturers are provided in Sections 2.2 and 2.3 of this chapter. Other manufacturer's literature that offers guidance in the proper use of all equipment is also provided in Sections 2.2 and 2.3.

A thorough inspection should be done at the completion of cargo operations to ensure that the containers are properly secured. The inspection shall include an assessment of any stevedore damage that might have occurred during cargo operations. Vessel's cargo gear not in use (lashing rods, rod extensions, turnbuckles, twist locks) and ships fixed equipment (D rings, lashing plates, deck sockets, pedestal catwalks and hand rails) should be maintained as required.

Any improper stowage must be reported and corrected before the vessel sails.

While the vessel is still in port, any damaged or missing lashing gear shall be reported to the attention of the Marine Manager. In all cases Stevedore Damage reports must be properly filled out and distributed.

The Master's periodic or relieving report in regard to lashing gear shall include an inventory of lashing gear. The inventory of lashing gear may be conducted on a periodic or continuous basis at the discretion of the Master and the contingencies of the vessel type and service. If the inventory is not a complete inventory, as will be the case when a continuous inventory is used, comments should indicate what type or section of gear was inventoried.

Any deficiencies in lashing gear inventory should be indicated. Losses due to failure or excessive wear should be documented. Lashing gear losses due to stevedore error or mishandling should be documented. Documentation shall include the port where such error or mishandling caused the loss.

Lashing gear inventory sheets, records and reports should not be forwarded unless specifically requested. Requisitions for lashing gear should be forwarded separately from the Master's report.

2.2 SPECIFICATIONS OF FIXED SECURING DEVICES

The fixed securing gear aboard the C9 class *George II* consists of:

- 1) Fixed welded deck sockets and welded lashing points.
- 2) Each hatch uses rectangular type sockets and lashing plates, since the older round sockets have been replaced.
- 3) Fixed securing devices are welded to hatch covers or for the two outboard most stacks of containers welded to pedestals. Hatch 9 and 10 has four outboard stacks on pedestals.

Maker Contact Information

For new deck sockets and lashing eye plates:

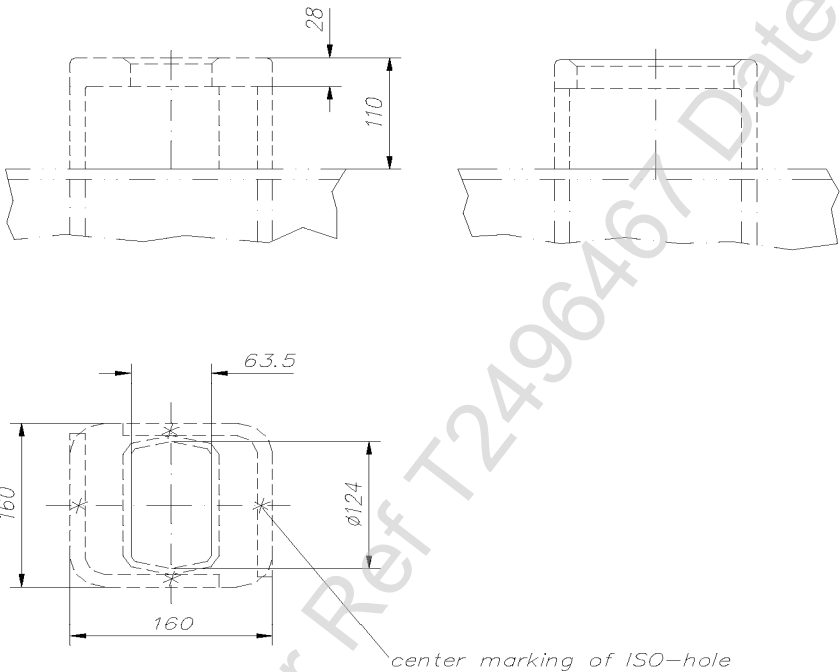
MacGregor Conver
Fritz-Thiele-Str 4
28279 Bremen
Germany
Tel 49-421-83-9183, Fax 49-421-83-91899

TWISTLOCK FOUNDATION

TF-11

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Order No.:



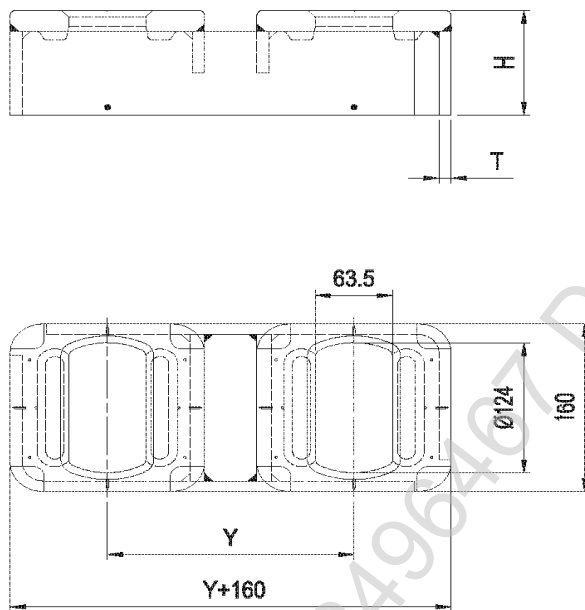
MSL = 25.0 t (Tens.) / 21.4 t (Shear)

Class approval	: ABS	Weight	: 7,1 kg
Breaking load F_B	$\geq$ 490 kN (Tens.) / 420 kN (Shear)	Index	: b
Testing load F_P	$\geq$ 269.5 kN (Tens.) / 231 kN (Shear)	Name	: SMU
Working load F_N	$\leq$ 245 kN (Tens.) / 210 kN (Shear)	Date	: 21.11.95
Finish	: untreated	Drawing No.	: A 1300.5100
Material	: High tensile steel	Date name	: B3005100

MacGREGOR
CONVER

TWISTLOCK FOUNDATION TF-12

TF-12



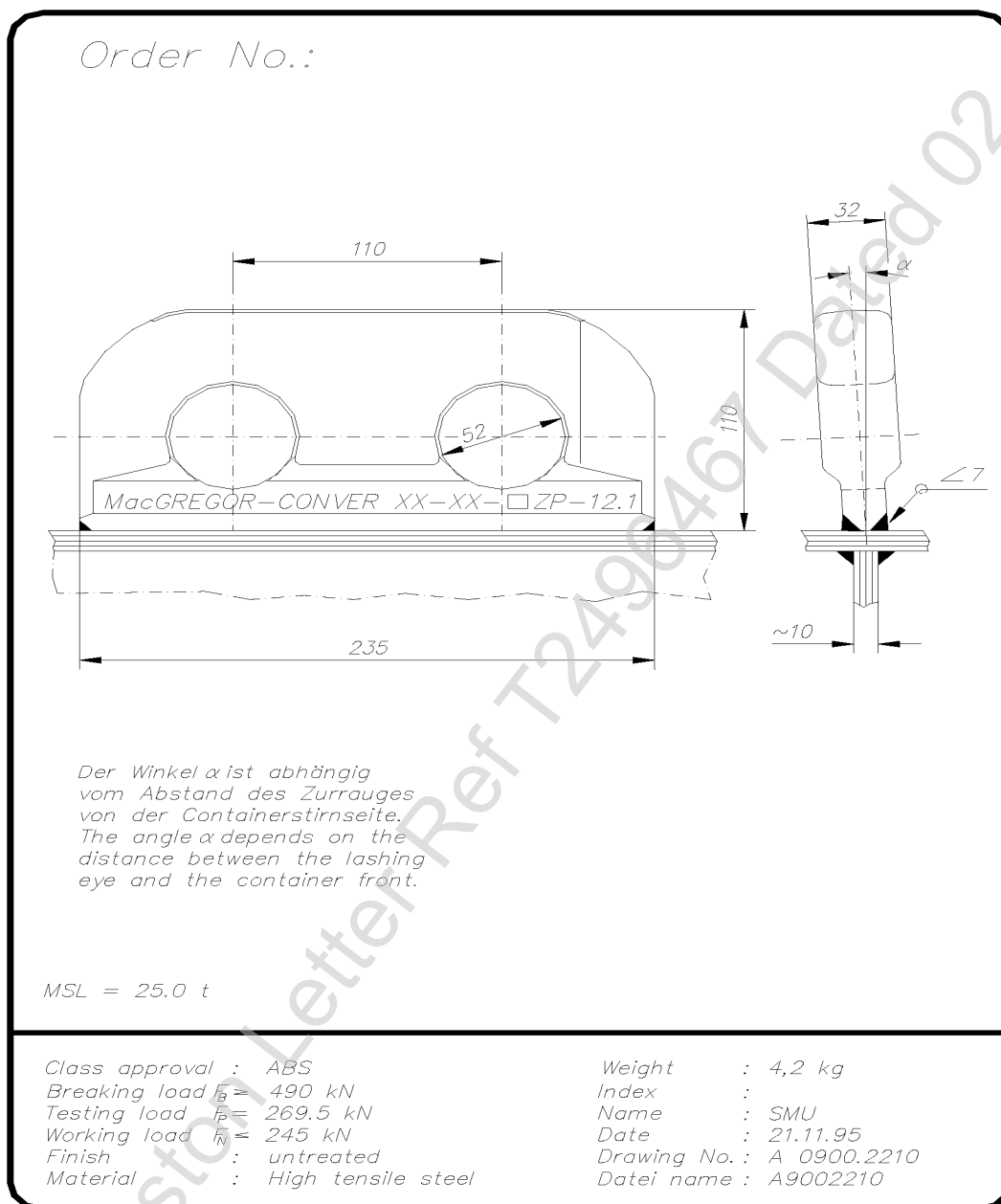
TF-11-12-13

Product	Height H mm	Distance Y mm	Length L mm	Hole length L1 mm	Hole length L2 mm	Sideplate thickness T mm	Material	Treatment	SWL Tension N	SWL Shear kN	Details	Counter part	Design
TF-12-1	70- 180	203,216,258,278, other hole distances Y also available	Y+160	124	124	10	High Tensile steel	Shotblasted and primed	250	210		Manual and semi- automatic twistlock	MacGregor

FIXED LASHING PLATE

ZP-12.1

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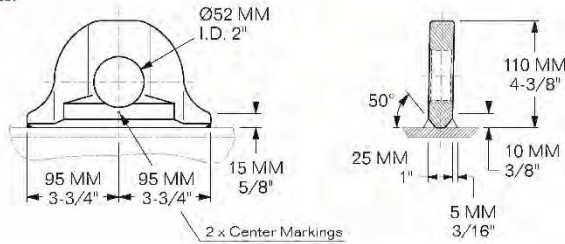
MacGREGOR
CONVER



LASHING EYES SINGLE & DOUBLE 52 MM HOLE, 50 M. TON B.L.



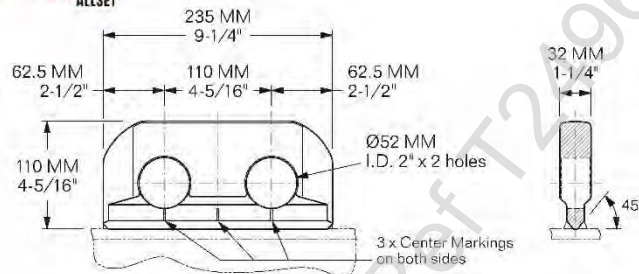
MODEL 5012-ZP-11



SUB FOR: MACGREGOR 1012760
MATERIAL: STRUCTURAL STEEL
FINISH: PRIMER
WEIGHT: 2.2 Kg / 4.9 LBS



MODEL 5012-ZP-12.1



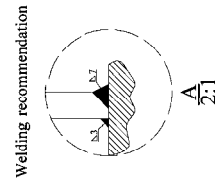
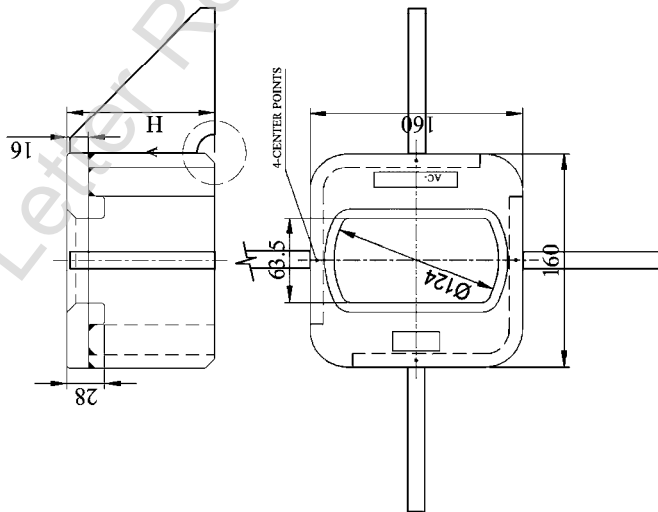
SUB FOR: MACGREGOR 1012769
MATERIAL: STRUCTURAL STEEL
FINISH: PRIMER
WEIGHT: 4.2 Kg / 9.3 LBS

www.pacificmarine.net

AG-A1

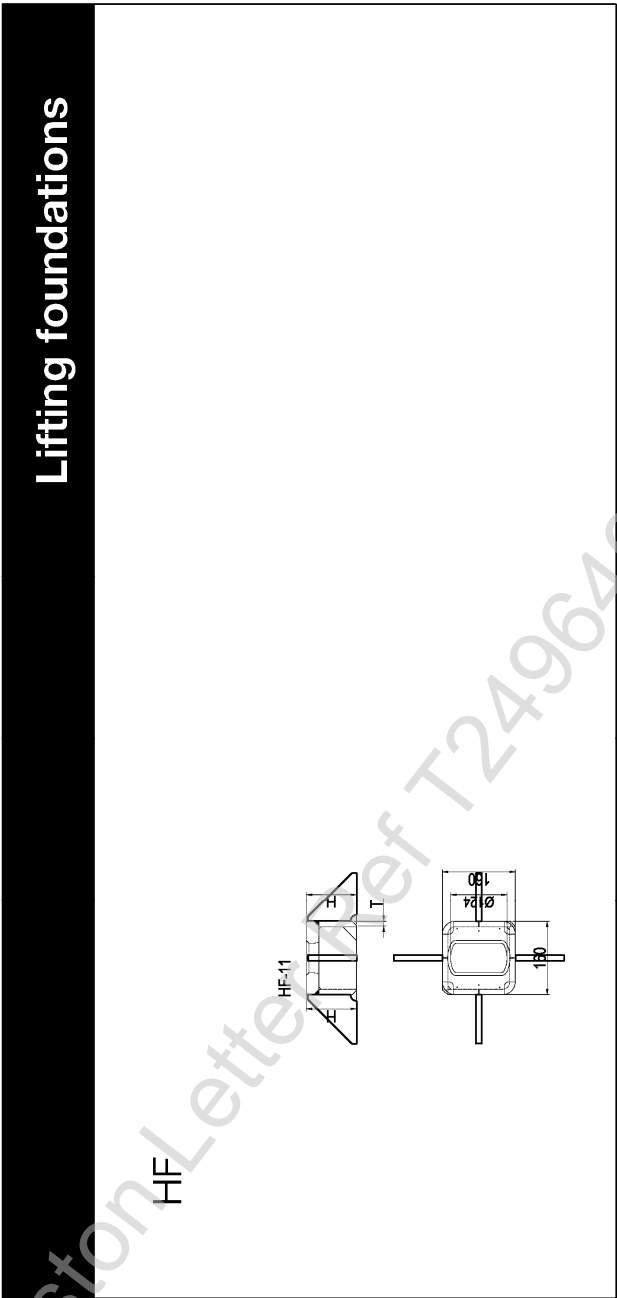
Pacific Marine & Industrial

LIFTING FOUNDATION NO EYE



TYPE	H	Weight
AG-A1/110	110	6.0 KG
Other height upon request		

LOADS	SHEAR [KN]	TENSION [KN]
BREAKING LOAD	420	500
SAFE WORKING	210	250



PRODUCT	HEIGHT H mm	DISTANCE Y mm	LENGTH L mm	HOLE LENGTH L1 mm	HOLE LENGTH L2 mm	SIDEPLATE THICKNESS T mm	MATERIAL	TREATMENT	SWL TENSION kN	SWL SHEAR kN	DETAILS	COUNTER PART	DESIGN
HF-11-1	110-160		160	124		12	HIGH TENSILE STEEL	SHOTBLASTED AND PAINTED	250	210		MANUAL, SEMI AUTOMATIC AND SPREADER TWISTLOCK	MACGREGOR
HF-11-2	110-160	PRODUCT	160	124	PRODUCT	12	HIGH TENSILE STEEL	SHOTBLASTED AND PAINTED	250	210	TOPPLATE HARDNESS MIN 235HB	MANUAL, SEMI AUTOMATIC AND SPREADER TWISTLOCK	MACGREGOR

Type Certificate

Certificate No.: 82731 HH 09



The Lashing Fitting **ZP-11**

of the manufacturer **Mac Gregor (Deu) GmbH**

was type tested according to the Rules and Guidelines of GL for Stowage and Lashing of Containers, edition 2007.

The main elements of the fitting are specified as follows:

Material: As specified in the drawing(s)

Part: Fixed Lashing Plate Drawing No.: 0900 2101k

Drawings approval No.: 04-105132 Date: 2004-08-12

Allowable working load: 245 KN

Place and date of test: Sundo, 2009-07-16

Breakload test results: 245 KN

245 KN

245 KN

At 306 KN test load no permanent deformation and no incipient cracks were detected.



Certificate expires on: 2014-09-07

Germanischer Lloyd

Hamburg, 2009-09-07

H. W. Sievers
Hans-Werner Sievers

F. Nitsch
Frank Nitsch

The latest edition of the „General Terms and Conditions for Classification“ and the „General Terms and Conditions for Activities other than Classification“ respectively are applicable. German law applies.

Type Certificate

Certificate No.: 01432 HH 2004



The Lashing Fitting **ZP-12.1**
of type Lashing Plate
of the manufacturer **Cargotec Finland Oy**

was type tested according to the Rules and Guidelines of GL for Stowage and Lashing of Containers, edition 2007.

The main elements of the fitting are specified as follows:

Material: as specified in the drawings
Part: Fixed Lashing Plate Drawing No.: 0900.2210h

Drawings approval No.: 04-105132 Date 2004-08-12

Allowable working load: 245 kN tension

Place and date of test: Dalian, P.R.China, 2012-04-18

Breakload test results: 490 kN
490 kN
490 kN

At 368 kN test load no permanent deformation and no incipient cracks were detected.



Certificate expires on: 2017-05-10

Germanischer Lloyd

Hamburg, 2012-05-10

i. A. Daniel Abt

i. A. Mathias Bohn

The latest edition of the „General Terms and Conditions for Classification“ and the „General Terms and Conditions for Activities other than Classification“ respectively are applicable. German law applies.

2.3 SPECIFICATIONS FOR PORTABLE SECURING DEVICES

During inspection of the *George II* during dry-dock of 2022 the all used portable devices are listed below.

Vessel is using Twistlock C5-AM-HC and BD-J2-HC for connection between hatch covers and pedestals to first tier of container, at all hatch covers.

For all other locations, vessel is using NA-3J and BD-J1 Twistlocks for connection between containers, above tier one.

Vessel is using Stacker Twistlocks on inboard, inaccessible ends of 20 feet containers.

Maker Contact Information:

Taiyo Seiki Iron Works Co., Ltd
10-2, 1-Chrome
Chiyosaki
Nishi-Ku, Osaka
Japan 550-0023
Tel 81-6-6582-1612, Fax 81-6-6584-7680

MacGregor Conver
Fritz-Thiele-Str4
28279 Bremen
Germany
Tel 49-421-83-9183, Fax 49-421-83-91899

BUFFERS, USA
PO Box 26947
Imeson Industrial Park
Jacksonville, FL 32218-0947
Tel. 904-696-0010, Fax. 904-696-0019

PACIFIC MARINE & INDUSTRIAL
1560 Jessie Street
Jacksonville, FL 32206
info@pacificmarine.net

Tayio Seiki Iron Works

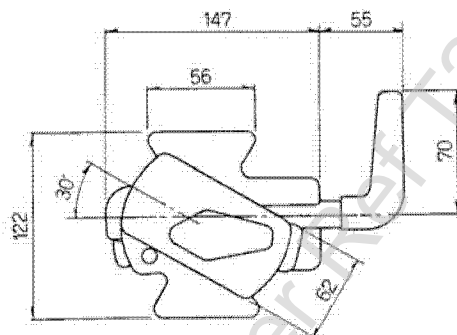
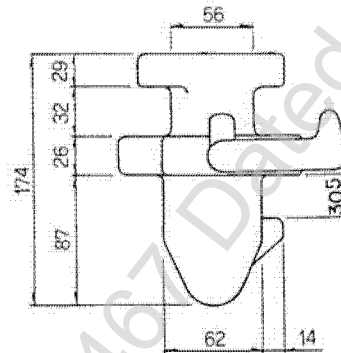
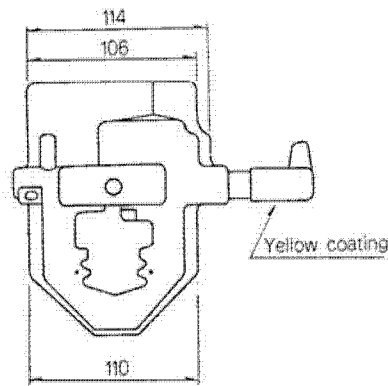
Model NA-7E Semi-Automatic Twist Lock

NAME

AUTO TWIST LOCK

MODEL NO.

Ⓐ NA-7 (SELF LOCK TYPE)



P.A.T

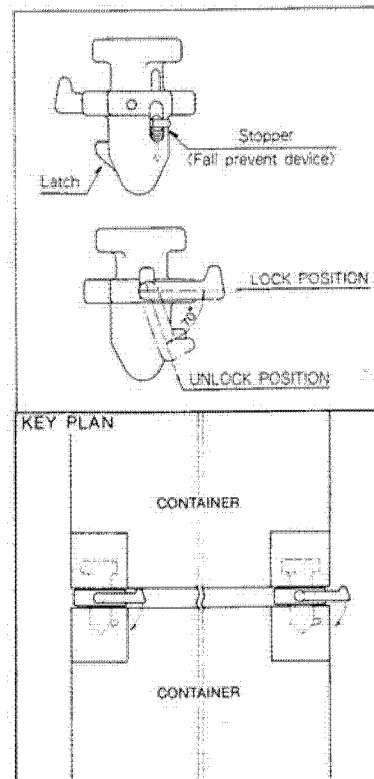
FINISH : Galvanized

WEIGHT : 6.9kg

SWL : 20 ton

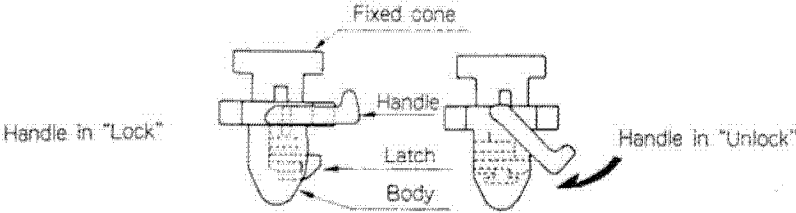
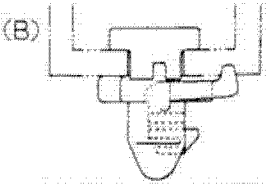
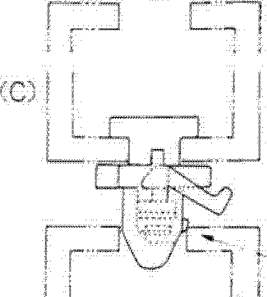
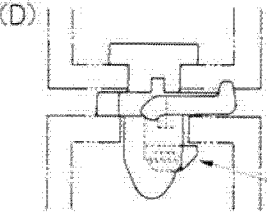
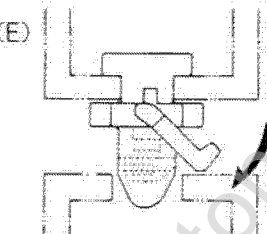
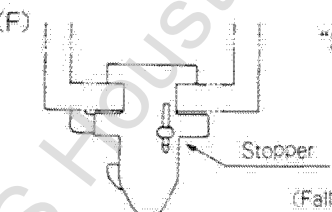
TL : 30 ton

(T : Type approved by Lloyd's Register of shipping.)



Maximum Securing Load (MSL) is equal to 20 tons.

OPERATION MANUAL FOR AUTO TWIST LOCK NA-7

- (A) 
- (B)  **"Set"** Setting the handle "Lock" position, insert fixed cone into bottom corner casting of a container to be stowed and twist it. Then fixed cone is locked and never fall down by any shock or vibration. Down the container with NA-7 on a container which already stowed.
- (C)  When a latch of NA-7 comes in touch with the top corner casting of lower container, the latch is pushed inwards.
- (D)  **"Lock"** When NA-7 comes in normal position, the latch is turned back with the internal spring and the both upper & lower containers are completely locked each other.
- (E)  **"Unlock"** For release, down the handle to the position of "Unlock". Lift up the upper container.
- (F)  **"Set by"** Pull down a "Stopper" and twist NA-7 clockwise. Draw it from corner casting.
 (Fall prevent device from corner casting)

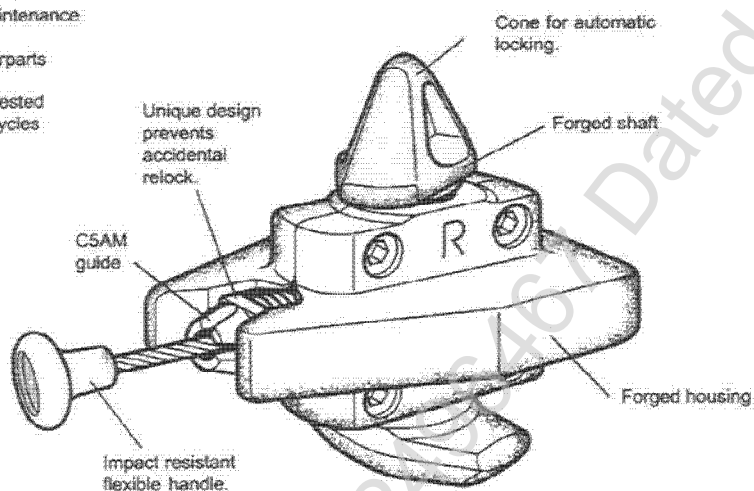
Semi-Automatic Twistlock – C5-AM-HC

Twistlock C5AM-HC Semi-automatic

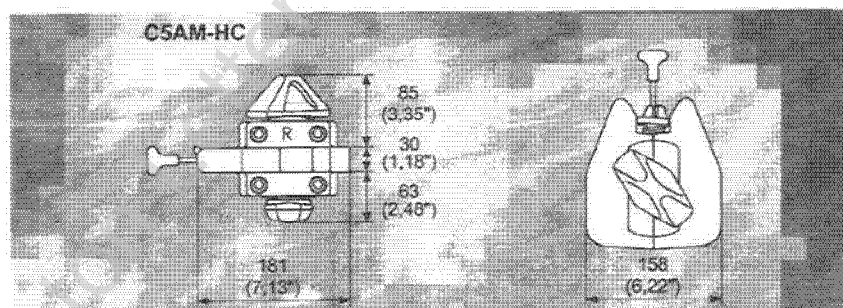
- Easy maintenance

- Few wearparts

- Fatigue tested 10 000 cycles



Item	Tension Breaking load (M.ton)	Shear Breaking load (M.ton)	Compression SWL (M.ton)	Weight (kg)
C5AM-HC	50	40	100	8,4



- Material: Forged shaft and housing. Stainless steel springs and wire
- Finish: Hot dip galvanized
- Class approval: All items can be supplied with the approval of any Classification Society

ALL SET MARINE LASHING

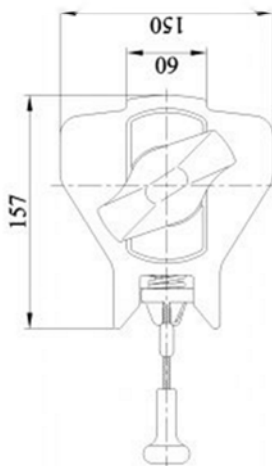
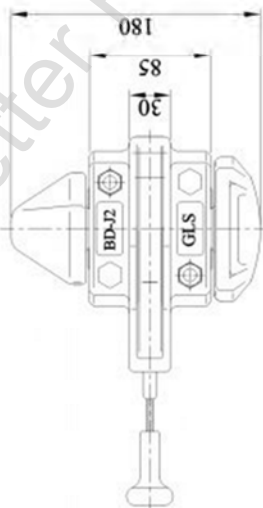
BUFFERS USA INC.
 Jacksonville, FL 32226
 Phone 904-596-0010
 Fax 904-596-0019

Semi-Automatic Twistlock – BD-J2-HC



BD-J2

SEMI-AUTOMATIC TWISTLOCK
(LARGE SEAT)

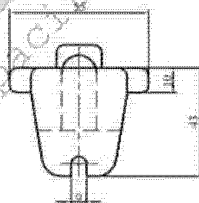
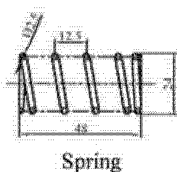
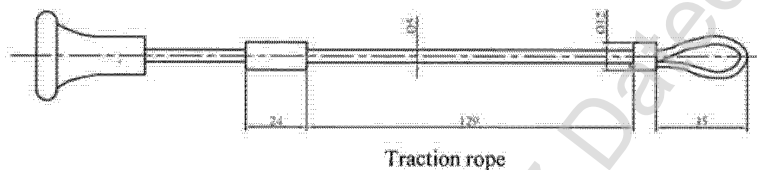


LOADS	SHEAR [KN]	TENSION [KN]
BREAKING LOAD	420	500
SAFE WORKING LOAD	210	250

TOTAL WEIGHT	5.45 KG
MATERIAL	FORGED STEEL
FINISH	HOT DIP GALVANISED

SEMI-AUTOMATIC TWISTLOCK

BD-J1 / BD-J2 Parts



Mouth Piece



Material: -

Finish: -

Weight: -

M.B.L: Tension: -

M.B.L: Shear: -

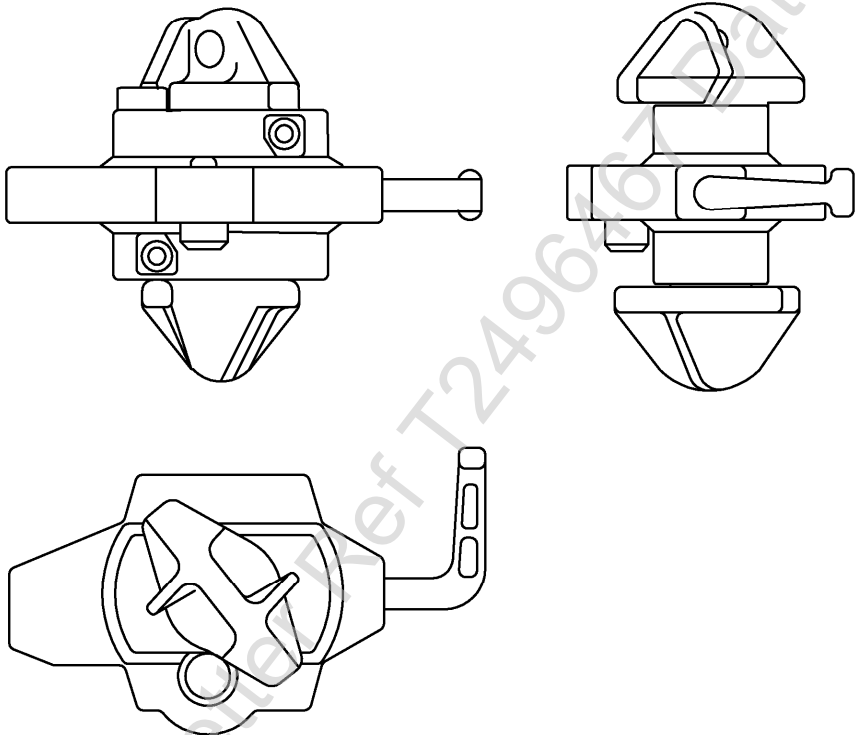
M.B.L: Compression: -

Pacific Marine & Industrial

www.pacificmarine.net

Semi-Automatic Twistlock NA-3J

NAME	MODEL NO.
SEMI AUTO TWISTLOCK	NA-3J



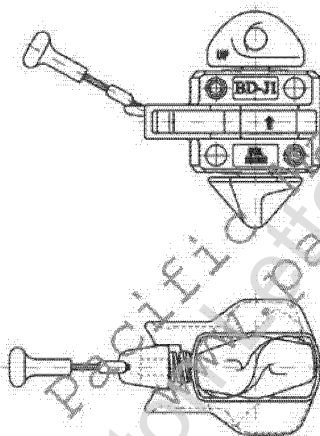
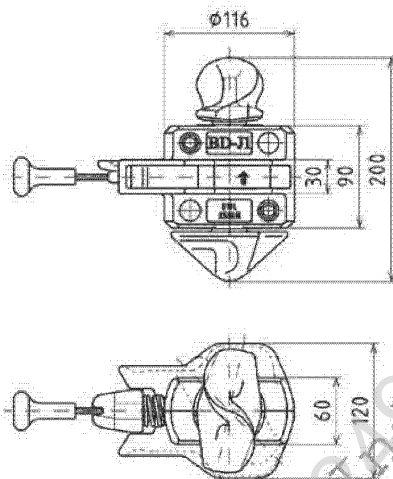
(P.A.T.)
FINISHED : Galvanized
WEIGHT : 7.2kg
B.L. : Tention 50.0 ton (490.3kN)
Shear 42.0 ton (411.9kN)



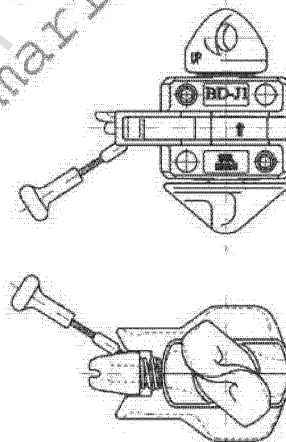
Semi-Automatic Twistlock BD-J1

Semi-automatic twistlock

TYPE:BD-J1



THE POSITION OF TOP CONE OPEN



THE POSITION OF BOTTOM CONE OPEN

Material: Forged steel
Finish: Hot dip glavanised
Weight: 6.0KG

M.B.L:Tension: 500KN
M.B.L:Shear: 420KN
M.B.L:Compression:2000KN

Stacker Twistlock – 5006-SDL-1

Terminal stackers

SDL-1

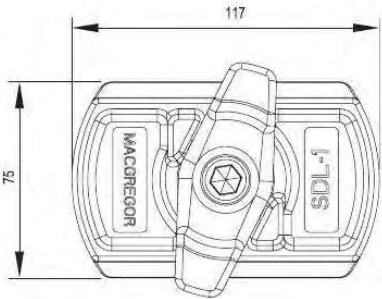
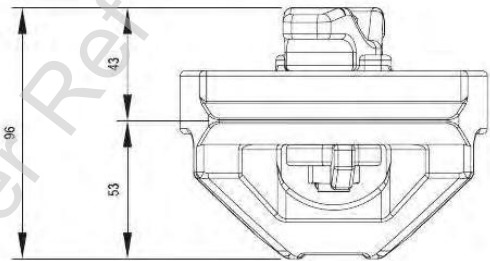
Terminal stacker for advanced mixed stowage systems

The flangeless MacGregor SDL-1 terminal stacker is a flexible, easy to operate device that will increase the cargo efficiency of your container vessel.

The simple 'one stacker per container end' system guarantees a simple, secure, timesaving method for container operations. This securing arrangement offers a high degree of flexibility in container stowage, employing a minimum number of fixed elements.

The flangeless stacker slightly decreases the height of the centre of gravity of a container stack, which is an advantage for the ship's stability.

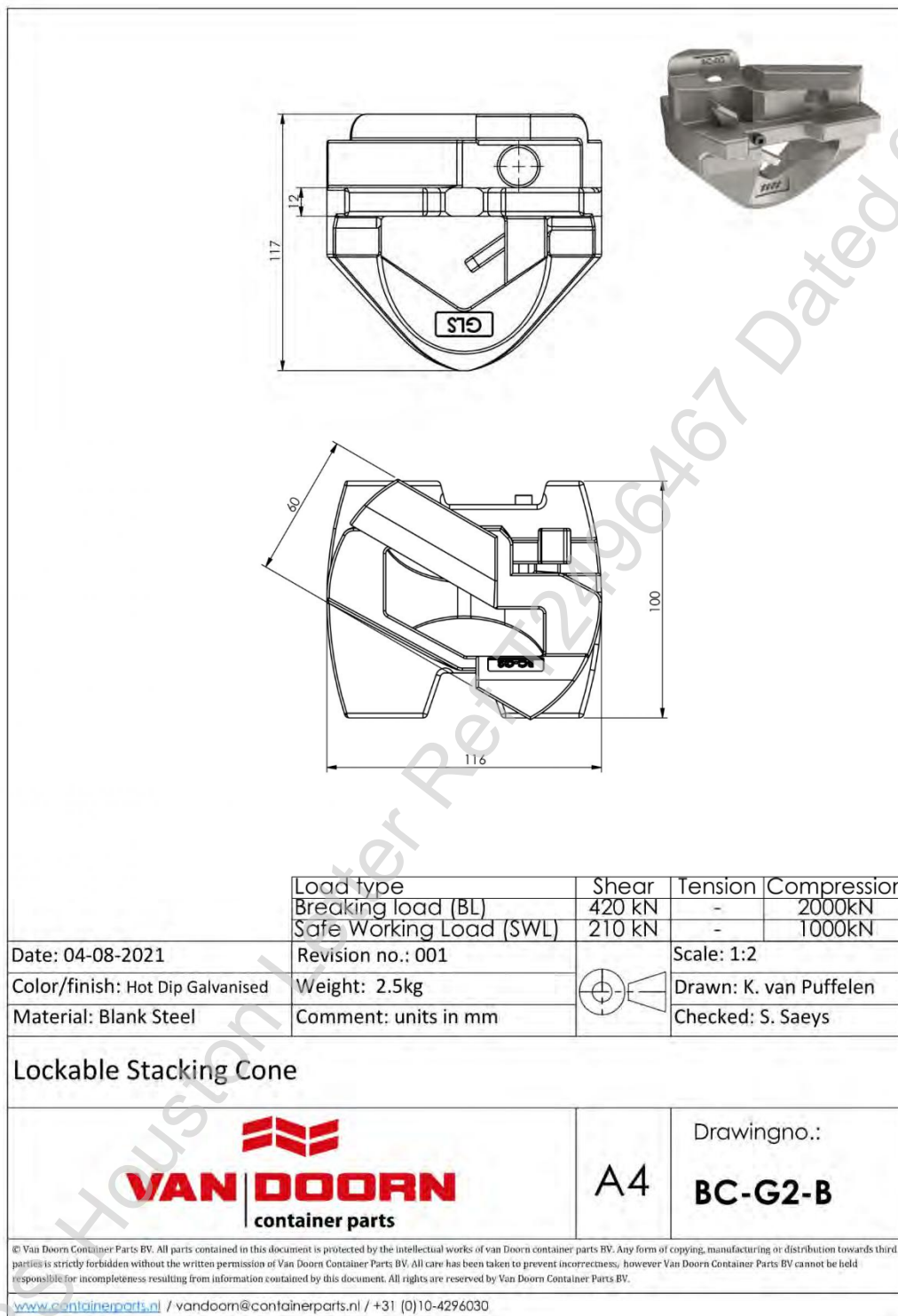
The cargo hold of the vessel can also be utilised more efficiently with the flangeless stacker system as containers are stowed more compactly. Two SDL-1 units are required per 20ft container: one fitted at the ISO gap end and one fitted at the cell guide end.



SDL-1

Order number	Product	Weight kg	Material	Treatment	Safe Working Load Shear, kN	Details	Counter part	Related system	Design
1008830	SDL-1	2.1	Precision cast steel	Hot-dip galvanised	210	No flange	Container corner casting	CSHA-compliant advanced mixed stowage system	MacGregor

Stacker Twistlock – BC-G2



Stacker Twistlock – AFC-1L

AFC-1L

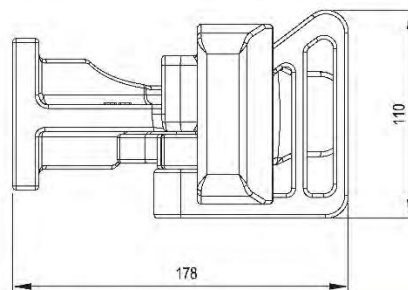
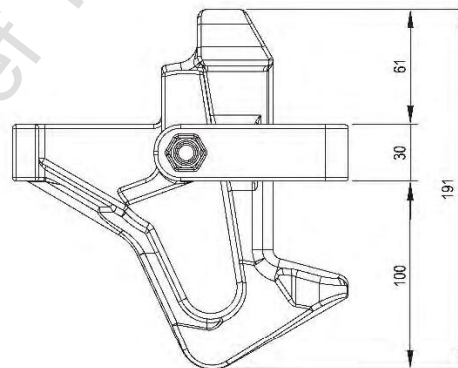
All-purpose midlock for 20' ISO gap and lashing gap

The second generation AFC-1L midlock was developed from the proven features of the world's first automatic fixing cone – the renowned AFC-1.

The AFC-1 was a pioneer product in modern container securing arrangements because it provided the opportunity to easily and efficiently stow 20' containers with minimum longitudinal clearance on board vessels.

The AFC-1L midlock is always used in conjunction with semi-automatic or manual twistlocks to ensure safely secured containers.

As for ship design, the AFC-1L provides the opportunity to either shorten a ship's overall length with a fixed number of container bays or accommodate additional container bays to a given ship length.



AFC-1L

Order number	Product	Weight kg	Material	Treatment	Safe Working Load Tension, kN	Safe Working Load Shear, kN	Details	Counter part	Design
1086914	AFC-1L	5.2	Precision cast high tensile steel	Hot dip galvanised	250	210	For use in ISO-gsp. only in combination with semi automatic twistlocks	Container castings, twistlock foundations	MacGregor

Stacker Twistlock – C19A

Midlocks

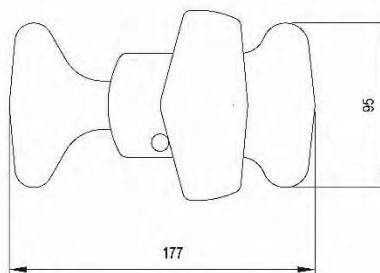
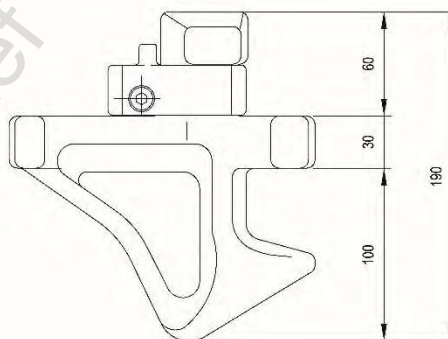
C19A

Midlock

The C19A fully forged midlock is easy to use and reliably designed to secure 20' containers with an ISO gap between units.

It is efficient at reducing loading and discharging times because no locking or unlocking is required. This adds an additional safety feature for the operator, as stevedores are not required to open the C19As on the top of the container stacks.

The C19A midlock is always used in conjunction with semi-automatic or manual twistlocks to ensure safely secured containers. The C19A is compliant with OSHA rules and with the latest requirements of international classification societies.



C19A

Order number	Product	Material	Safe Working Load Tension, kN	Safe Working Load Shear, kN	Details	Counter part	Design
1120845	C19A	Forged steel	250	210	For use in ISO-gap, only in combination with semi-automatic twistlocks	Container castings, twistlock foundations	Allset

Turnbuckle – TBS-3G1 and TBS-3G1-RBSR and TBS-3G1-RB-SR



Container Turnbuckle Jaw to Knob Type

Jaw to Knob Type Turnbuckle Open Body

Counter Parts: Knob Type Lashing Bar and D Rings or Lashing Plate

Tension B.L.(KN):500KN

SWL (KN): 250KN

Weight: 12.5kgs

Materials: high tensile steel-drop forged

Finish: hot dip galvanized

Certificate: GL. CCS. ABS, BV, etc

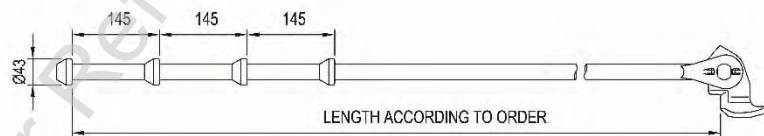
Lashing Rod (Short) – LB 11

LB-11

for Multilash lashing system

The MacGregor LB-11 knob-type lashing bars are used together with the TBS-3G family of turnbuckles in the MacGregor Multilash system.

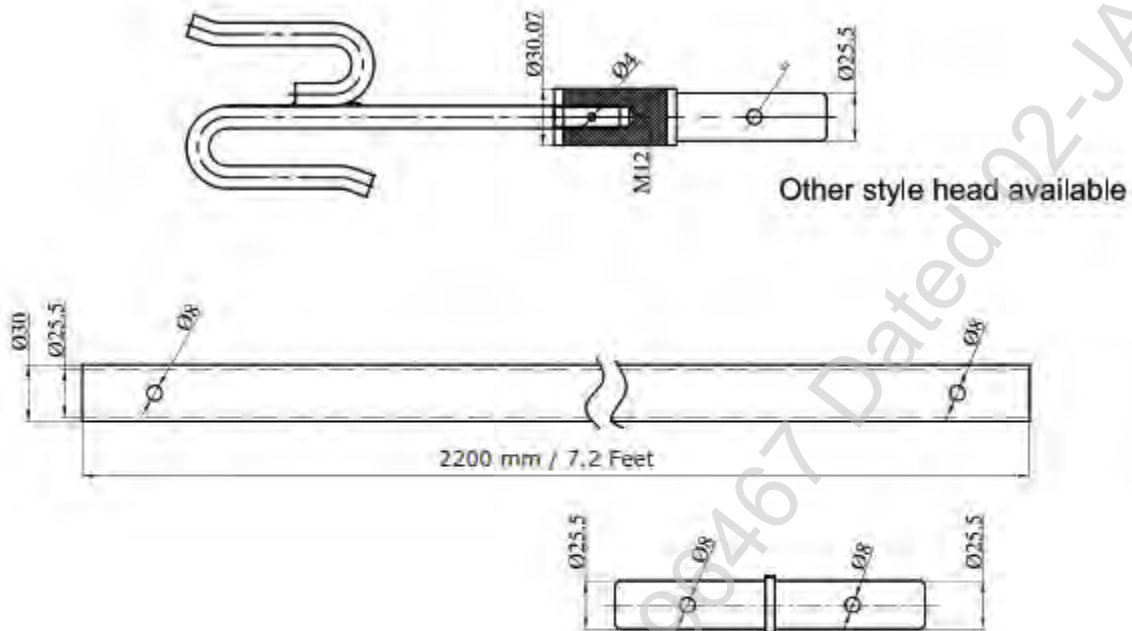
The LB-11 is available in various lengths according to the vessel's needs. In its standard version, the LB-11 is designed for 50-tonne breaking loads, but it can also be supplied in a lighter 36-tonne breaking load version, the LB-11/22.



LB-11

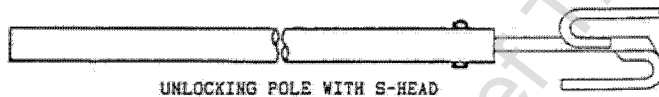
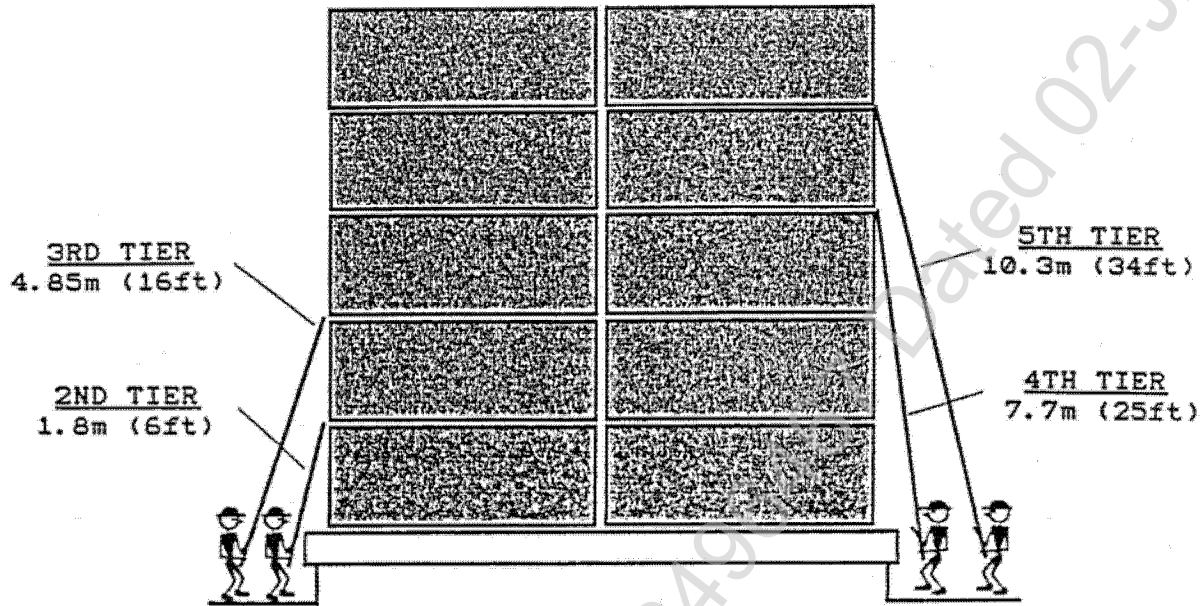
Product	Length, mm	Treatment	Safe Working Load Tension, kN	Counter part	Related system	Design
LB-11	According to order	High tensile steel Hot-dip galvanised	245	TBS 3G/11 turnbuckle, LB-14 Extension bar	Multilash	MacGregor

Unlocking Pole



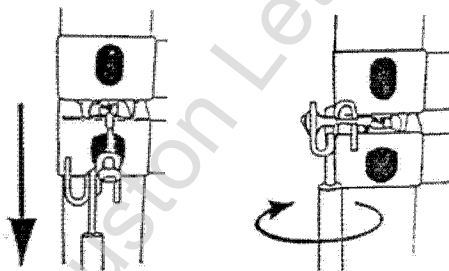
Unlocking Pole Operation

UNLOCKING POLES

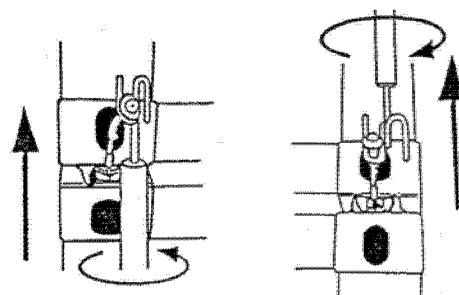


S-HEAD CAN BE USED FOR ALL POSSIBLE UNLOCKING POSITIONS: 6 O'CLOCK, 12 O'CLOCK OR 9 O'CLOCK, IN BETWEEN 20' AND ON 40' ON TOP OF 45'. ONE TYPE DOES IT ALL.

UNLOCK LOWER CONE FOR DE-CONING ON DOCK



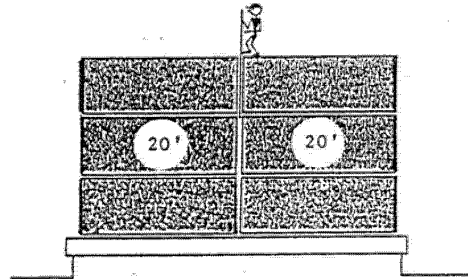
UNLOCK UPPER CONE FOR DE-CONING ON TOP



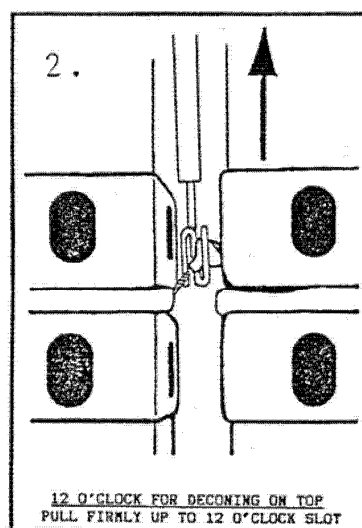
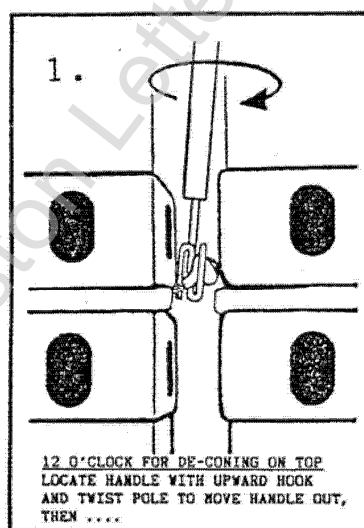
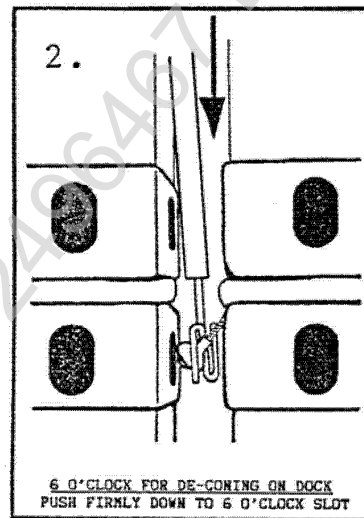
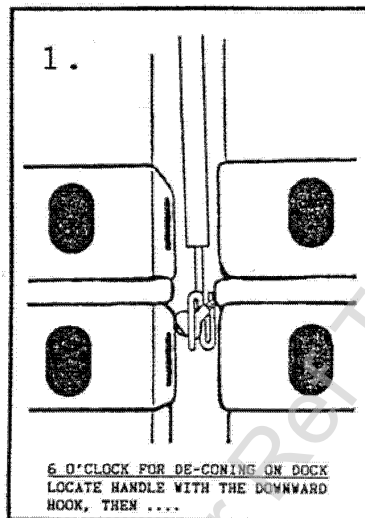
Unlocking Pole Operation

UNLOCKING POLES

TO UNLOCK BETWEEN TWO 20' WHEN IN A 40' SPACE



UNLOCK THE INNER LOCKS FROM THE TOP
BY PUSHING DOWN TO 6 O'CLOCK OR PULL
UP TO 12 O'CLOCK

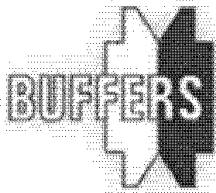


C9 Class - GEORGE II**Loose Lashing Gear**

Location On Deck	Hatch	1st Tier					
		Semi-Auto Twist Lock	Twist Lock C5AM-HC	20' Mid Stacker	Short Rod	Long Rod	Turn Buckle
	1	176	44		88	28	88
	2	192	48		96		96
	3	192	48	72	96		96
	4	192	48	72	96		96
	5	192	48		96		96
	6	192	48		96		96
	7	192	48		96		96
	8	192	48		96		96
	9	192	48		84	12	96
	10	128	32		52	12	64
	11	192	48		96		96
	12	192	48		96		96
Sub Total:		2,224	556	144	1,088	52	1,112
10% Spares:		222	56	14	109	5	111
Total:		2,446	612	158	1,197	57	1,223

Notes:

- Twistlocks - vessel uses:
C5-AM-HC between 1st tier and hatch cover or pedestal
NA-3J or BD-J1 in all other locations
- Multi stop turnbuckle and lashing rods used
Hatch 1 aft, 9 fwd & 10 aft require additional long rods
- Hanging stackers used
- See Section 2.3 for items MSL



Securing Your Containers

Protecting Your Cargo

At Sea, Over Land, On Rail

CERTIFICATE OF COMPLIANCE

THIS IS TO CERTIFY THAT THE:

BUFFERS PART NO. 5002-C5AM-HC

SUPPLIED TO: M/V HORIZION RELIANCE
DOCKSIDE MACHINE AND SHIP REPAIR
211 N Main Avenue
WILMINGTON, CA 90744
USA

ON M/V HORIZION RELIANCE PURCHASE ORDER: RGRF 156784 AND
SHIPPED ON 2/17/2009 CONFORM TO THE ATTACHED MATERIAL
CERTIFICATIONS.

QTY SHIPPED

100

ANY RESULTS OF INSPECTIONS OR TEST REQUIRED BY THE
PURCHASE ORDER ARE ON FILE AND AVAILABLE FOR REVIEW
BY AUTHORIZED PERSONNEL.

Simmie Stewart

simmie@BuffersUSA.com

BUFFERS USA, INC.

17-Feb-09



Certificate No.:
ASM 08-10-29/01
Date:
OCT. 29, 2008

CERTIFICATE OF STRENGTH TEST AND THOROUGH EXAMINATION

Article No.	Description	Quantity total (pcs)	Quantity tested (pcs)	Date of test and examination (yy-mm-dd)	Test load applied (kN)	Max. securing load (kN)	Test mode
004170	TWISTLOCK C5AM-EC	500	2 2	08-10-29	375 315	250 210	TENSION SHEAR

Name and Address of Owner:

The test was carried out by:

We certify that the above particulars are correct and that the test and examination were carried out by a competent person and that the items described herein were tested and thereafter examined and were found to be free from cracks, flaws or other defects.

Name in print



CROWLEY CSAM TWISTLOCK
RCD JAX
NOV 2017

DET NORSKE VERITAS

TYPE APPROVAL CERTIFICATE

CERTIFICATE NO. S-7013

This is to certify that the
Cargo Securing Devices

with type designation(s)

Twistlock, semi-auto, for use on H/C Type: C5AM-HC

Liftlock, semi-auto w dual function, Type: C5AM-DF

Twistlock, semi-auto w dual function, Type: C5AM-DF

Liftlock, semi-auto w dual function, low temperature, Type: C5AM-DF/LT

Manufactured by

Cargotec Finland Oy
KAARINA, Finland

is found to comply with
Det Norske Veritas' Rules for Classification of Ships
Container Securing Devices

Application

Type	MSL, tension (kN)	MSL, shear (kN)	MSL, compr. (kN)
Twistlock, semi-auto, for use on H/C Type: C5AM-HC	250	210	-
Liftlock, semi-auto w dual function, Type: C5AM-DF	250	210	-
Twistlock, semi-auto w dual function, Type: C5AM-DF	250	210	-
Liftlock, semi-auto w dual function, low temperature, Type: C5AM-DF/LT	250	210	-

Høvik, 2012-05-10
for Det Norske Veritas AS

This Certificate is valid until
2016-06-30

Morten Bentzon
Head of Section

DNV local office:
Turku

Nina Thorvaldsen
Surveyor

This Certificate is subject to terms and conditions overleaf. Any significant change in design or construction may render this Certificate invalid.
The validity date relates to the Type Approval Certificate and not to the approval of equipment/systems installed.
If any person suffers loss or damage which is proved to have been caused by any negligent act or omission of Det Norske Veritas, then Det Norske Veritas shall pay compensation to such person for his proved direct loss or damage. However, the compensation shall not exceed an amount equal to ten times the fee charged for the service in question, provided that the maximum compensation shall never exceed USD 2 million. In this provision "Det Norske Veritas" shall mean the Foundation Det Norske Veritas as well as all its subsidiaries, directors, officers, employees, agents and any other acting on behalf of Det Norske Veritas.

DET NORSKE VERITAS AS, Veritasveien 1, NO-1322 Høvik, Norway, Tel.: +47 67 57 99 00, Fax: +47 67 57 99 11, Org.No. NO 945 748 931 MVA www.dnv.com
Form No.: TA 1411a Issue: October 2009 Page 1 of 3

Type Certificate

Certificate No.: 75571 HH 13



The Lashing Fitting

BD-J1

of type

Semi Automatic Twistlock

was type tested according to the Rules and Guidelines of GL for Stowage and Lashing of Containers, edition 2007.

The main elements of the fitting are specified as follows:

Material: As specified in the drawing(s)

Part:	Shaft	Drawing No.:	BD-J1/01
	Body R		BD-J1/R
	Body L		BD-J1/L

Drawings approval No.: 07-134241 Date: 2007-11-20

Allowable working load: 250 kN tension / 210 kN shear

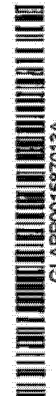
Place and date of test: Ningbo, P.R.China, 2013-04-01

Breakload test results: 500 kN / 420 kN

500 kN / 420 kN

500 kN / 420 kN

At 313 kN / 263 kN test load no permanent deformation and no incipient cracks were detected.



Certificate expires on: 2018-05-28

Germanischer Lloyd

Hamburg, 2013-05-28

I.V. Daniel Abt
I.A. Mathias Bohn

The latest edition of the „General Terms and Conditions for Classification“ and the „General Terms and Conditions for Activities other than Classification“ respectively are applicable. German law applies.



AMERICAN BUREAU OF SHIPPING

Customer Name KUNSHAN YONGTAI MACHINE & TRADE CO., LTD. Purchase Order No.
Attending Office Shanghai Report Number SQ2743864-A
First Visit Date 31-Oct-2014 Last Visit Date 04-Nov-2014

Certification Of: CONTAINER SECURING DEVICES FOR Quantity: One(1)
STOCK
Manufacturer: KUNSHAN YONGTAI MACHINE
& TRADE CO., LTD.

Survey Location: KUNSHAN CITY, JIANGSU, P. R. CHINA

Equipment Data

Model Number NA-3J
Designer Name TAIYO SEIKI IRON WORKS CO., LTD.
Purchaser Name HORIZON LINES LLC

Design Details

Design State Product Design Assessed
ABS Reviewing Organization Yokohama Ship Eng - Structure
Drawing Number Under PDA certificate: 06-YO173663/1-PDA, Date: 10 August 2011

This is to Certify that the undersigned surveyor(s) to this Bureau did, at the request of the customer, carry out the following survey and report as follows:

Traceability of materials used on this project has been verified.

All testing (pressure/load/operational/etc.) has been carried out as applicable and verified in accordance with the applicable Rules/specifications.

Testing machines are maintained in a satisfactory condition and records of their recheck or calibration dates confirmed.

All parts of the machinery/equipment satisfactorily complied with the approved drawings. Amendments, if any, verified to be rectified and considered satisfactory.

Asbestos-free declaration verified and supporting documentation reviewed

Final markings for identification confirmed.

1. Principal Particulars:

Equipment/Type	Pieces	S.W.L.(ton)	Proof Load (ton)	Break Load(ton)
Semi-auto Twistlock / NA-3J	1350	30.0(T)/21.0(S)	33.0(T)/23.1(S)	50.0(T)/42.0(S)

Note: T - Tension; S - Shear

2. The following test was carried out and found satisfactory

- Breaking test (1 piece in 250) (W)
- Proof test (1 piece in 50) (W)
- Visual examination & dimensions check (A)

Note: W - Witnessed; A - Audited and randomly check witnessed

NOTE: This report evidences that the survey reported herein was carried out in compliance with one or more of the Rules, guides, standards or other criteria of the American Bureau of Shipping and is issued solely for the use of the Bureau, its committees, its clients or other authorized entities. This Report is a representation only that the vessel, structure, item or material equipment, machinery or any other item covered by this Report has been examined for compliance with, or has met one or more of the Rules, guides, standards or other criteria of American Bureau of Shipping. The validity, applicability and interpretation of this report is governed by the Rules and standards of American Bureau of Shipping who shall remain the sole judge thereof. Nothing contained in this Report or in any notation made in the contemplation of this Report shall be deemed to relieve any designer, builder, owner, manufacturer, seller, supplier, repeller, operator or other entity of any warranty express or implied.



Securing Your Containers

Protecting Your Cargo

At Sea, Over Land, On Rail

CERTIFICATE OF COMPLIANCE

THIS IS TO CERTIFY THAT THE:

BUFFERS PART NO.: 5007-TBS-3G/1-SR/RB-1086963

DESCRIPTION: Turnbuckle, Multilash, 1080-1485mm stroke, round bar body w slack reducer, 50 Mton BL, TC, 1236914

SUPPLIED TO: Crowley Maritime Corporation
9487 Regency Square Blvd
Jacksonville, FL 32225
USA

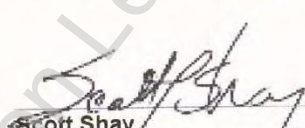
FOR: *MV Horizon Spirit*

ON CROWLEY MARITIME CORPORATION PURCHASE ORDER:
3139187 (BUFFER'S ORDER 20072971.1) AND SHIPPED ON 11/30/2016
CONFORM TO THE ATTACHED MATERIAL CERTIFICATIONS.

QTY SHIPPED

100

**ANY RESULTS OF INSPECTIONS OR TEST REQUIRED BY THE
PURCHASE ORDER ARE ON FILE AND AVAILABLE FOR REVIEW
BY AUTHORIZED PERSONNEL.**


Scott Shay
Scott@BuffersUSA.com
BUFFERS USA, INC.
30-Nov-2016



Printed On: 30-Nov-2016

**This is to certify:****That the Cargo Securing Devices**

with type designation(s)

Lashing Bar (Knob Type) BB-C1

Issued to

**General Lashing System (China) Co., Ltd.
Ningbo, China**

is found to comply with

DNV GL rules for classification – Ships**Application :****Product(s) approved by this certificate is/are accepted for installation on all vessels classed by DNV GL.****SWL, tension (kN) 250**Issued at **Hamburg** on **2019-08-22**This Certificate is valid until **2024-08-21**.DNV GL local station: **Ningbo**Approval Engineer: **Mark-Oliver Wobig**for **DNV GL****Andree Orth
Head of Section**

This Certificate is subject to terms and conditions overleaf. Any significant change in design or construction may render this Certificate invalid.
The validity date relates to the Type Approval Certificate and not to the approval of equipment/systems installed.



Form code: TA 251

Revision: 2016-12

www.dnvgl.com

Page 1 of 2

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Job Id: **262.1-031265-1**
Certificate No: **TAS0000254**

Product description

Lashing Bar

Application/Limitation

Load	Tension [kN]
Test Load	375
Breaking Load	500

Tests to be carried out:

- Production testing: At least 2% of the items delivered are to be selected and subjected to the test load.
- Material tests are to be included energy testing at -20°C, with the impact energy requirements acc. to DNVGL-CP-0068. The material shall be according to the approved drawings. Material certificates shall be 3.1 acc. to ISO 10474:2013.

Type Approval documentation

Drawing No.	Rev.	Title
BB-C1		Lashing Bar (Knob Type)
BB-B1/02		Jaw
BB-B1/01		Swivel Head
BB-C1/06		Multiple Section Rod
BB-B1/05		Rod
BB-B1/03		Bar Pin

Drawing approval No. GL Ref.-No.: 14-010515 Date: 2014-04-29

Tests carried out

The prototype test was carried out in Ningbo, P.R. China, dated 2019-07-02 in accordance with the DNVGL-CP-0068.

Marking of product

Each item shall be marked by manufacturer for traceability.

Periodical Assessment

For retention of the Type Approval, a DNVGL surveyor shall perform a survey after 2 and 3.5 years, to verify that the Type Approval is complied with.

2.5 INSPECTION AND MAINTENANCE PROCEDURES

2.5.1 Regular inspections and maintenance are carried out under the responsibility of the Master. Cargo securing device inspections should include as a minimum:

1 Routine visual examinations of components being utilized:

All fixed eyeplates for cargo securing are to be regularly examined for damage and any necessary repairs and appropriate testing as carried out prior to re-use. This examination should be particularly stringent following heavy weather passages.

All portable securing devices are to be thoroughly examined by longshoremen and lashers prior to their being reused. Bad gear shall be separated from serviceable gear. The vessel shall ensure bad gear is properly repaired by the vessel's crew, the shore gang or an approved vendor, prior to being placed back into service.

All portable securing devices shall be annually greased as necessary.

2 Any significant deformation of the ship's structure in way of securing points is to be reported to the vessel's superintendent and written up in an appropriate MAXIMO Work Order to effect repairs.

Fixed fittings are regarded as part of the ship structure and will be surveyed for damage at routine hull surveys.

2.5.2 Inspection of the vessel's cargo securing devices.

- 1 Each device should be checked for damage and wear which could affect the ability of the device to adequately and safely perform its designated function or which could lead to physical injury of persons handling the devices. The devices should also be examined prior to being used for a particular purpose to determine that they are suitable for that purpose, with regard to both strength and efficacy.

Container sockets, twistlock sockets, fixed cones and other securing sockets

These devices should be inspected in way of the welds connecting them to the ship's structure (whether surface-mounted or flush-fitting) and any fractures or tearing should be gouged out and rewelded. If the underlying structure of the deck, tank top or hatch cover is deformed to such an extent that an uneven stow would result, the structure should be repaired by the most appropriate method.

The aperture of the pocket should be examined for excess wear, deformation or fractures. The sockets' dimensions should be checked against a standard cone or twistlock. If there is any doubt as to the security of the socket it should be replaced.

The presence of light corrosion, not affecting the ability of the socket to perform its function, may be accepted without remedial action.

Any scale that has formed within the aperture of the socket should be removed and the socket re-examined. The dimensions of the aperture should be carefully checked for satisfactory fit.

Likewise, any dust and debris, previous cargo material, etc., should be removed from the socket before it is put back into service.

A thorough examination of these devices is to be conducted during each shipyard availability.

Lashing plates

These devices should be inspected in way of the welds connecting them to the ship's structure and any fractures or tearing should be gouged out and rewelded. The structure to which the device is attached (whether deck, tank top, hatch cover, bulkhead, web frame, stanchion, stay, bulwark or deck head) should be examined for deformation and/or fractures and if any are found the structure is to be repaired by the most appropriate method.

The material of the device should be examined for excess wear and/or fractures and if any are found the device should be replaced with one of at least the same strength. Welding of the device to the structure should be carried out by approved personnel in accordance with appropriate welding practice.

A thorough examination of these devices is to be conducted during each shipyard availability.

Turnbuckles

These devices should be inspected by the longshoremen/lashers prior to each use for excessive wear, deformation and/or fractures. If any are found the device should be rejected. The threaded section of the eyebolts should be examined for damage to the screw threads or 'cross-threading' and should be adequately greased and free from corrosion. The eye should be examined for excessive wear and/or fractures and, if any are found, the device should be rejected.

Lashings rods/ extension rods

These should be inspected by the longshoremen/lashers prior to each use for excessive wear, deformation and/or fractures. If any are found the device should be rejected. Particular attention should be given to swivel type heads on lashing rods.

Inter-layer stackers, twist-locks.

These are subject to deterioration or physical damage of one kind or another and should always be inspected by the longshoremen/lashers before use. Operational structural failure of twist-lock inter-layer stackers may well result from them being weakened by rough handling at discharge ports. Such rough handling could also cause defects in other components that may then fail under operational stress.

The provision of suitable bags and boxes that can be used to collect components before lowering them to deck level helps to prevent such misuse.

Inter-layer stackers and twistlocks are to be examined by the longshoremen/lashers under a Continuous Examination Program that ensures they are inspected prior to each use to verify their safe condition. This examination is a component of the process of using the locks.

The examination program shall ensure:

- Each semi-automatic lock's heads or latches are moved to determine adequate spring tension;
- Each stacker and lock is inspected visually for obvious structural defects;
- Each stacker and lock is exercised to ensure it is functional; and
- Each stacker and lock is checked for corrosion and deterioration.

Each stacker or lock that is not considered to be safe after examination is to be immediately removed from service. Suitable bins or other receptacles are to be provided during longshoring operations to facilitate the segregated return of bad-order stackers and locks to the ship for maintenance and repair. All defective stackers and twistlocks are to be repaired as required and restored to safe operating condition before they are returned to service.

In addition to their regular examination each lift lock (SATL) shall be regularly maintained. As a minimum all stackers and twistlocks must be serviced/maintained during each shipyard availability. All maintenance activities are to be recorded on a copy of the Maintenance Record Log in Appendix C. Completed Maintenance Record Logs are to be maintained in Appendix C for at least 3 years from the last dated entry in the log.

2.5.3 Information for the Master regarding inspections and adjustments of securing arrangements during the voyage.

- 1 The integrity of the securing arrangements should be maintained throughout the voyage.
- 2 Lashings should be regularly checked and re-tightened.

Particular attention should be paid to lashings that may become slack due to the cargo deforming or compacting during the voyage. Lashings may also become slack when cargoes are loaded and secured in conditions of low ambient temperature and the vessel then proceeds to areas of significantly higher ambient temperature. Adjustments of securing devices include re-tightening of lashings by use of turnbuckles, if fitted, otherwise by remaking the lashing. If necessary, additional lashings should be fitted. Inspections should be documented by an appropriate entry into Appendix C or the deck logbook.

- 3 Greasing the thread of clips and turnbuckles increases working life and prevents corrosion.
- 5 Action which may be taken in heavy weather:

General

The purpose of this section is not to usurp the responsibilities of the master, but rather to offer some advice on how stresses induced by excessive accelerations caused by bad weather conditions could be avoided.

Excessive accelerations

Measures to avoid excessive accelerations are:

- 1 alteration of course or speed, or a combination of both;
- 2 heaving to;
- 3 early avoidance of areas of adverse weather and sea conditions;
- 4 timely ballasting or deballasting to improve the behavior of the ship, taking into account the actual stability conditions (see also 6 below).

Voyage planning

One way of reducing excessive accelerations is for the master, as far as possible and practical, to plan the voyage of the ship carefully so as to avoid areas with severe weather and sea conditions. The master should always consult the latest available weather information.

- 6 Actions which may be taken once cargo has shifted

The following actions may be considered:

- 1 alterations of course to reduce accelerations
- 2 reductions of speed to reduce accelerations and vibration
- 3 monitoring the integrity of the ship
- 4 restowing or resecuring the cargo and, where possible, increasing the friction
- 5 diversion of route in order to seek shelter or improved weather and sea conditions

Tank ballasting or deballasting operations should be considered only if the ship has adequate stability.

3.1 Allowable Stowage Patterns

The drawings in this section show container stowage plans for each hatch. The drawings consist of a color-coded drawing showing the section views of all hatches and individual section views of each hatch with similar hatches grouped together. All section views are from aft looking forward. The views indicate the possible types of containers that can be stowed at each location on board the vessel. The types of containers are 20' standard height (8'-6"), 40' standard height (8'-6"), 40' HC (high cube 9'-6"), 45' (9'-6" high), and 2x20' (two twenties in a forty-foot slot).

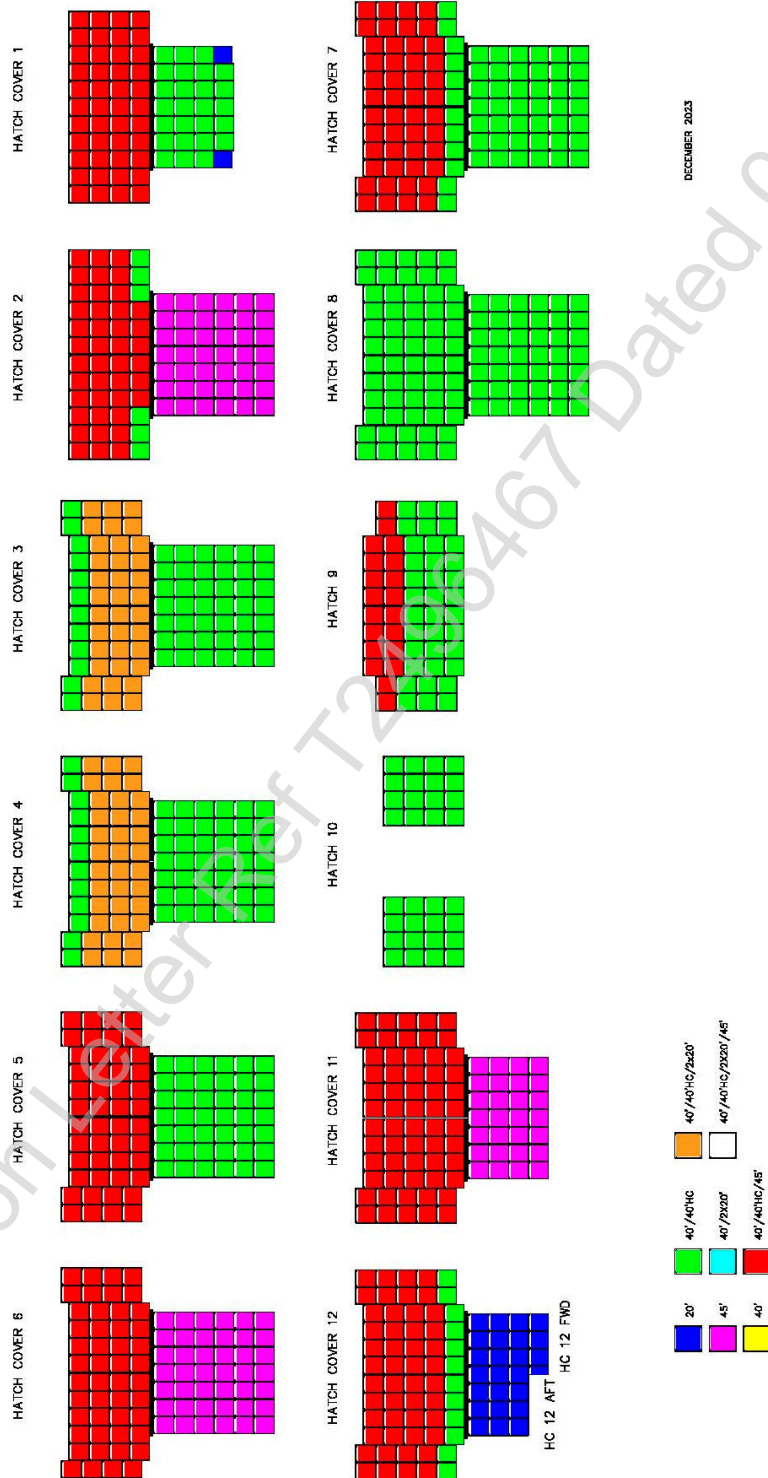
Below deck high cube locations are determined by under hatch cover clearance and structural configuration of the cell guides. The stowage of high cube containers below deck must agree with the patterns indicated on the color drawing and section views.

45' containers can be stowed as indicated on the drawings and are high cube containers.

The stowage of 20' containers on deck must agree with the patterns indicated on the color drawing and section views. Stack weights of 20' containers on deck are set by the strength of the hatch covers and lashing system. 40' containers must be stowed over two 20' containers on deck. 20' containers can be stacked one, two or three tiers high, for this class of vessel.

Allowable stack weights for all on deck container locations are given in Section 3.4. Allowable stack weights depend upon; the structural strength of the hatch cover and pedestals, the number of lashing rods used, the number of tiers of containers in stack, the number of high cubes stowed in stack and the vessels stability condition (GM).

MV GEORGE II



LEGEND

H – HI CUBE 40 FT CONTAINER SPACE
 W – DEDICATED 20 FT CONTAINER SPACE
 FF – 40 FT (STANDARD OR HI CUBE)
 OR 45 FT CONTAINER SPACE
 2W – 2x20 FT OR 1x40 FT CONTAINER SPACE
 45 – DEDICATED 45 FT CONTAINER SPACE

HI-CUBE NOTE

ABOVE DECK HI-CUBE STOWAGE DEPENDS ON
 TIER, WEIGHTS, SEE CONTAINER
 STOWAGE INFORMATION, (CSW)

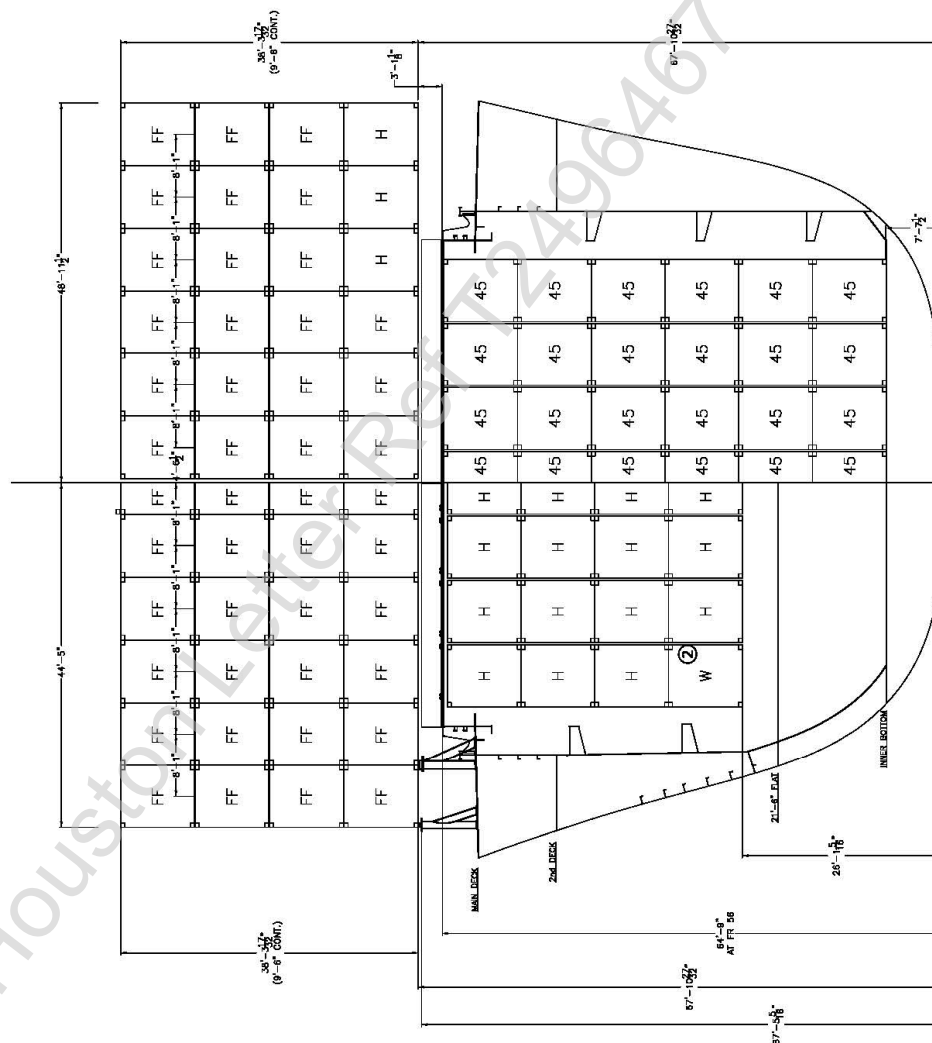
NOTE

- ① TWO (2) 20 FT CONTAINER SPACE ONLY
 ② ONE (1) 20 FT CONTAINER SPACE, AFT

MV GEORGE II

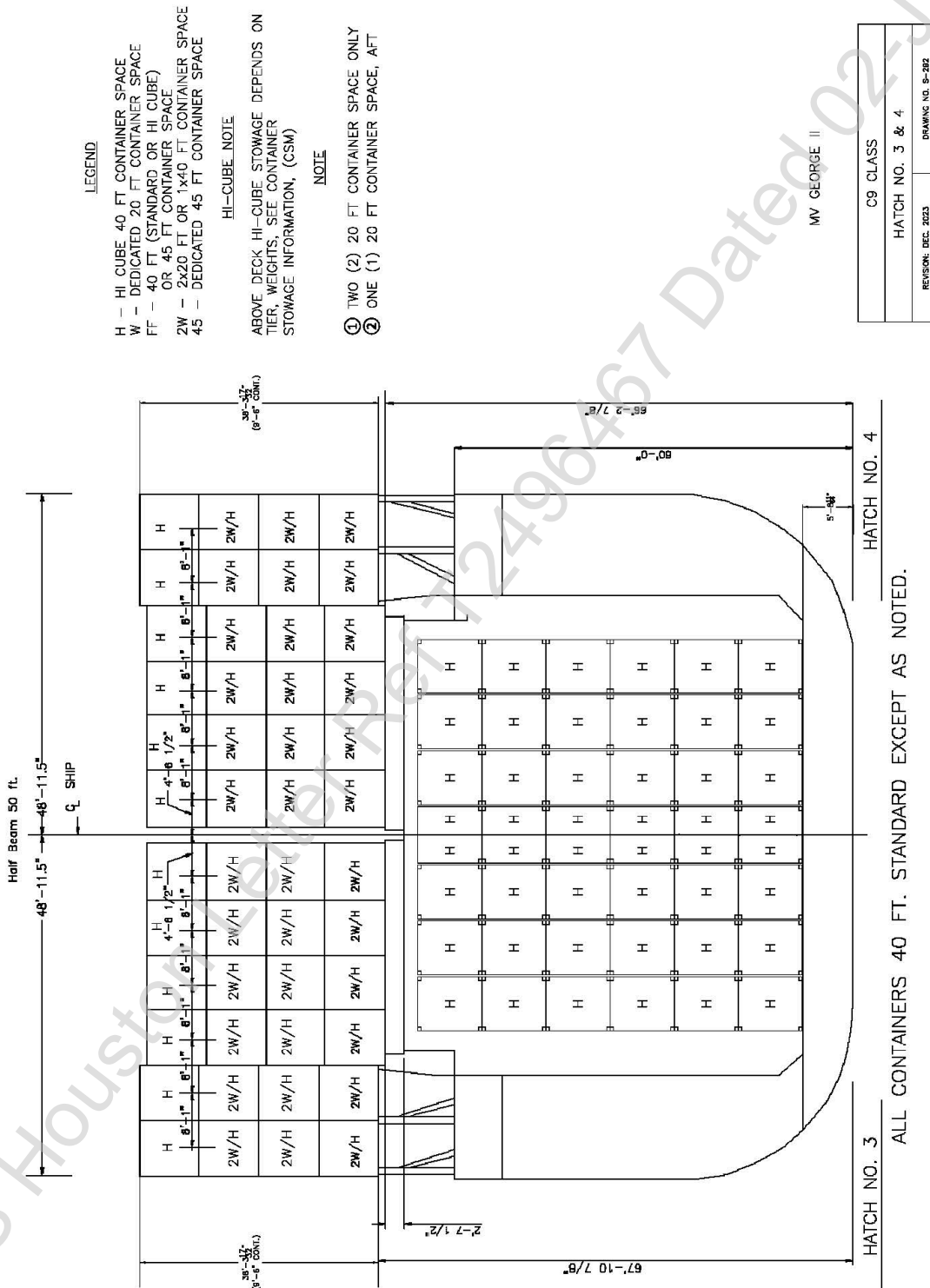
C9 CLASS	
HATCH NO. 1 & 2	
REVISION: DEC. 2023	DRAWING NO. 9-281

HALF BEAM 50 FT



HATCH NO. 1 HATCH NO. 2

ALL CONTAINERS 40 FT STANDARD EXCEPT AS NOTED



LEGEND

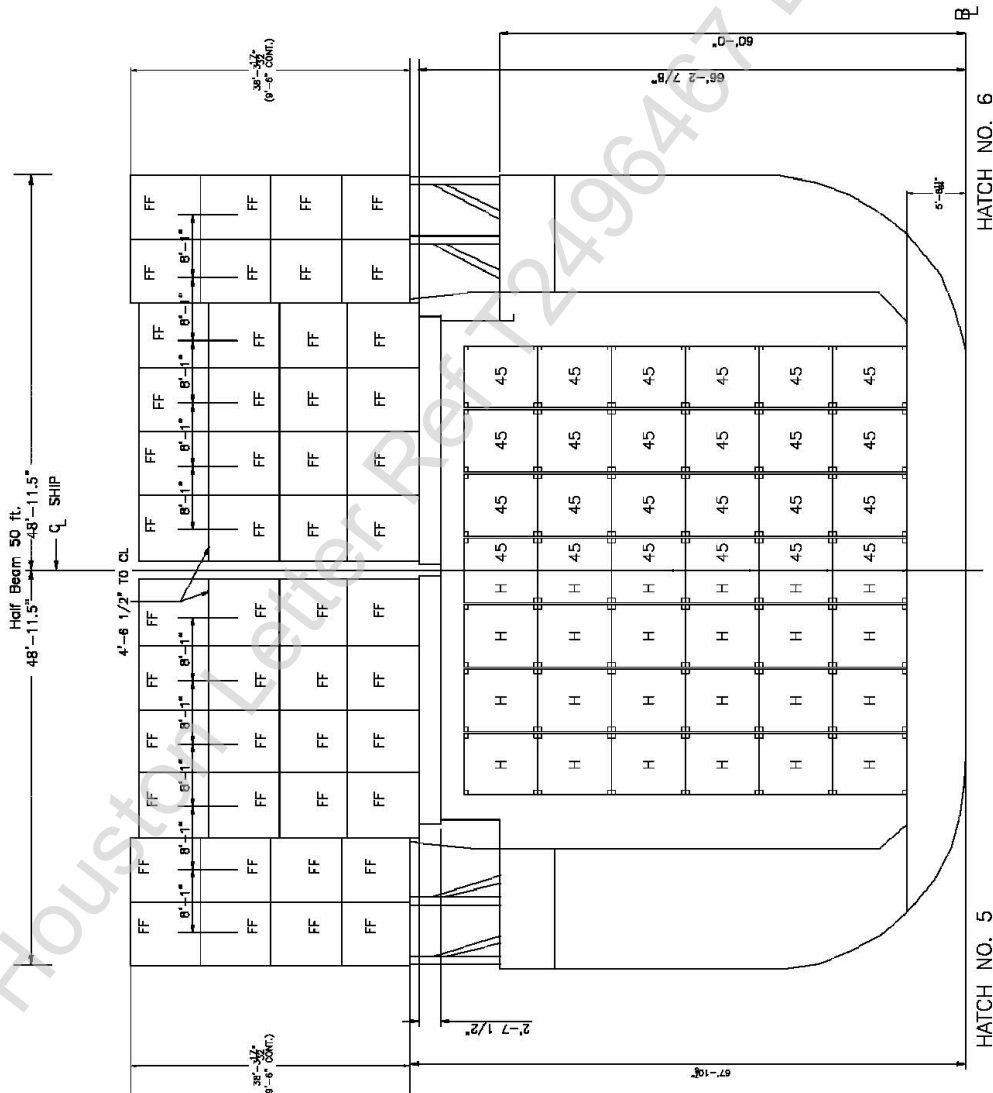
H – HI CUBE 40 FT CONTAINER SPACE
 W – DEDICATED 20 FT CONTAINER SPACE
 FF – 40 FT (STANDARD OR HI CUBE)
 OR 45 FT CONTAINER SPACE
 2W – 2x20 FT OR 1x40 FT CONTAINER SPACE
 45 – DEDICATED 45 FT CONTAINER SPACE

HI-CUBE NOTE

ABOVE DECK HI-CUBE STOWAGE DEPENDS ON
 TIER, WEIGHTS, SEE CONTAINER
 STOWAGE INFORMATION, (CSM)

NOTE

- ① TWO (2) 20 FT CONTAINER SPACE ONLY
- ② ONE (1) 20 FT CONTAINER SPACE, AFT



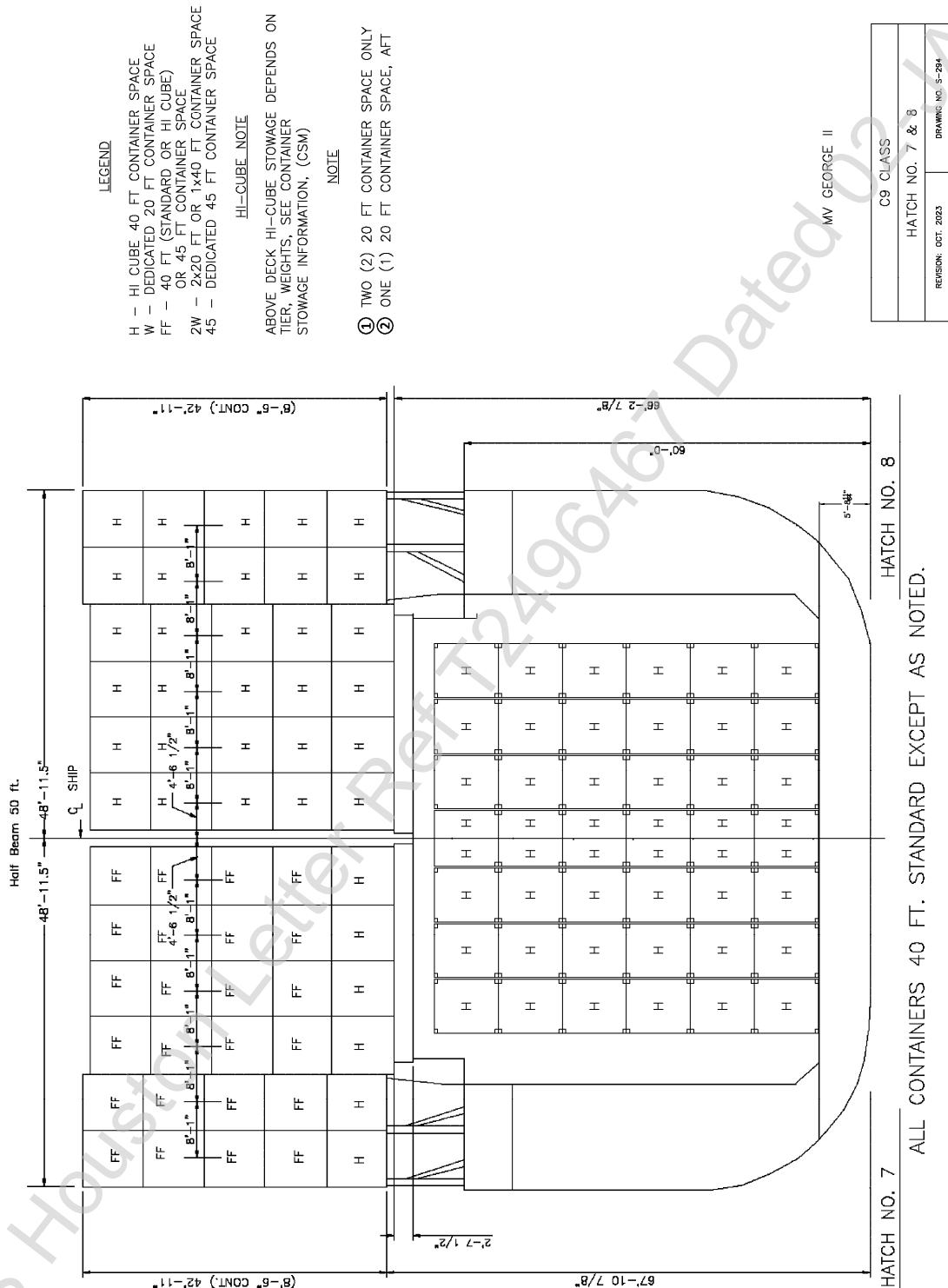
MV GEORGE II

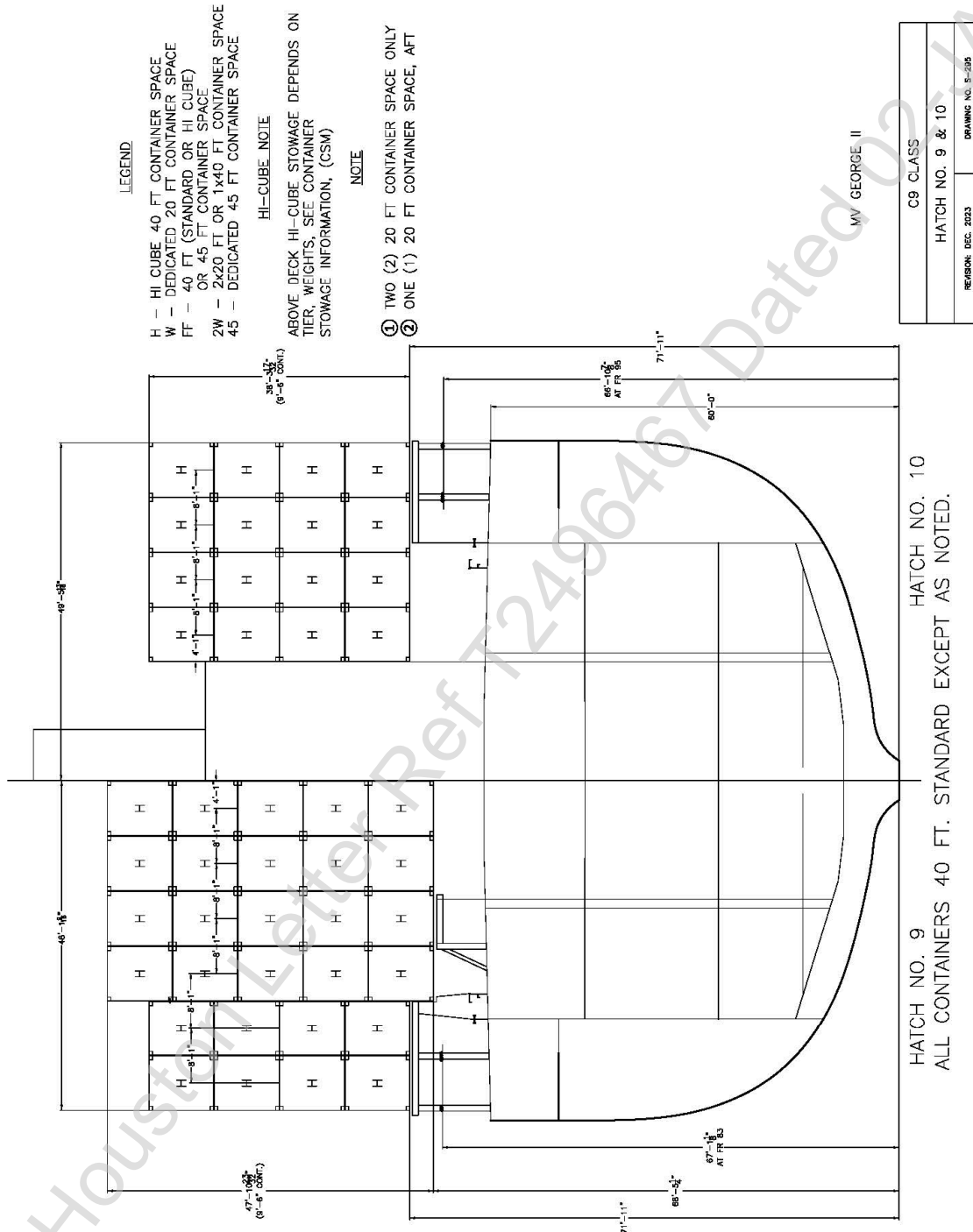
C9 CLASS
HATCH NO. 5 & 6
REVISION: DEC. 2023
DRAWING NO. 5-203

HATCH NO. 6

HATCH NO. 5

ALL CONTAINERS 40 FT. STANDARD EXCEPT AS NOTED.







MV GEORGE II

ABOVE DECK

H 1 40' ISO / 40' high cube and 45' containers may be stowed 11 across on cover and secured by twistlocks up to four high. Parallel lashed rods can be used, as needed, depending on tier weight. At aft end of the hatch, due to lashing plates being lower than sockets, long lashing rods will be required for stack 00, 01, 02, 03, 04, 05 & 06 only for reefer containers, next to reefer platform.

H 2 40' ISO / 40' high cube and 45' containers may be stowed 12 across on cover and secured by twistlocks up to four high. Upper and lower cross-lashed rods can be used, as needed, depending on tier weight. Parallel lashed rods can be used, as needed, depending on tier weight. No 45' containers can be stowed in the first tier 3 outboard cells.

H 3 20' & 40' ISO / 40' high cube containers may be stowed 12 across on cover and secured by twistlocks up to four high. Parallel lashed rods can be used, as needed, depending on tier weight. 2x20' container can be stowed in the first three tiers and topped by a 40' container. 20' containers are stowed close together longitudinally are lashed and twistlocked only on the forward and aft ends, stackers are used at the midpoints.

H 4 20' & 40' ISO / 40' high cube containers may be stowed 12 across on cover and secured by twistlocks up to four high. Parallel lashed rods can be used, as needed, depending on tier weight. 2x20' container can be stowed in the first three tiers and topped by a 40' container. 20' containers are stowed close together longitudinally are lashed and twistlocked only on the forward and aft ends, stackers are used at the midpoints.

BELOW DECK

40' ISO / 40' high cube are stowed in fixed cell guides below deck. 20' ISO needs to be loaded at first tier, outboard. If 20' ISO is not available, stopper has to fitted at aft cell guides, before 40' container can be loaded in outboard cell. Total 26x40' and 2x20' containers can be stowed below. Each container may weigh up to 30.0 LT (30.5 MT)

45' containers are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 42 can be stowed.

40' ISO / 40' high cube are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 42 can be stowed.

40' ISO / 40' high cube are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 42 can be stowed.

MV GEORGE II

ABOVE DECK

H 5 40' ISO / 40' high cube and 45' containers may be stowed 12 across on cover and secured by twistlocks up to four high. Parallel lashed rods can be used, as needed, depending on tier weight.

H 6 40' ISO / 40' high cube and 45' containers may be stowed 12 across on cover and secured by twistlocks up to four high. Parallel lashed rods can be used, as needed, depending on tier weight.

H 7 40' ISO / 40' high cube and 45' containers may be stowed 12 across on cover and secured by twistlocks up to five high. Parallel lashed rods can be used, as needed, depending on tier weight. No 45' containers can be stowed in the first tier.

H 8 40' ISO / 40' high cube containers may be stowed 12 across on cover and secured by twistlocks up to five high. Parallel lashed rods can be used, as needed, depending on tier weight.

H 9 40' ISO / 40' high cube and 45' containers may be stowed 12 across on deck and pedestals and secured by twistlocks up to five high. Outboard two stacks can be four high only. 45' containers can be stowed at upper two tiers only. Parallel lashed rods can be used, as needed, depending on tier weight. At forward end of the hatch, due to lashing plates being lower than sockets, long lashing rods will be required for stack 09, 10, 11, 12.

BELOW DECK

40' ISO / 40' high cube are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 42 can be stowed.

45' containers are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 42 can be stowed.

40' ISO / 40' high cube are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 42 can be stowed.

40' ISO / 40' high cube are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 42 can be stowed.

No cargo below.

MV GEORGE II

ABOVE DECK

H 10 40' ISO / 40' high cube containers may be stowed 8 across on deck and pedestals and secured by twistlocks up to four high. Parallel lashed rods can be used, as needed, depending on tier weight. At aft end of the hatch, due to lashing plates being lower than sockets, long lashing rods will be required for stack 09, 10, 11, 12.

H11 40' ISO / 40' high cube and 45' containers may be stowed 12 across on cover and secured by twistlocks up to five high. No 45' containers at first tier. Parallel lashed rods can be used, as needed, depending on tier weight.

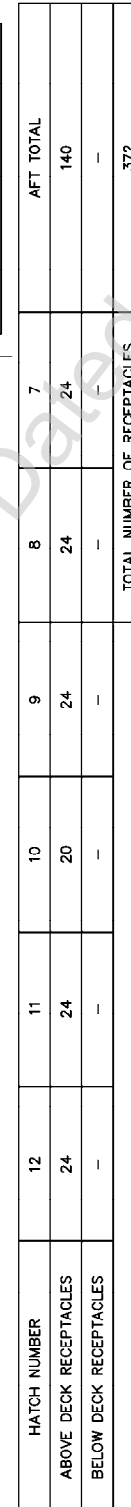
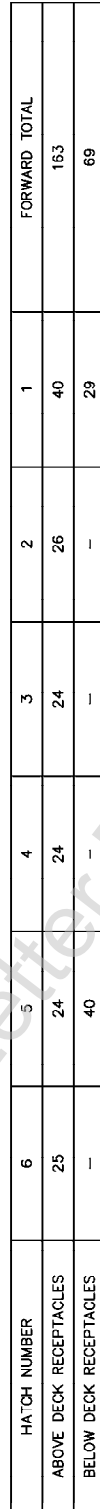
H12 40' ISO / 40' high cube and 45' containers may be stowed 12 across on cover and secured by twistlocks up to five high. Parallel lashed rods can be used, as needed, depending on tier weight. No 45' containers can be stowed in the first tier.

BELOW DECK

No cargo below.

45' containers are stowed in fixed cell guides below deck. Each container may weigh up to 30.0 LT (30.5 MT); a total of 28 can be stowed.

20' ISO are stowed in fixed cell guides below deck fwd. In Hatch 12A four tiers high and aft in Hatch 12B three tiers high, seven across. Total 49 can be stowed. Each container may weigh up to 30.0 LT (30.5 MT).



MV GEORGE II
REEFER OUTLET DIAGRAM
C9 CLASS

DECEMBER, 2022

3.4 ALLOWABLE STACK WEIGHTS

This section contains guidance sheets of allowable stack and tier weights for the *MV George II*, C9 Class Vessel. All possible stowage patterns are indicated for 40', 40' high cube, 45' containers and 20' containers, by hatch.

Container stack and tier weights are pre calculated for several GM's of 2.0 feet, 4.0 feet, 6.0 feet and 10.0 feet, for inboard and outboard stacks using the ABS LASH program.

Weights indicated are maximum allowable tier and stack weight. The weights indicated must not be exceeded.

For a GM between those given, a linear interpolation of stack and tier weight can be used. For GM's below 2.0 feet, the 2.0 feet weight should be used.

NOTES: 1. With four tier 40'/45' stacks and only a single rod lashing from the deck to the bottom of the 2nd tier, THE BOTTOM BOX, (first tier), MUST BE THE HEAVIEST BOX IN THE STACK.

2. With five tier 40'/45' stacks paralashed and two rod lashing, THE BOTTOM BOX, (first tier), MUST BE HEAVIER THAN THE SECOND HIGH, (second tier), BOX IN THE STACK.

3. With four or five tier 40' over 20' stacks and one or two rod lashing, THE BOTTOM 20', (first tier), MUST BE HEAVIER THAN THE SECOND HIGH 20', (second tier), BOX IN THE STACK.

4. With 45' in 2nd tier above a 40' in first tier – use the paralash system – DO NOT BEND RODS.

5. A 25 LT MSL should be used between 1st Tier and Hatch Cover/Pedestal.

6. The OUTBOARD STACK should always use PARALASH, if more than 2 high stacks.

7. IN/W (NOTE): Stacks inboard of IN/W, add two tons to stack at the fourth tier.

FOR A VIEW OF LASHING ARRANGEMENT AND STOWAGE PLEASE CLICK **BLUE NUMBERS AND LETTER** FOR DESIRED STOW

Views also printed at the end of this section.

ALLOWABLE TIER and STACK WEIGHTS, HATCHES 1 AND 2.

Two High - 40', 40'HC or 45': Twist locked only. 2A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	IN	IN	OUT	IN	OUT	IN	OUT
	2 nd Tier	30	30	30	30	30	28	29	24	24	20
	Stack	60	60	60	60	60	58	59	54	54	50

Three High - 40': Twist-locked only. 3A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	23	18	18	13	16	12	13	10	10	8
	2 nd – 3 rd	47	37	37	28	33	25	28	21	24	18
	Stack	77	67	67	58	63	55	58	51	54	48

Three High - 40'HC, or 45': Twist locked only. 3A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	23	15	18	12	16	10	13	10	10	7
	2 nd – 3 rd	47	35	37	26	33	23	28	20	24	17
	Stack	77	65	67	56	63	53	58	50	54	47

Three High - 40': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 3B	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	28	27	22	24	20	21	18	17	13
	2 nd – 3 rd	60	58	55	48	49	41	43	36	37	31
	Stack	90	88	85	75	79	71	73	66	67	61

Three High - 40'HC, or 45': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 3B	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	27	26	21	24	18	20	16	16	14
	2 nd – 3 rd	60	55	53	43	48	39	41	34	36	29
	Stack	90	85	83	73	78	69	71	64	66	59

Four High - 40': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 4A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	20	11	12	5	10	5	5	5	5	5
	3 rd – 4 th	40	27	30	19	25	14	20	10	15	10
	2 nd – 3 rd – 4 th	65	53	55	43	50	39	45	34	37	28
	Stack	90	80	84	68	79	65	72	63	67	53

Four High - 40'HC or 45': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 4A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	10	5	5	5	5	5	5	5	5	5
	3 rd – 4 th	36	21	25	16	20	10	15	10	15	10
	2 nd – 3 rd – 4 th	63	51	53	40	45	36	42	30	35	17
	Stack	90	81	82	70	75	66	70	60	65	47

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ALLOWABLE TIER and STACK WEIGHTS, HATCHES 3 Through 12.

Two High - 40', 40'HC or 45': Twist locked only. 2A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	2 nd Tier	30	30	30	30	30	29	29	24	24	20
	Stack	60	60	60	60	60	59	59	54	54	50

Three High - 40': Twist locked only. 3A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	23	18	18	13	16	12	13	10	10	8
	2 nd – 3 rd	47	37	37	28	33	25	29	21	24	18
	Stack	77	67	67	58	63	55	59	51	54	48

Three High - 40'HC or 45': Twist locked only. 3A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	23	15	18	12	16	11	13	10	10	7
	2 nd – 3 rd	47	35	37	26	33	23	29	20	24	17
	Stack	77	65	67	56	63	53	59	50	54	47

Three High - 40': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 3B	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	28	27	22	24	20	22	18	18	14
	2 nd – 3 rd	60	58	56	48	50	41	44	36	37	31
	Stack	90	88	86	75	80	71	74	66	67	61

Three High - 40'HC or 45': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 3B	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	27	26	21	24	18	20	16	16	14
	2 nd – 3 rd	60	55	54	43	48	39	42	34	36	29
	Stack	90	85	84	73	78	69	72	64	66	59

Four High - 40': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 4A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	20	11	12	5	10	5	5	5	5	5
	3 rd – 4 th	40	27	30	19	25	14	20	10	15	10
	2 nd – 3 rd – 4 th	68	53	55	43	50	39	45	35	37	28
	Stack	90	80	85	68	80	65	73	64	67	58

Four High - 40'HC or 45': Twist locked and single rod lashing, deck to bottom of 2 nd tier. 4A	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	10	5	5	5	5	5	5	5	5	5
	3 rd – 4 th	36	21	25	16	20	10	15	10	15	10
	2 nd – 3 rd – 4 th	63	51	53	40	45	36	42	31	36	18
	Stack	90	81	82	70	75	66	70	61	66	48

ALLOWABLE TIER and STACK WEIGHTS, HATCHES 3 Through 12 (Continued).

Five High - 40': Twist locked and one rod lashing, deck to bottom of 2 nd tier, INBOARD ONLY. <u>5A</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	5 th Tier	10	-	5	-	5	-	5	-	NA	-
	4 th -5 th	24	-	15	-	15	-	10	-	5	-
	3 rd - 4 th -5 th	44	-	32	-	27	-	15	-	15	-
	2 nd -3 rd -4 th -5 th	65	-	57	-	47	-	42	-	37	-
	Stack	90	-	82	-	77	-	70	-	67	-

Five High - 40'HC or 45': Twist locked and one rod lashing, deck to bottom of 2 nd tier, INBOARD ONLY. <u>5A</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	5 th Tier	5	-	5	-	5	-	5	-	NA	-
	4 th -5 th	12	-	10	-	10	-	10	-	5	-
	3 rd -4 th -5 th	32	-	19	-	15	-	15	-	15	-
	2 nd -3 rd -4 th -5 th	59	-	46	-	37	-	24	-	35	-
	Stack	87	-	76	-	67	-	54	-	65	-

NOTE: Five High stowage to agree with Section 3.1 & 3.2.

IN/W (NOTE): Stacks inboard of IN/W, add two tons to stack.

**ALLOWABLE TIER and STACK WEIGHTS, HATCHES 3 and 4,
with two 20' containers in a 40' slot.**

Maximum 20' stack weight is 24 Long Tons. Maximum stack weight: 90 Long Tons.

Maximum 20' container weight: 24 Long Tons each.

Two High - 20': Twist locked only. 2B	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	2 nd Tier	19	19	19	19	19	19	19	19	19	19
	Stack	24	24	24	24	24	24	24	24	24	24

Three High - 20': Twist-locked only. 3C	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	14	14	14	14	14	14	14	14	14	14
	2 nd – 3 rd	19	19	19	19	19	19	19	19	19	19
	Stack	24	24	24	24	24	24	24	24	24	24

Three High - 20' on 1 st tier: Twist locked and single rod lashing, deck to bottom of 2 nd tier. 3D	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	30	28	20	23	25	18	14	15	10
	2 nd – 3 rd	60	60	58	49	53	44	47	39	40	33
	Stack	84	84	82	73	77	68	71	63	64	57

Three High - 20' on 1 st and 2 nd tiers: Twist locked and single rod lashing, deck to bottom of 2 nd tier. 3E	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	30	30	30	30	28	29	22	22	16
	2 nd – 3 rd	49	49	49	49	49	47	48	41	41	35
	Stack	54	54	54	54	54	52	43	46	46	40

Four High - 20' on 1 st tier: Twist locked and single rod lashing, deck to bottom of 2 nd tier. 4B	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	20	10	10	5	10	5	5	5	5	5
	3 rd – 4 th	40	26	28	20	27	15	22	15	18	10
	2 nd – 3 rd – 4 th	66	56	58	45	52	40	47	31	38	25
	Stack	90	80	82	69	76	64	71	55	62	39

Four High - 20' on 1 st and 2 nd tiers: Twist locked and single rod lashing, deck to bottom of 2 nd tier. 4C	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	15	10	10	7	10	5	5	5	5	5
	3 rd – 4 th	45	25	33	18	27	16	24	11	17	10
	2 nd – 3 rd – 4 th	64	44	52	37	46	35	43	30	36	21
	Stack	69	49	56	42	56	40	47	35	40	26

ALLOWABLE TIER and STACK WEIGHTS, ALL HATCHES,
40’/45’ Containers Maximum
Stack Weight 90 LT Year
Round

Four High - 40’: Twistlocks only, no rod lashings. <u>4D</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	15	10	14	5	10	5	10	5	5	5
	3 rd – 4 th	34	24	29	16	25	14	20	10	14	10
	2 nd – 3 rd – 4 th	53	40	44	32	40	29	35	26	30	18
	Stack	72	56	59	48	56	45	50	42	47	42

Four High - 40’HC or 45’: Twistlocks only, no rod lashings. <u>4D</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN/W	OUT	IN/W	OUT
	4 th Tier	15	5	10	5	10	5	5	NA	5	NA
	3 rd – 4 th	30	15	22	10	20	10	14	14	10	11
	2 nd – 3 rd – 4 th	50	36	42	29	36	22	29	29	23	23
	Stack	75	57	63	48	59	49	51	44	49	40

Five High - 40’: Twistlocks only, no rod lashings. <u>5B, 4D</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN/W	OUT	IN/W	OUT	IN/W	OUT	IN/W	OUT	IN	OUT
	5 th Tier	5	NA	5	NA	5	NA	5	NA	NA	NA
	4 th - 5 th	15	10	10	5	10	5	10	5	5	5
	3 rd - 4 th - 5 th	30	24	22	16	18	14	15	10	14	10
	2 nd - 3 rd - 4 th - 5 th	50	40	37	32	32	29	23	26	30	18
	Stack	70	56	57	48	52	45	43	42	47	42

Five High - 40’HC or 45’: Twistlocks only, no rod lashings. <u>5B, 4D</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN/W	OUT	IN/W	OUT	IN/W	OUT	IN/W	OUT	IN/W	OUT
	5 th Tier	5	NA	5	NA	5	NA	NA	NA	NA	NA
	4 th - 5 th	13	5	10	5	10	5	5	NA	5	NA
	3 rd - 4 th - 5 th	25	15	15	10	15	10	14	14	10	11
	2 nd - 3 rd - 4 th - 5 th	40	36	32	29	22	22	29	29	23	23
	Stack	70	57	62	48	42	49	51	44	49	40

NOTE: Five High stowage to agree with Section 3.1 & 3.2.

IN: Inboard Stack not Exposed to Wind.

IN/W: Inboard Stack / Inboard Stack Top Tier Exposed to Wind.

OUT: Stack Exposed to Wind.

ALLOWABLE TIER and STACK WEIGHTS, HATCHES 1 Through 12. Paralash.

Three High - 40', 40' HC or 45': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. 3F	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	30	30	30	30	25	28	22	21	17
	2 nd – 3 rd	60	60	60	60	60	55	55	47	46	37
	Stack	90	90	90	90	90	85	85	77	76	67

Three High - 45' on 40': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. 3G	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	3 rd Tier	30	30	30	30	30	25	25	22	21	17
	2 nd – 3 rd	60	60	60	60	60	55	55	47	46	37
	Stack	90	90	90	90	90	85	85	77	76	67

Four High - 40', 40' HC OR 45': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. 4E	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	23	15	11	5	10	5	5	5	5	5
	3 rd – 4 th	45	30	37	25	30	19	24	14	16	10
	2 nd – 3 rd – 4 th	65	60	63	48	60	45	54	34	46	24
	Stack	90	90	90	78	90	75	84	64	74	54

Four High - 45' on 40': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. 4F	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	4 th Tier	25	15	11	5	10	5	5	5	5	5
	3 rd – 4 th	45	30	37	25	30	19	24	14	16	10
	2 nd – 3 rd – 4 th	65	60	63	48	60	45	54	34	46	24
	Stack	90	90	90	78	90	75	84	64	76	54

Five High - 40', 40' HC or 45': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. 5C, 5E	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN/W	OUT	IN/W	OUT	IN/W	OUT
	5 th Tier	5	NA	5	NA	5	NA	5	NA	5	NA
	4 th – 5 th	15	10	13	5	10	5	10	5	10	5
	3 rd – 4 th – 5 th	38	27	23	25	21	18	15	10	15	10
	2 nd – 3 rd – 4 th – 5 th	62	56	53	45	50	46	37	27	25	24
	Stack	87	86	83	75	80	76	67	45	55	54

Five High - 45' on 40': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. 5D, 5F	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN/W	OUT	IN/W	OUT	IN/W	OUT	IN/W	OUT	IN	OUT
	5 th Tier	5	NA	5	NA	5	NA	5	NA	NA	NA
	4 th – 5 th	15	15	13	5	11	5	10	5	5	5
	3 rd – 4 th – 5 th	38	30	23	25	21	18	15	10	16	10
	2 nd – 3 rd – 4 th – 5 th	62	60	53	48	50	46	37	27	46	24
	Stack	87	90	83	78	80	76	67	45	76	54

NOTE: Five High stowage to agree with Section 3.1 & 3.2.

IN/W (NOTE): Stacks inboard of IN/W, add two tons to stack.

ALLOWABLE TIER and STACK WEIGHTS, HATCHES 1 Through 12,

World Wide Trading

Extra Vertical Lash on Outboard Weather Stack.

Maximum Stack Weight 90 LT.

Five High - 40': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. <u>5D</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	5 th Tier	5	5	5	5	5	5	5	5	5	5
	4 th - 5 th	20	15	15	15	10	10	10	10	10	10
	3 rd - 4 th - 5 th	40	30	35	25	30	15	23	20	15	15
	2 nd -3 rd -4 th -5 th	65	60	61	45	58	45	52	40	45	35
	Stack	90	90	87	75	88	75	82	70	75	65

Five High - 40'HC or 45': Twist locked and two parallel rod lashings, deck to top of 1 st tier and bottom of 2 nd tier. <u>5E, 5F</u>	GM	2.0 Feet		4.0 Feet		6.0 Feet		10.0 Feet		15.0 Feet	
		IN	OUT	IN	OUT	IN	OUT	IN	OUT	IN	OUT
	5 th Tier	5	5	5	5	5	5	5	5	NA	NA
	4 th - 5 th	20	15	15	15	13	10	12	10	5	5
	3 rd - 4 th - 5 th	43	25	26	25	23	20	17	15	16	10
	2 nd -3 rd -4 th -5 th	68	55	56	40	53	40	39	35	46	24
	Stack	90	85	86	70	83	70	69	65	76	54

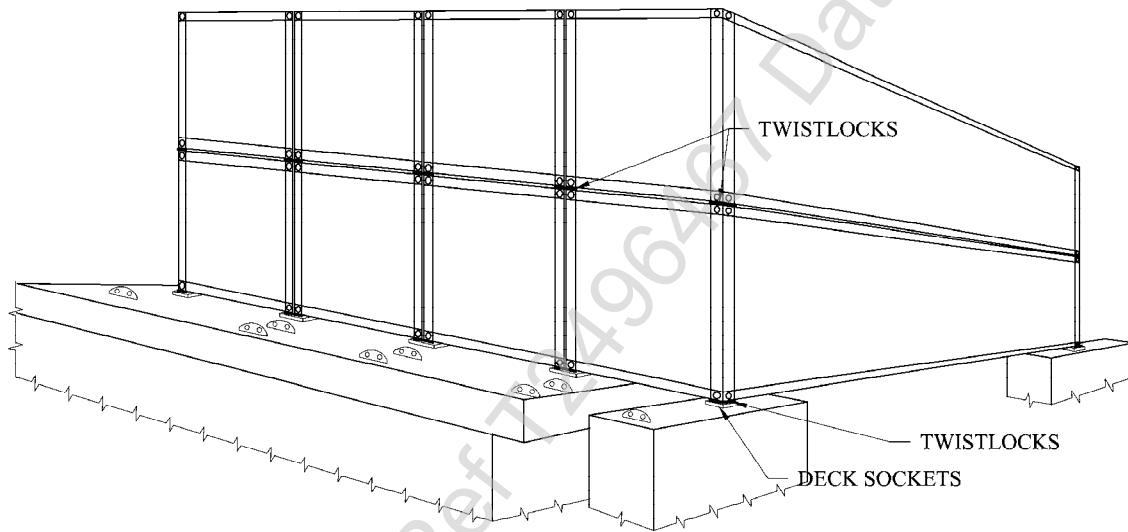
NOTE: Where first inboard stack's tier 5 is subject to wind load (four high outboard stack) the 4th and 5th tier shall be reduced by 2.0 tons each. For stacks not subject to any wind, use weights shown.

NOTE: Five High stowage to agree with Section 3.1 & 3.2.

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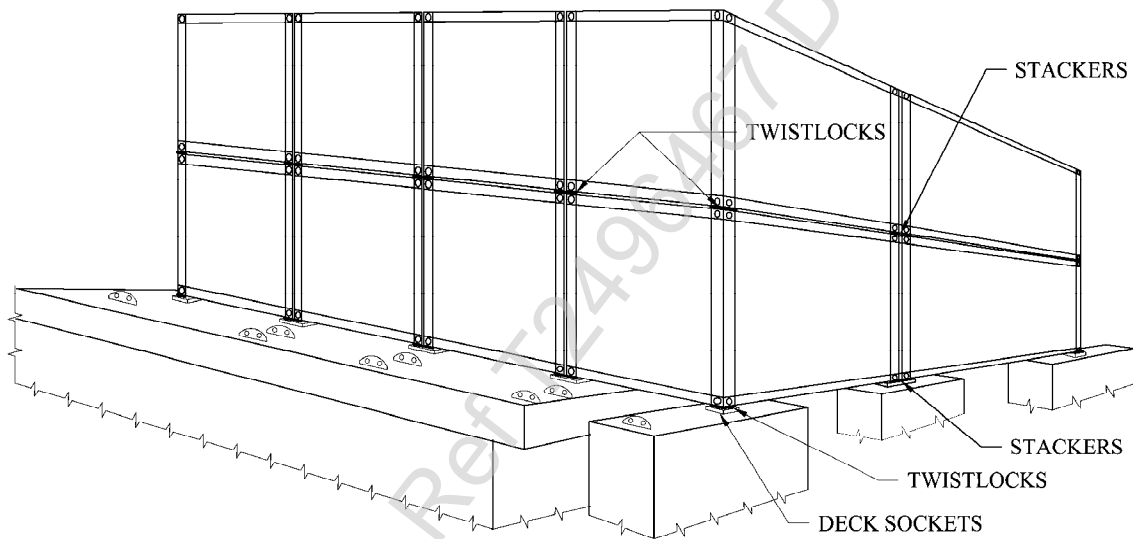
Stowage 2A

(40 FT. STOWAGE – STD/HC & 45 FT.)



LASHING SYSTEM
2 HIGH
TWIST LOCKS ONLY

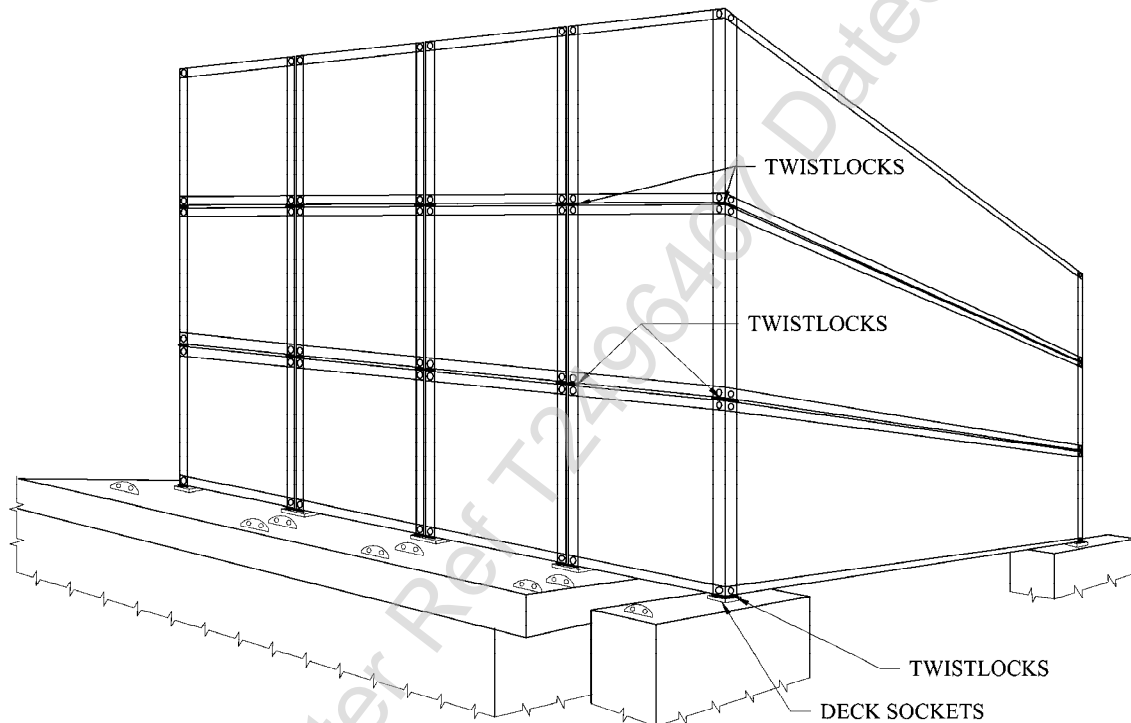
Stowage 2B (TWO 20 FT. IN 40 FT. SPACE)



LASHING SYSTEM
2 HIGH
TWIST LOCKS ONLY

Stowage 3A

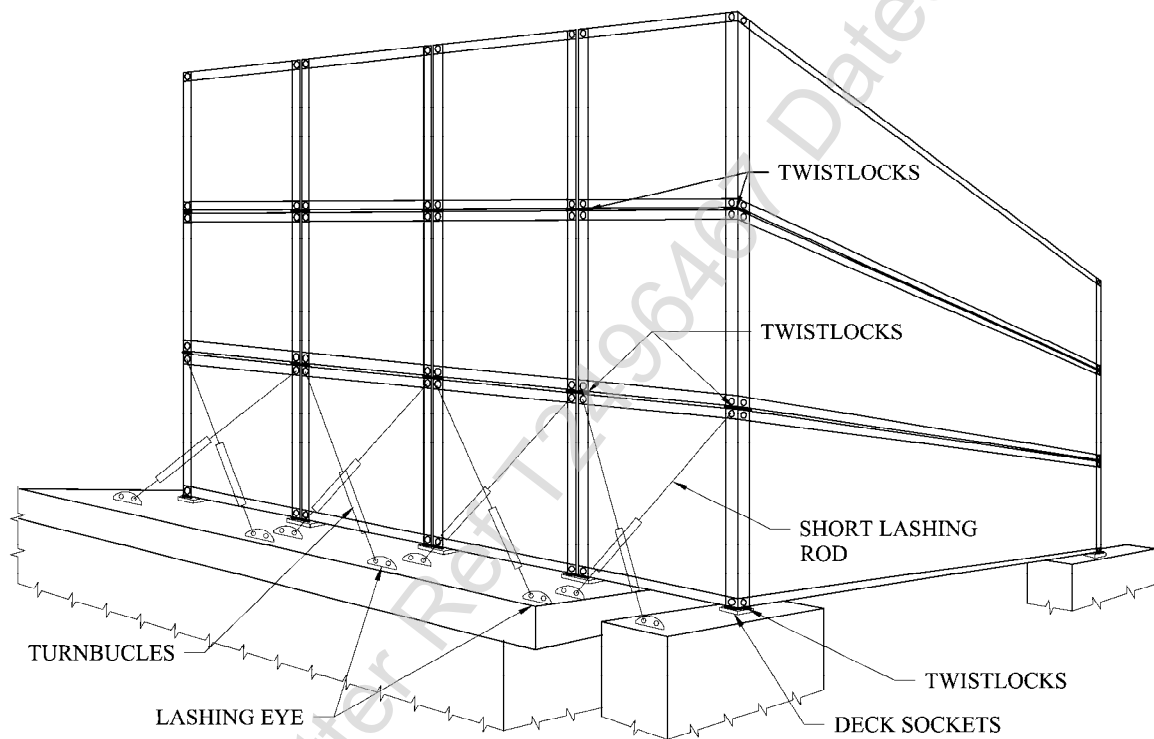
(40 FT. STOWAGE STD/HC & 45 FT.)



LASHING SYSTEM
3 HIGH
TWIST LOCKS ONLY

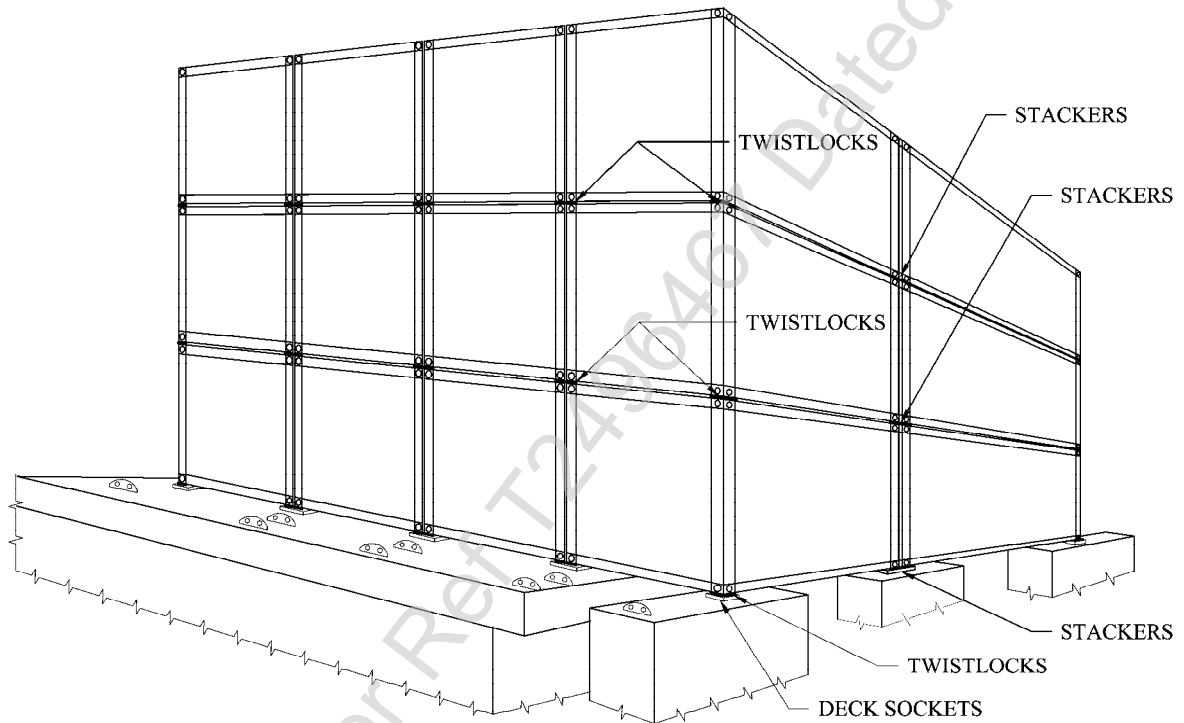
Stowage 3B

(40 FT. STOWAGE STD/HC & 45 FT.)



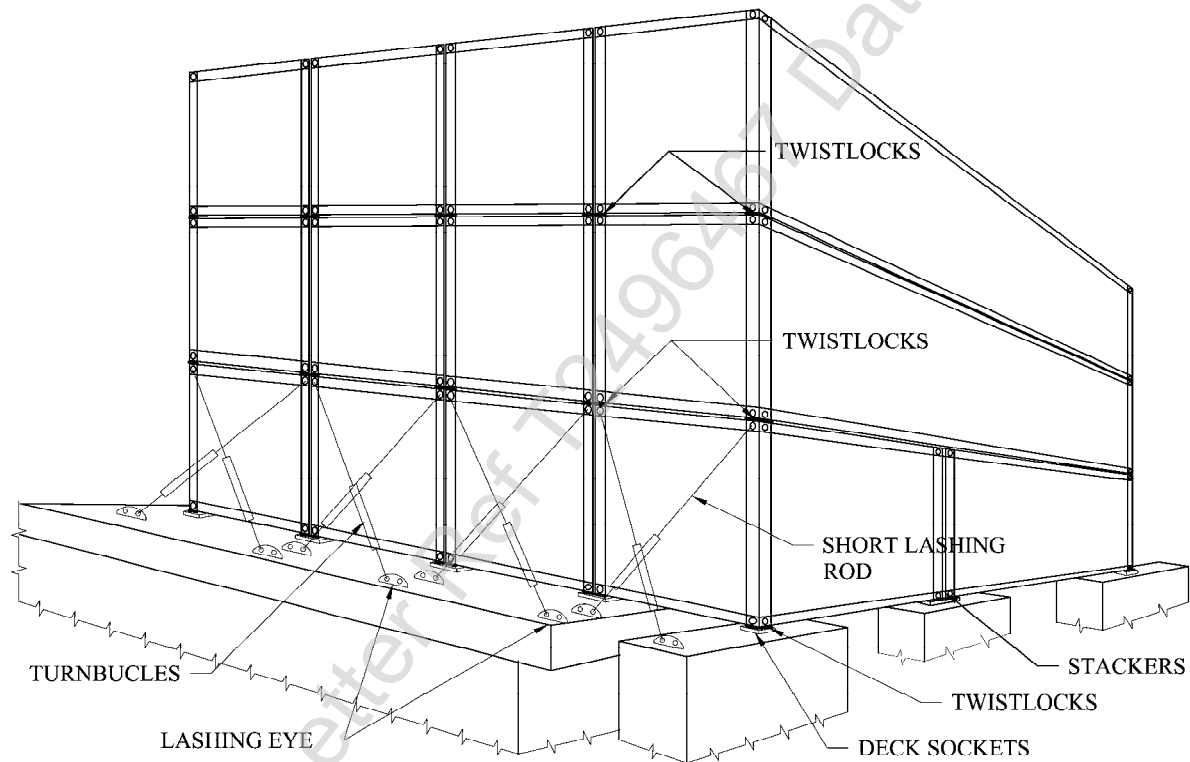
ROD LASHING SYSTEM
3 HIGH - TWIST LOCKS WITH
LOWER ROD ONLY

Stowage 3C (TWO 20 FT. IN 40 FT. SPACE)



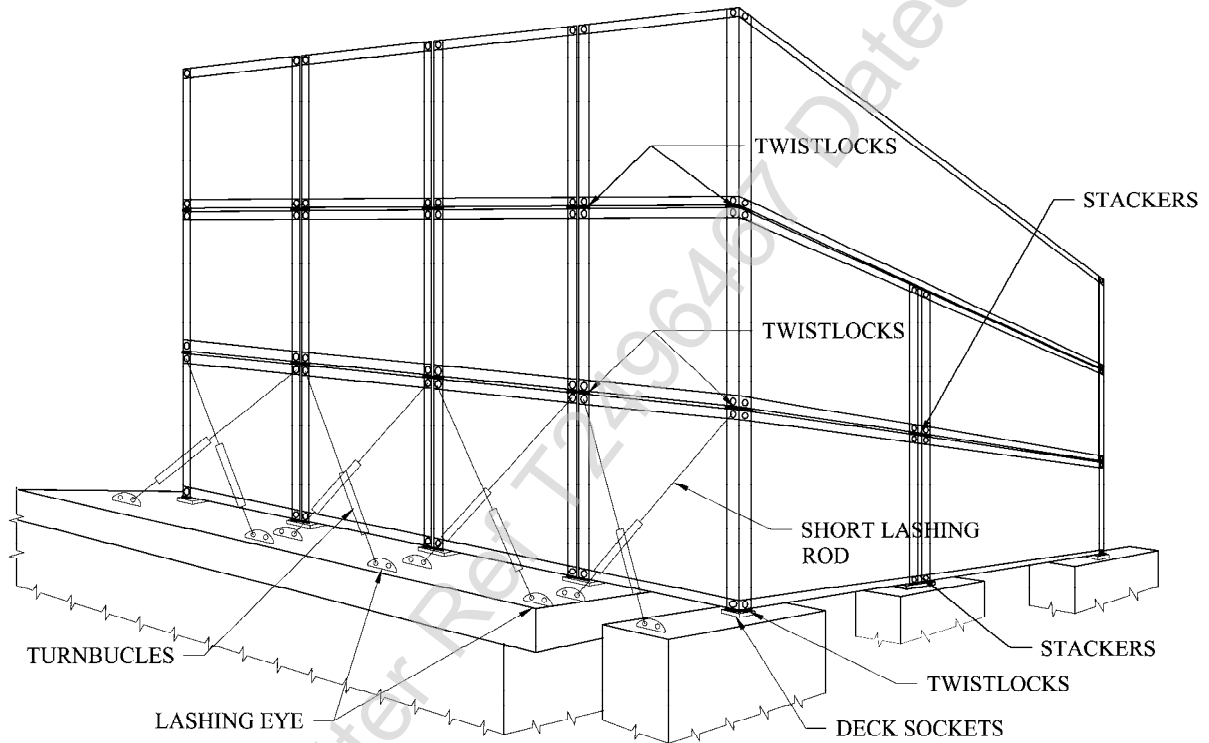
LASHING SYSTEM
3 HIGH
TWIST LOCKS ONLY

Stowage 3D (40 FT. STOWAGE OVER 20 FT.)



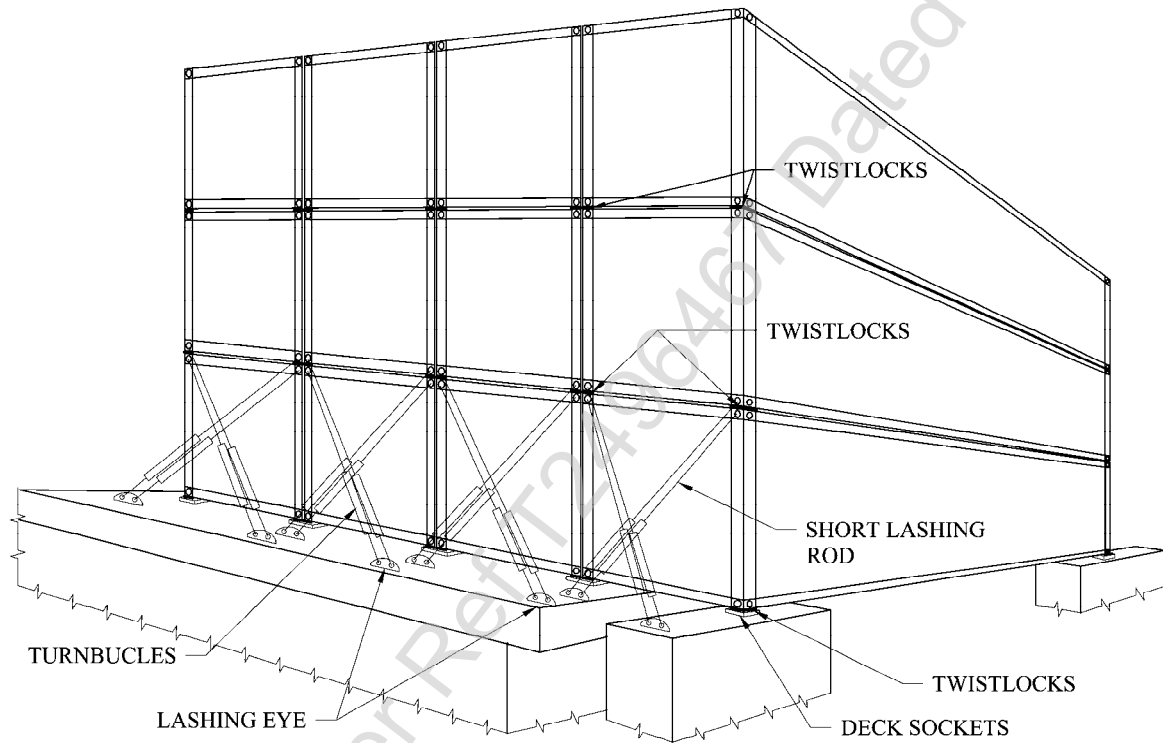
ROD LASHING SYSTEM
TWIST LOCKS & LOWER ROD
3 HIGH – 20' ON 1ST TIER

Stowage 3E (40 FT. STOWAGE OVER 20 FT.)



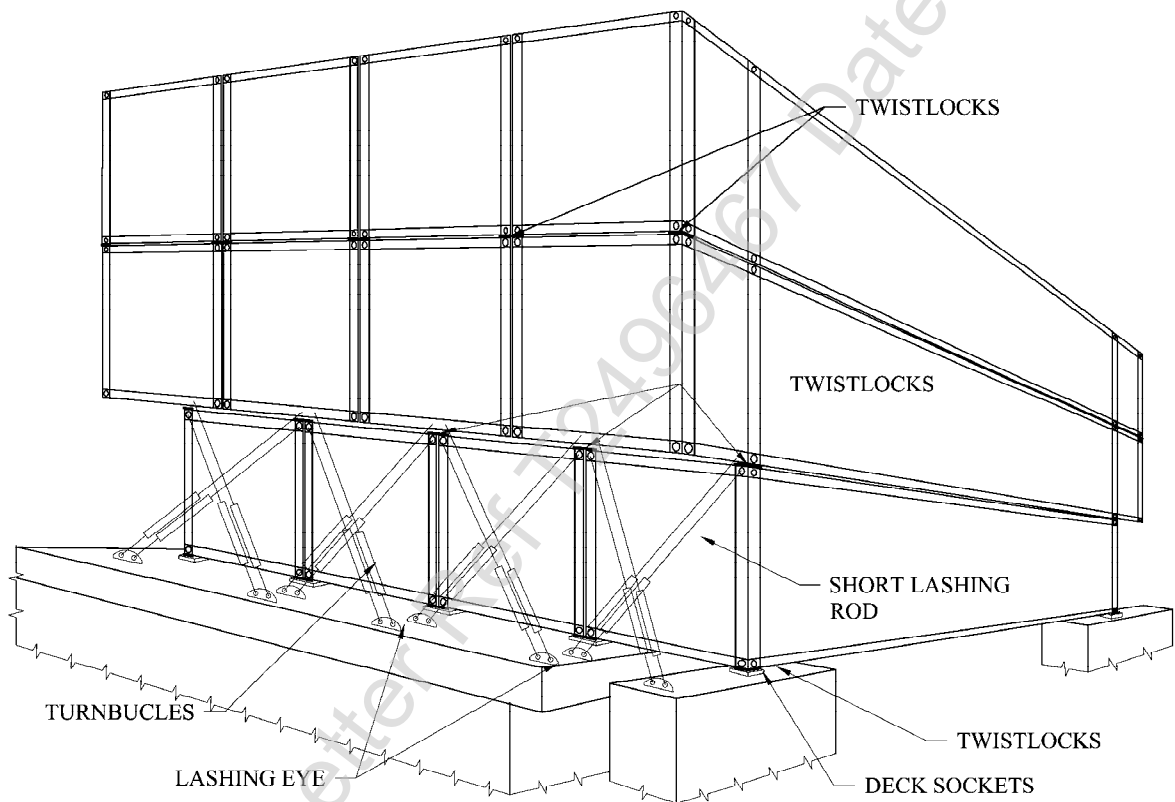
ROD LASHING SYSTEM
3 HIGH – TWIST LOCKS WITH
LOWER RODS
20' ON 1ST & 2ND TIER

Stowage 3F (40 FT. STD/HC STOWAGE & 45 FT.)



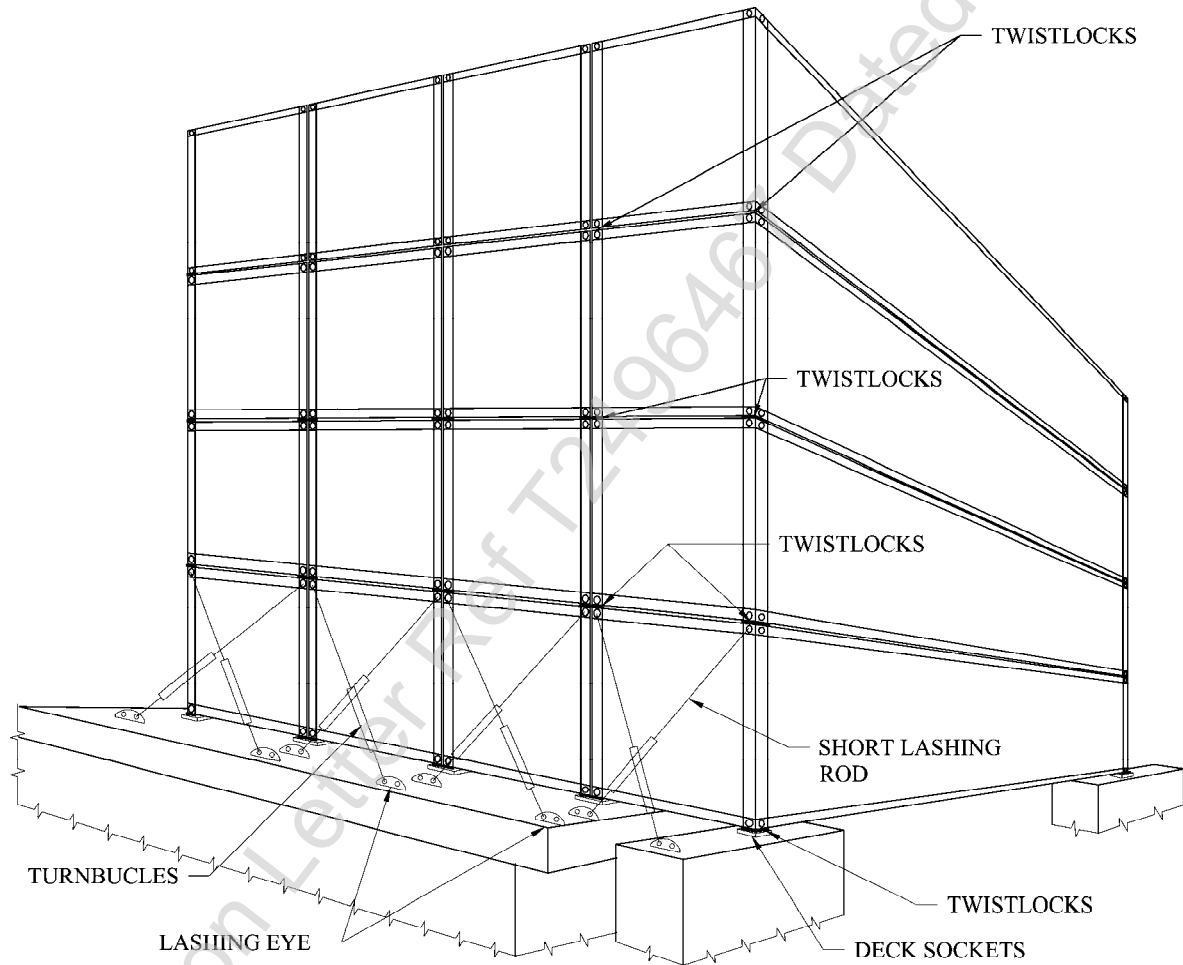
ROD LASHING SYSTEM
3 HIGH
TWIST LOCKS AND LOWER RODS
PARALLEL LASHING

Stowage 3G (45 FT. STOWAGE on 40 FT.)



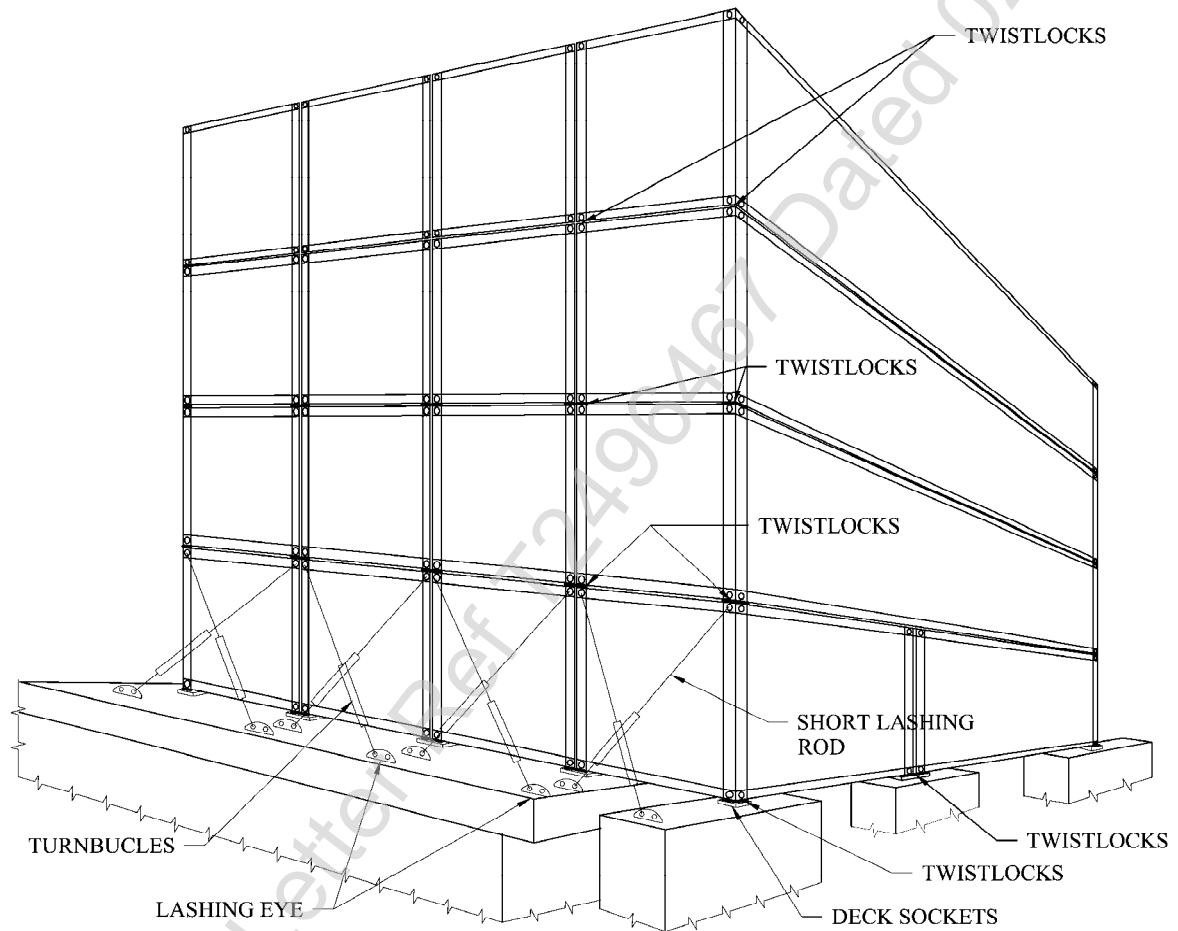
ROD LASHING SYSTEM
2 – 45' OVER 1 – 40'
TWIST LOCKS AND PARALLEL
LASHING

Stowage 4A (40 FT. STOWAGE STD/HC & 45 FT.)



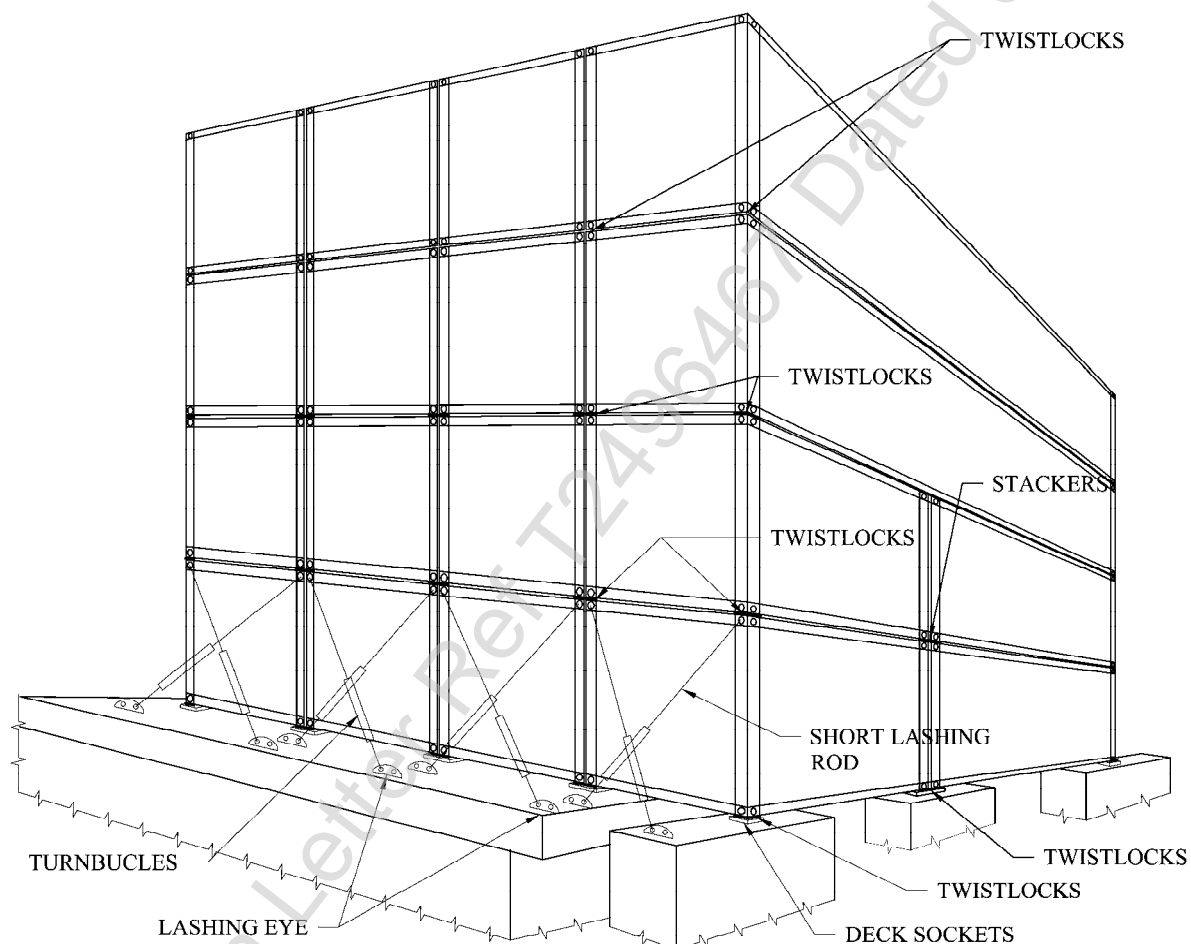
ROD LASHING SYSTEM
4 HIGH - TWIST LOCKS WITH
LOWER ROD ONLY

Stowage 4B (40 FT. STOWAGE OVER 20 FT.)



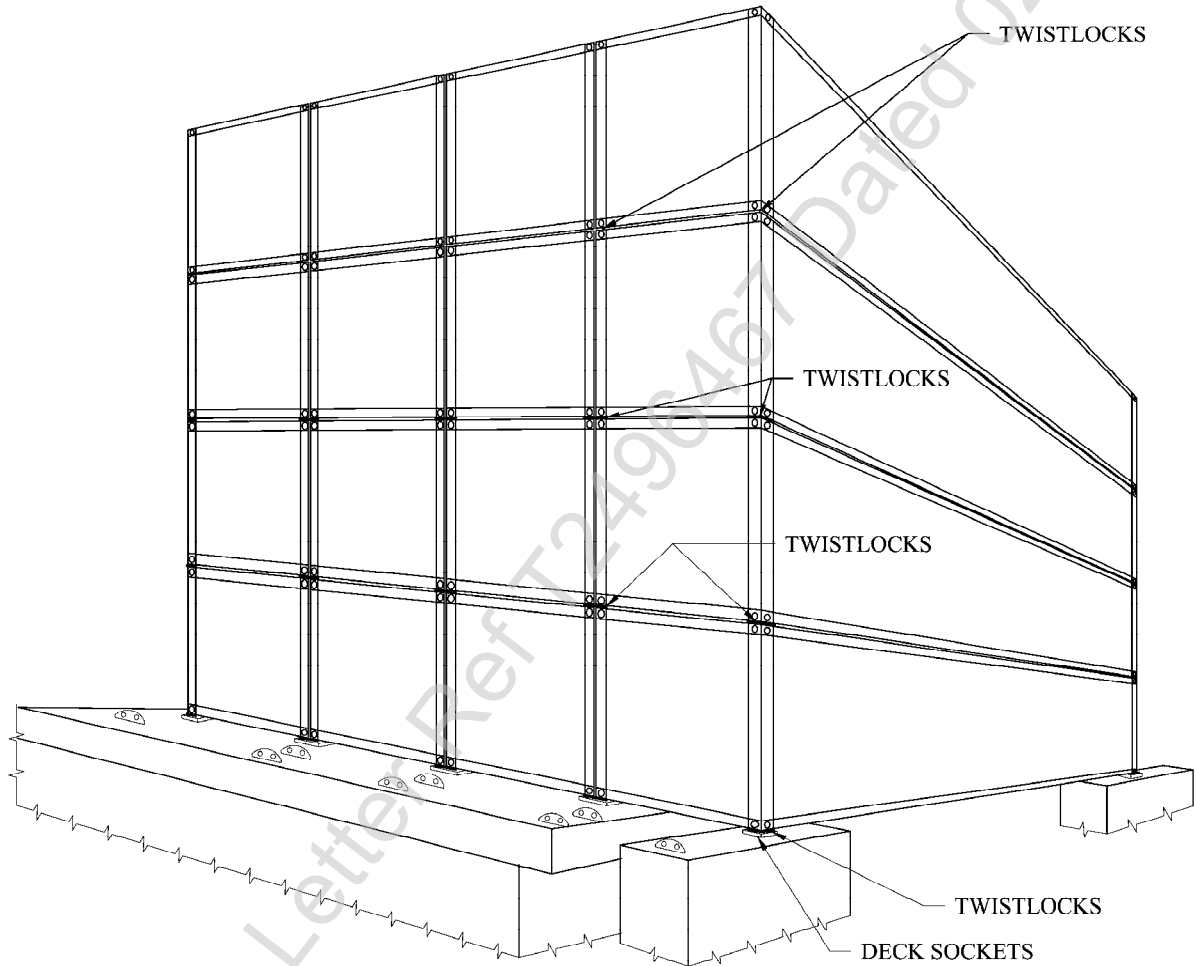
ROD LASHING SYSTEM
4 HIGH – TWIST LOCKS WITH
LOWER RODS
20' ON 1ST TIER

Stowage 4C
(40 FT. STOWAGE OVER 20 FT.)



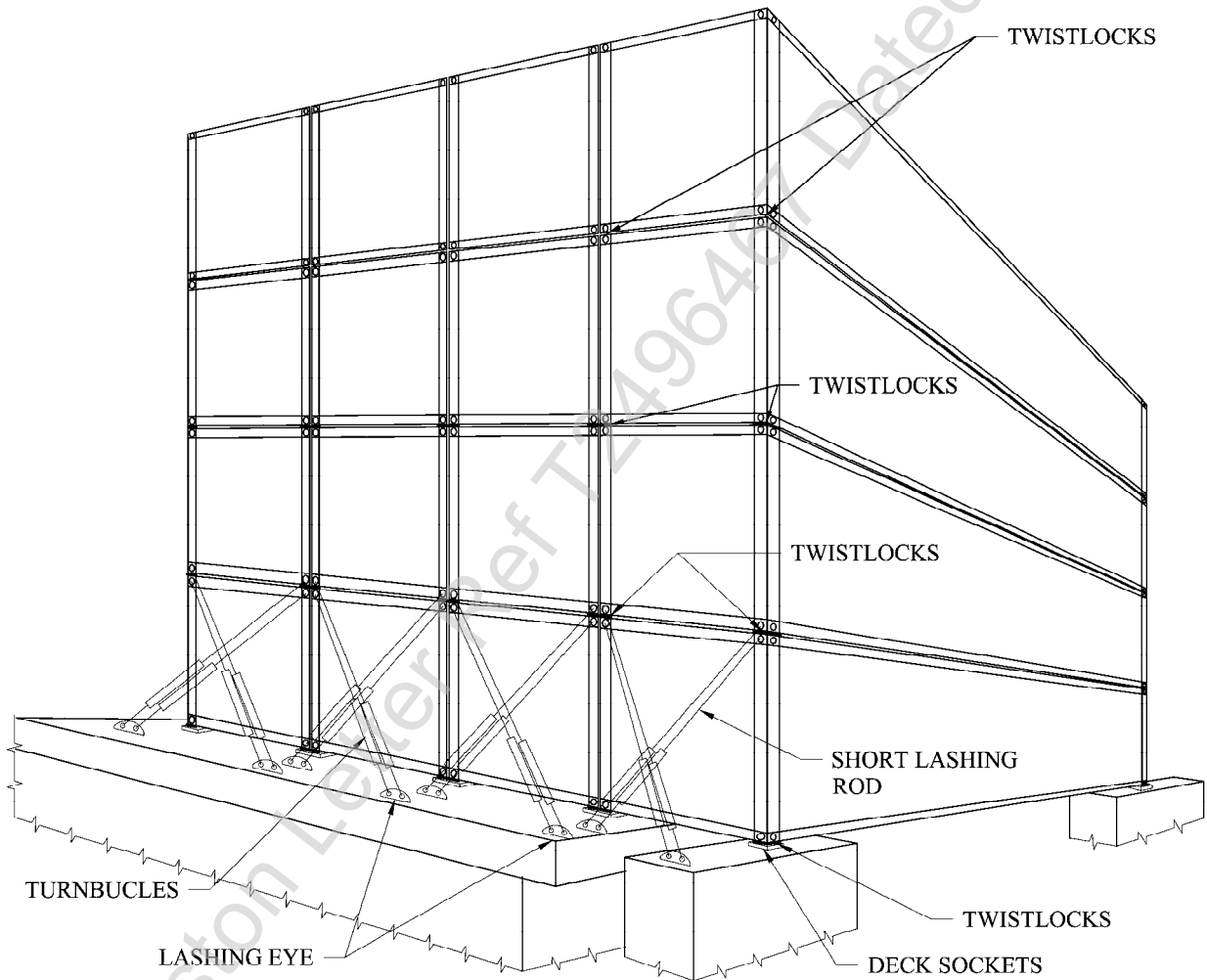
ROD LASHING SYSTEM
4 HIGH – TWIST LOCKS WITH
LOWER RODS
20' ON 1ST & 2ND TIER

Stowage 4D (40 FT. STOWAGE STD/HC & 45 FT.)



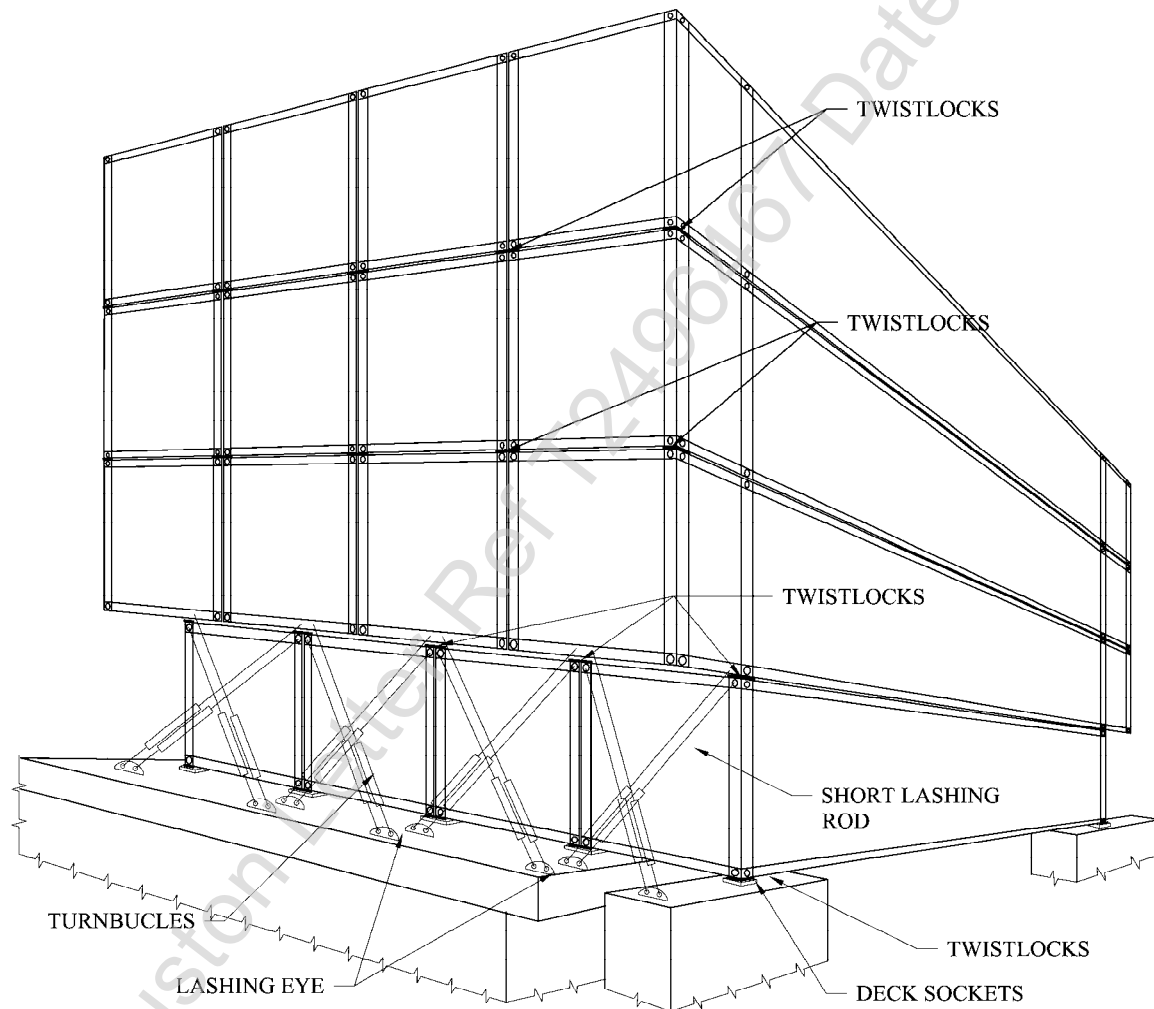
LASHING SYSTEM
4 HIGH
TWIST LOCKS ONLY

Stowage 4E (40 FT. STD/HC STOWAGE & 45 FT.)



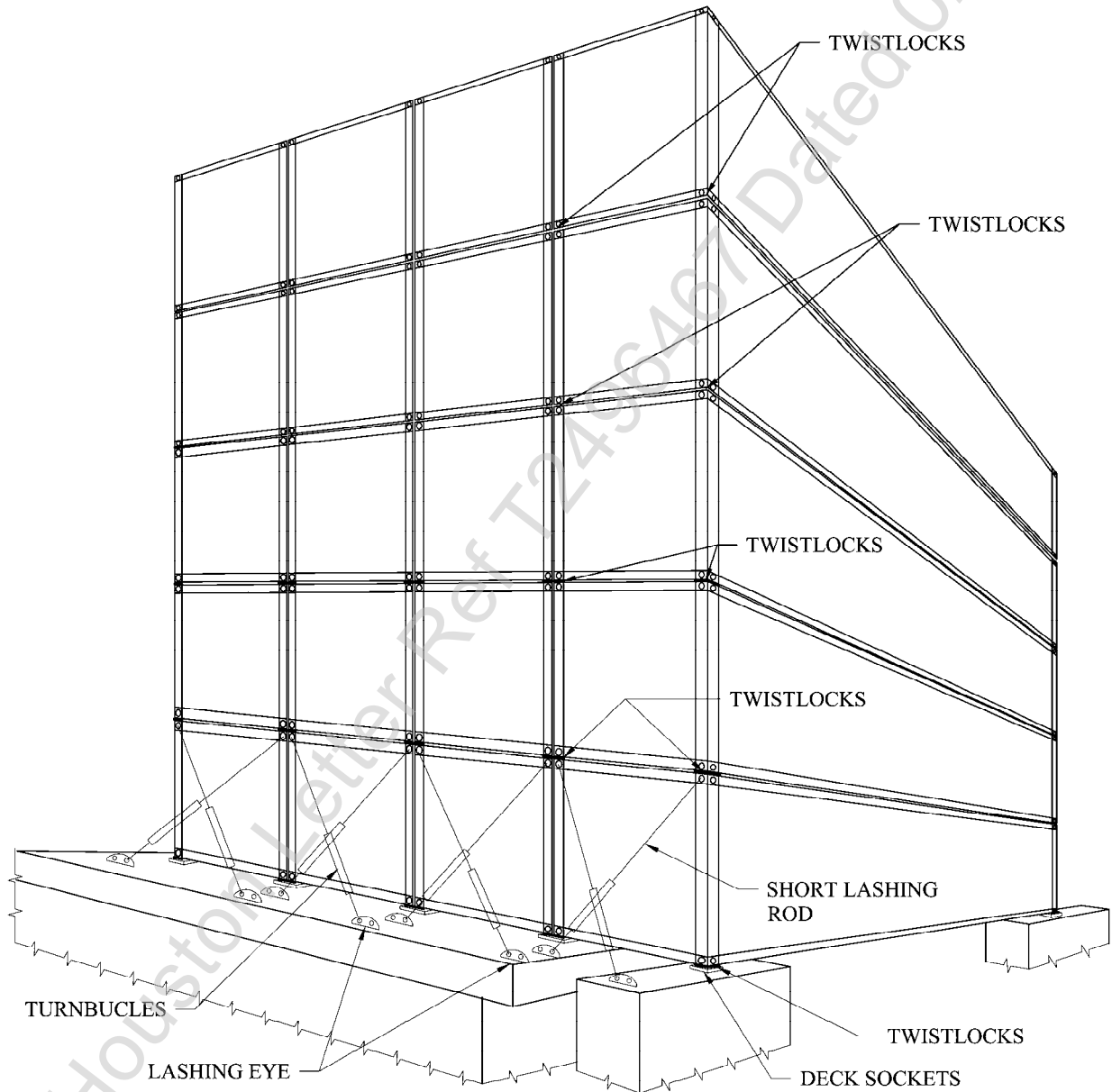
ROD LASHING SYSTEM
4 HIGH
TWIST LOCKS AND LOWER RODS
PARALLEL LASHING

Stowage 4F (45 FT. STOWAGE on 40 FT.)



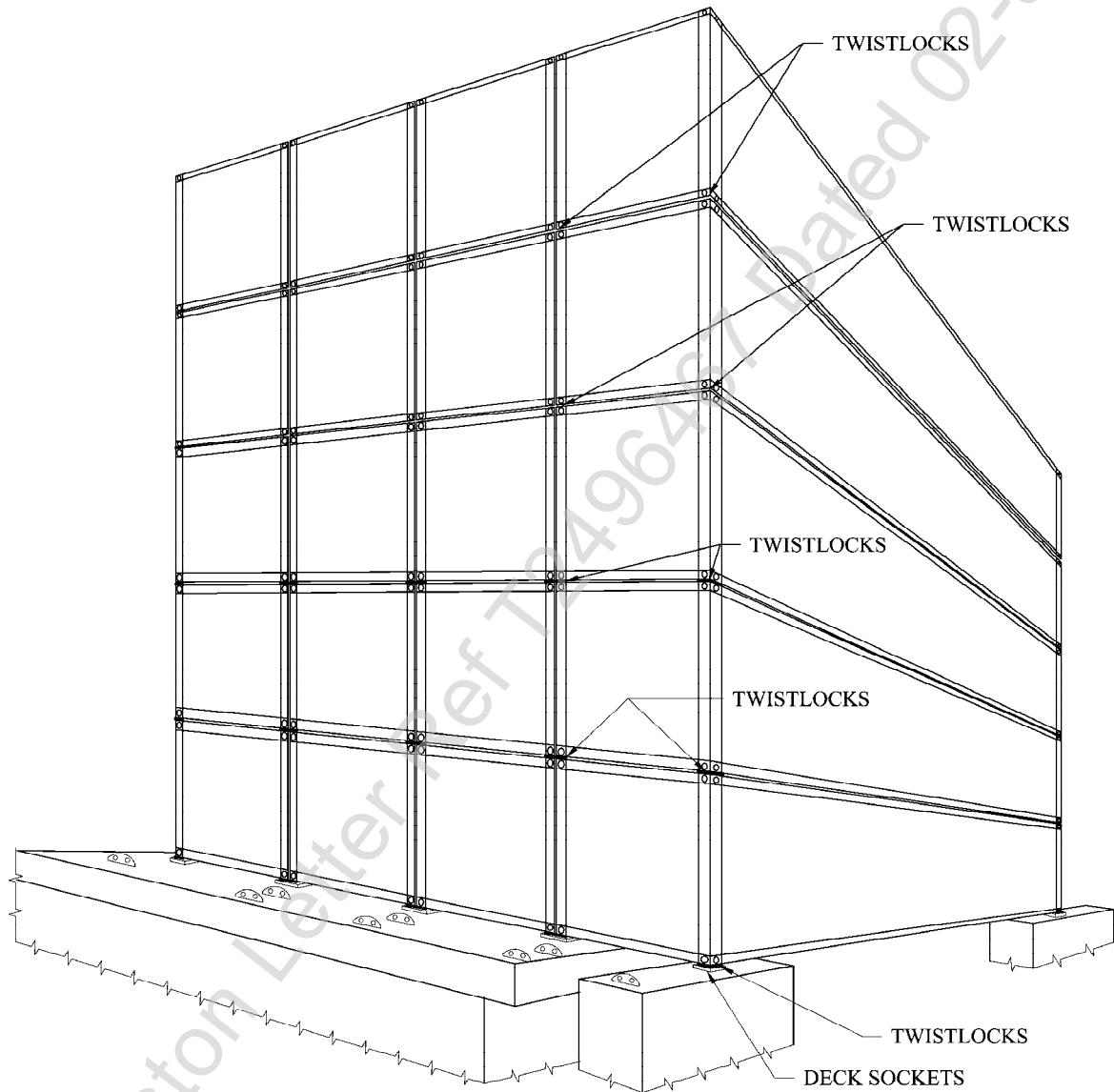
ROD LASHING SYSTEM
3 – 45' OVER 1 – 40'
TWIST LOCKS AND PARALLEL
LASHING

Stowage 5A (40 FT. STD/HC STOWAGE & 45 FT.)



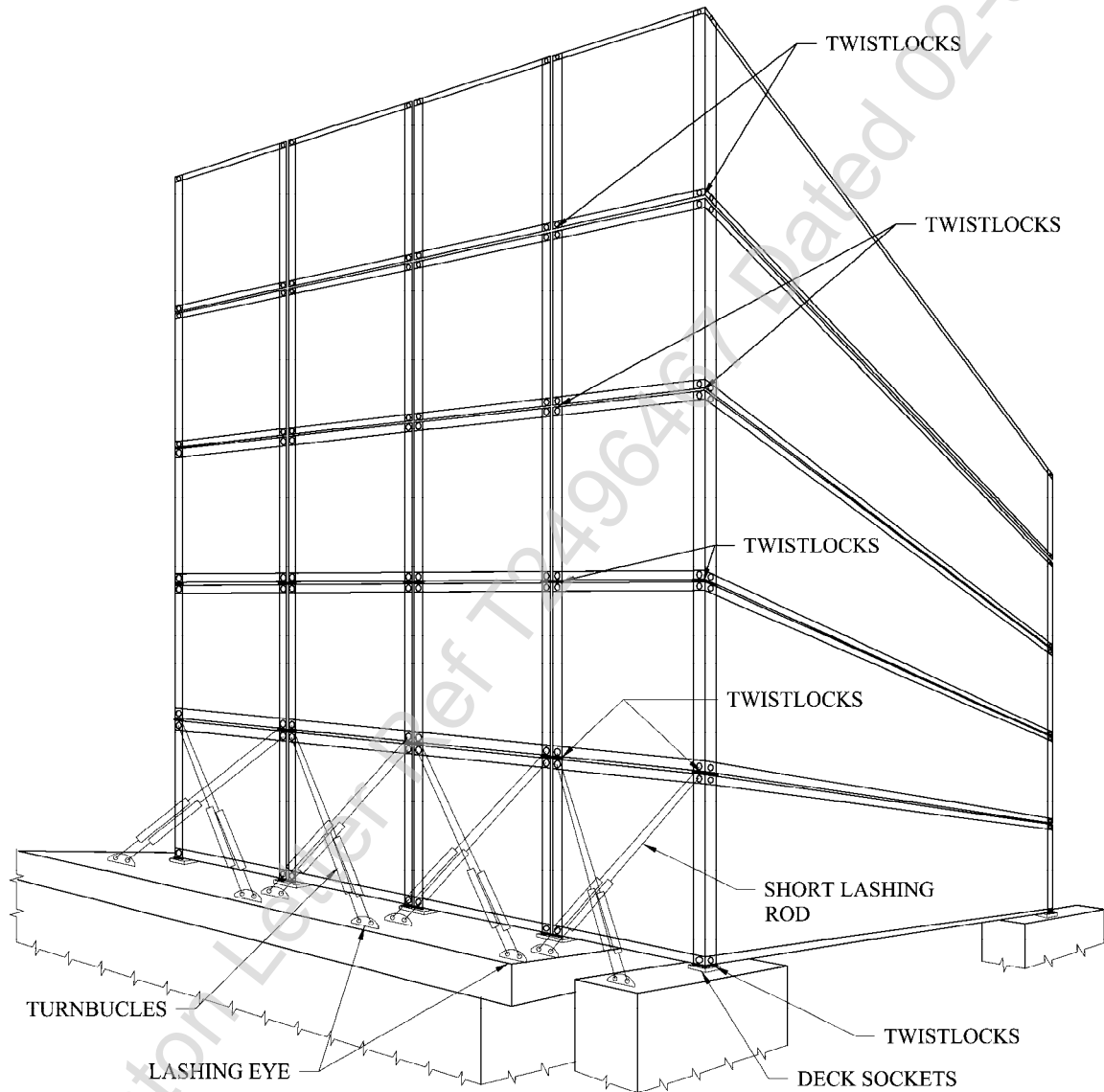
ROD LASHING SYSTEM
5 HIGH - TWIST LOCKS WITH
LOWER ROD ONLY

Stowage 5B (40 FT. STD/HC STOWAGE & 45 FT.)



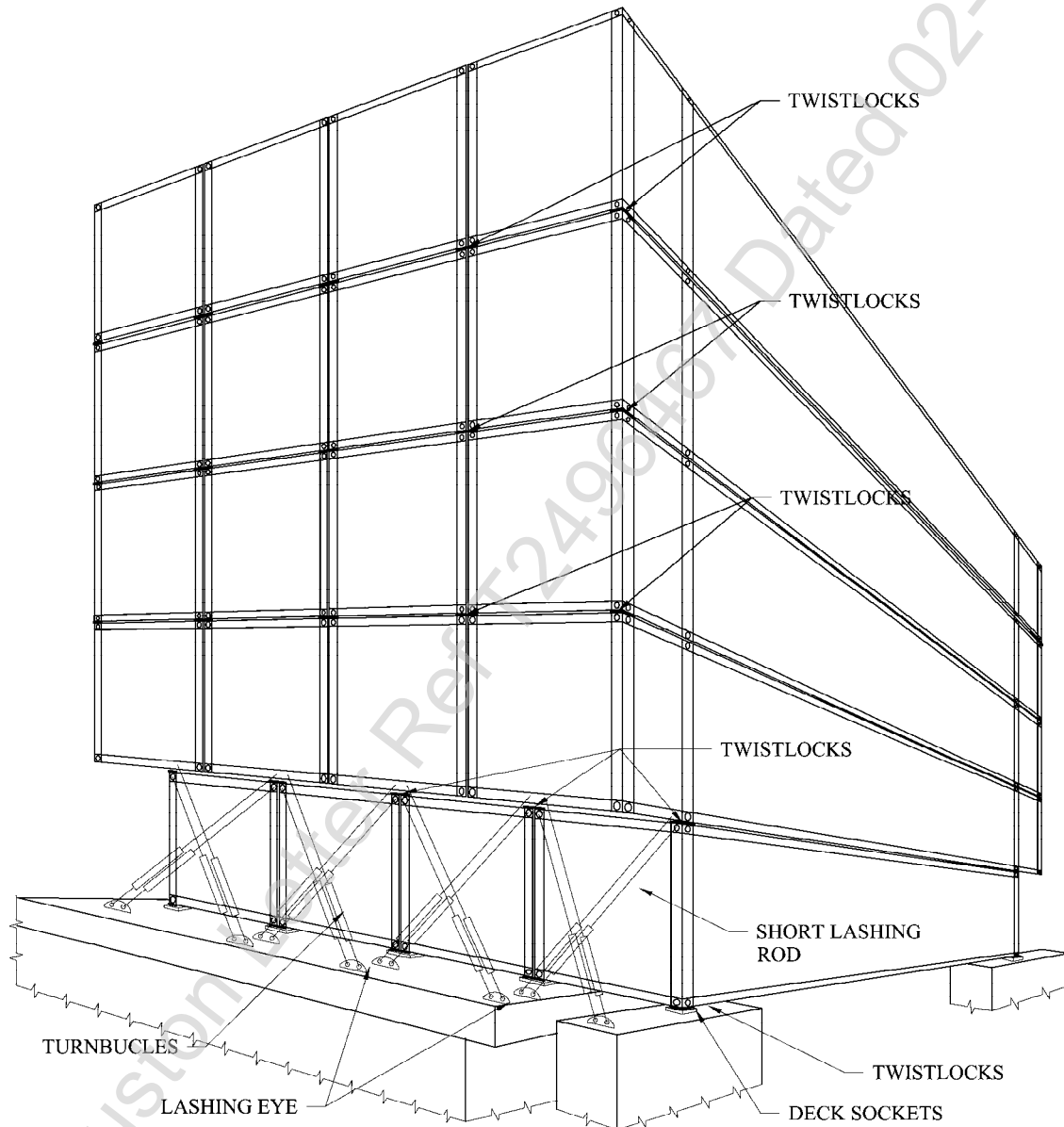
LASHING SYSTEM
5 HIGH
TWIST LOCKS ONLY

Stowage 5C (40 FT. STD/HC STOWAGE & 45 FT.)



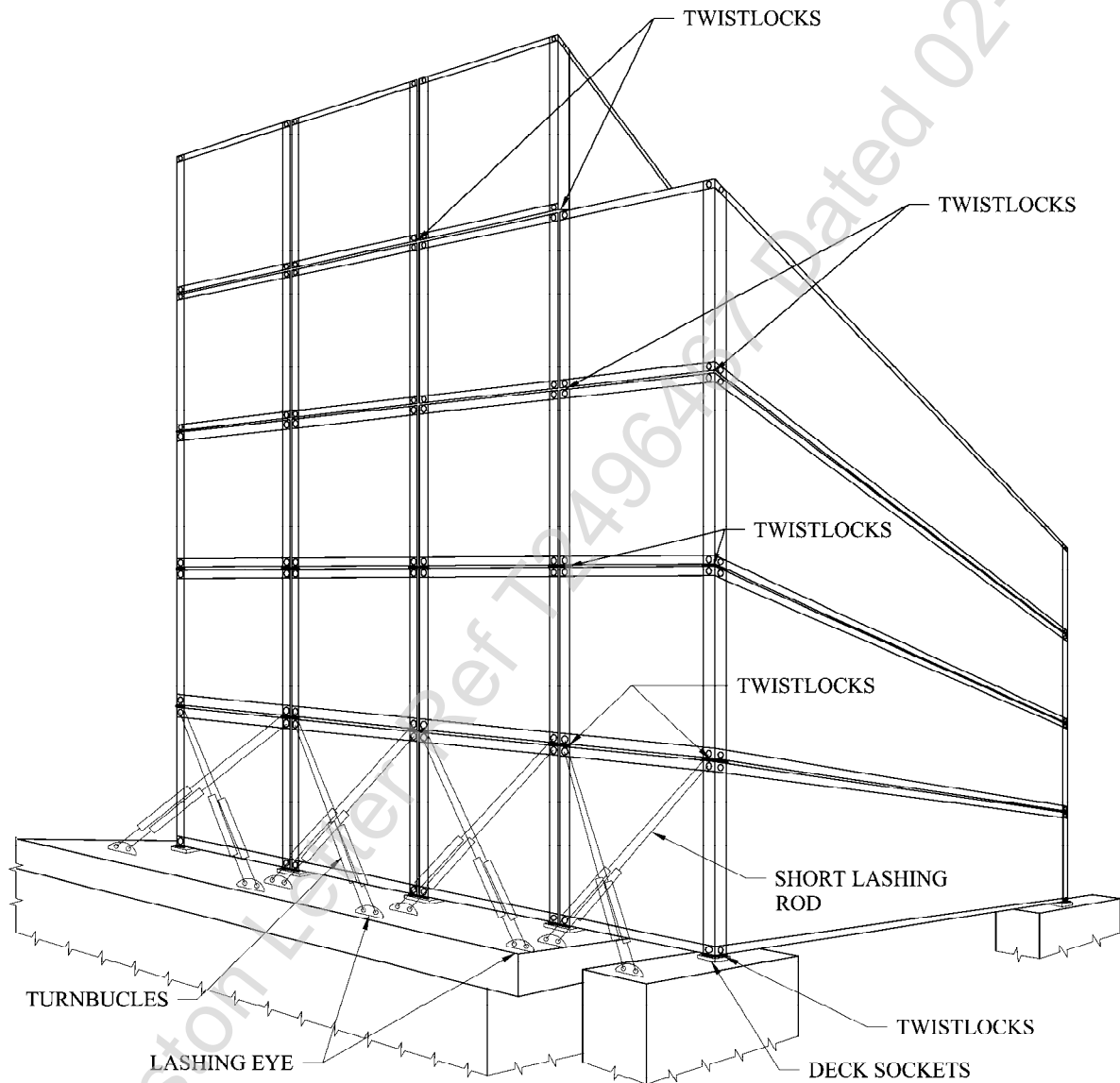
ROD LASHING SYSTEM
5 HIGH
TWIST LOCKS AND LOWER RODS
PARALLEL LASHING

Stowage 5D (45 FT. STOWAGE on 40 FT.)



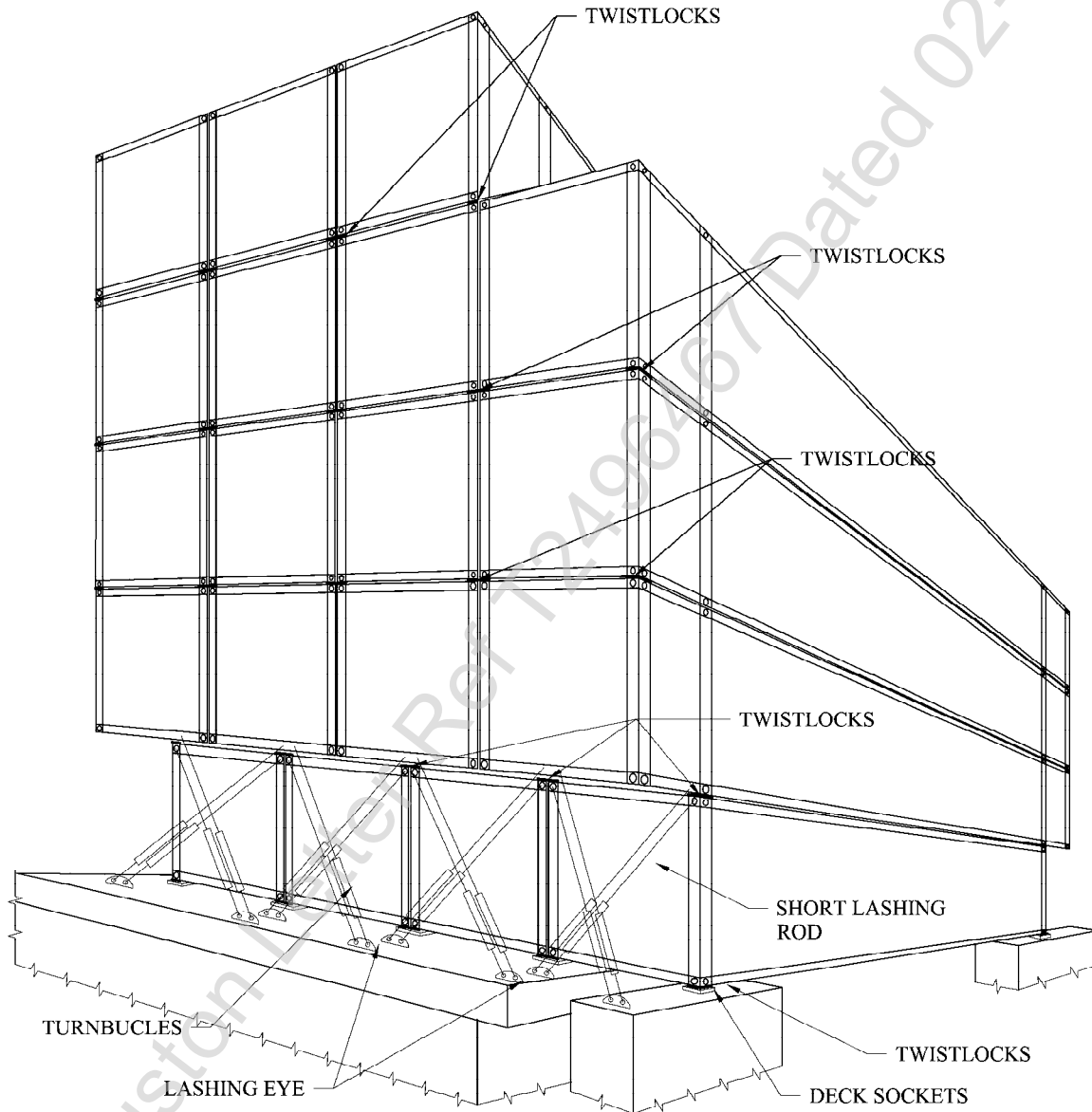
ROD LASHING SYSTEM
4 – 45' OVER 1 – 40'
TWIST LOCKS AND PARALLEL
LASHING

Stowage 5E (40 FT. STD/HC STOWAGE & 45 FT.)



ROD LASHING SYSTEM
5 HIGH
TWIST LOCKS AND LOWER RODS
PARALLEL LASHING

Stowage 5F (45 FT. STOWAGE on 40 FT.)



ROD LASHING SYSTEM
4 – 45' OVER 1 – 40'
TWIST LOCKS AND PARALLEL
LASHING

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3.5 ALLOWABLE STACK WEIGHTS – 20’ Below Deck in 40’ Cell Guides

For the C9 Class Vessels 20’ containers are NOT stowed in 40’ cell guides below deck.

Stack weight limitations for 20’ containers below deck are given in Section 3.2.

3.6 EVALUATION OF FORCES ACTING ON CONTAINERS

3.6.1 General

Basic Loads

The basic loads to be taken into account in container securing calculations are as follows: static gravitational forces, dynamic forces associated with ship motions, wind forces, lashing or other securing forces.

Pretensioning

Pretensioning of lashings should be kept to a minimum. Where pretensioning is an integral part of a securing system, it will be, specially considered.

Wind Forces

Wind forces are to be applied to exposed stacks of containers.

8.5-foot height:	Side of 20' Container	1.0 M. Ton (0.98 L. Ton)
	Side of 40' Container	2.0 M. Ton (1.96 L. Ton)

For containers of other dimensions, the wind forces may be based on a wind pressure of 0.0633 M. Ton/m² (0.0058 L. Ton/ft²).

When linkage plates or double stacking cones are used, wind loads may be equally divided between the connected container stacks.

Container Blocks In Holds

A container block is formed when individual stacks of containers are interconnected by double stacking cones or linkage plates. As a minimum, the double stacking cones are to be provided throughout the container block in line with the hull support points. Where the clearances at the double stacking cones can be neglected, a container block may be analyzed by the methods indicated under '*b Bridge Strut and Shoring System*' and '*c Bridge Strut and Shoring Spring Constants*', as appropriate, in Section 3.2.3.

3.6.2 Ship Motions and Forces on Containers

Design Considerations

All contemplated container loading conditions of the vessel are to be investigated and the container securing system is to be designed to withstand the worst anticipated loading. This loading condition generally is the one which results in the greatest vessel transverse metacentric height. Alternatively, the securing system may be approved for specific container loading conditions. Special consideration may be given to the required container securing gear where it can be shown that seasonal or trade route conditions significantly affect the vessel's dynamic motions and the loads on the containers.

Unless otherwise specified on stack arrangement drawings or stack weight tables, the maximum gross weights for which containers are certified are to be used in the calculations.

Ship Motions

The dynamic forces associated with the worst roll, pitch and heave motions for the particular loading condition of the vessel are to be used for the design of a container securing system. Where detailed studies of long term ship motion response to irregular seas are not submitted, the following values and equations will be used.

a Maximum Roll Amplitude

Metric Units

Inch/Pound Units

$$\theta_R = \frac{3150C}{(B+75)} \quad (\text{degrees; single amplitude})$$

$$\theta_R = \frac{10340C}{(B+246)} \quad (\text{degrees; single amplitude})$$

- $C = 1.0$ for ships without bilge keel when $T_R < 20$ sec
 $C = 0.85$ for ships without bilge keel when $T_R > 30$ sec
 $C = 0.9$ for ships with bilge keel when $T_R < 20$ sec
 $C = 0.75$ for ships with bilge keel when $T_R > 30$ sec
 $B =$ moulded breadth of the vessel, in m or ft

- Notes 1 C may be interpolated between 0.85 and 1.0, or between 0.75 and 0.9, as applicable, when T_R falls between 20 and 30 seconds.
 2 For ships with active roll damping, C will be specially considered.

b Roll Period

Metric Units

Inch/Pound Units

$$T_R = \frac{2k}{\sqrt{(GM)}}$$

$$T_R = \frac{1.108k}{\sqrt{(GM)}}$$

Where k is not submitted, the following expression for roll period will be used for vessels with even transverse mass distribution.

Metric Units

Inch/Pound Units

$$T_R = \frac{0.8B}{\sqrt{(GM)}}$$

$$T_R = \frac{0.442B}{\sqrt{(GM)}}$$

c Maximum Pitch Amplitude

- $\varphi = 7^\circ$ $L \leq 120\text{m}$ ($L \leq 394$ ft)
 $\varphi = 6^\circ$ $120\text{m} < L < 275\text{m}$ (394 ft $< L < 902$ ft)
 $\varphi = 5^\circ$ $L \geq 275\text{m}$ ($L \geq 902$ ft)
 (single amplitude)

d Pitch Period

Metric Units

$$T_p = 7 - (183 - L)(0.0123)$$

$$T_p = 7 + (L - 183)(0.0123)$$

$$183\text{m} > L \geq 60\text{m}$$

$$L \geq 183\text{m}$$

Inch/Pound Units

$$T_p = 7 - (600.4 - L)(0.00375)$$

$$T_p = 7 + (L - 600.4)(0.00375)$$

$$600$$
 ft $> L \geq 197$ ft
 $L \geq 600$ ft

e Heave Period Metric Units Inch/Pound Units

$$T_H = 0.5 \sqrt{L}$$

$$T_H = 0.277 \sqrt{L}$$

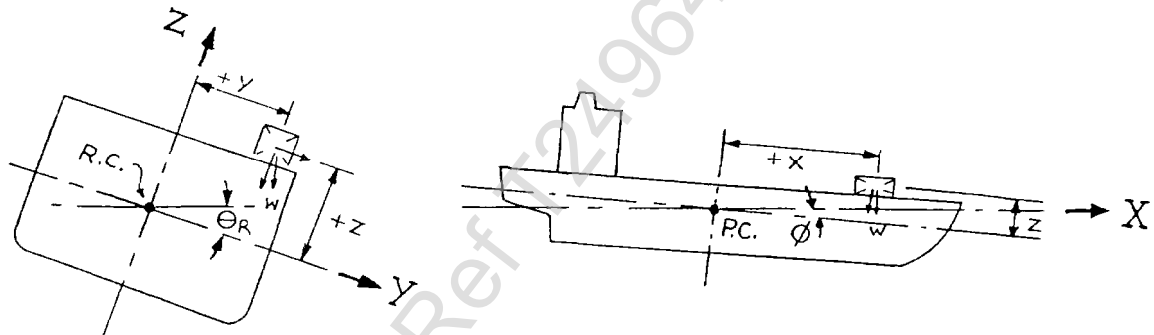
- k = calculated transverse radius of gyration for the vessel, in m or ft
 GM = maximum transverse metacentric height, in m or ft
 L = length between perpendiculars of the vessel, in m or ft

Loading Cases

Refer to Figure 3.1 for a diagram of lever arms x , y and z . These lever arms are the distances from the vessel's center of motion to the center of gravity of the container in question.

FIGURE 3.1

Lever Arms for Acceleration Terms



The roll center (R.C.) of the vessel is to be taken as the vertical center of gravity of the vessel for the particular loading condition. Where the calculated vertical center of gravity of the vessel is not submitted, the distance of the roll center above baseline will be estimated by the following equation.

$$\text{Roll Centre} = \text{R.C.} = 0.5 [(D/2) + d]$$

- D = depth of the vessel, in m or ft
 d = draft, in m or ft

The pitch center (P.C.) of the vessel is to be taken as the center of flotation. Where the calculated center of flotation is not submitted, the pitch center will be estimated as $0.45L$ forward of the aft perpendicular. The lever z is always measured from a plane passing through the roll center. The following loading combinations, Condition A and Condition B, are to be applied separately to the containers.

Metric Units

CONDITION A - Roll and Heave

$$\text{Vertical Load} \quad W [A_V] = W [\cos \theta_R \pm \frac{0.00226 \pi^3 \theta_R}{T_R^2} y + 0.0051 \frac{\pi^2 L}{T_H^2} \cos \theta_R]$$

$$\text{Transverse Load} \quad W [A_T] = W [\sin \theta_R + \frac{0.00226 \pi^3 \theta_R}{T_R^2} z + 0.0051 \frac{\pi^2 L}{T_H^2} \sin \theta_R]$$

Wind forces given in Section 3.2.1 are to be included where applicable.

CONDITION B - Pitch and Heave

$$\text{Vertical Load} \quad W [A_V] = W [\cos \phi \pm \frac{0.00226 \pi^3 \phi}{T_P^2} x + 0.0051 \frac{\pi^2 L}{T_H^2} \cos \phi]$$

$$\text{Longitudinal Load} \quad W [A_L] = W [\sin \phi + \frac{0.00226 \pi^3 \phi}{T_P^2} z + 0.0051 \frac{\pi^2 L}{T_H^2} \sin \phi]$$

Inch/Pound Units

CONDITION A - Roll and Heave

$$\text{Vertical Load} \quad W [A_V] = W [\cos \theta_R \pm \frac{0.00069 \pi^3 \theta_R}{T_R^2} y + 0.00155 \frac{\pi^2 L}{T_H^2} \cos \theta_R]$$

$$\text{Transverse Load} \quad W [A_T] = W [\sin \theta_R + \frac{0.00069 \pi^3 \theta_R}{T_R^2} z + 0.00155 \frac{\pi^2 L}{T_H^2} \sin \theta_R]$$

Wind forces given Section 3.2.1 are to be included where applicable.

CONDITION B - Pitch and Heave

$$\text{Vertical Load} \quad W [A_V] = W [\cos \phi \pm \frac{0.00069 \pi^3 \phi}{T_P^2} x + 0.00155 \frac{\pi^2 L}{T_H^2} \cos \phi]$$

$$\text{Longitudinal Load} \quad W [A_L] = W [\sin \phi + \frac{0.00069 \pi^3 \phi}{T_P^2} z + 0.00155 \frac{\pi^2 L}{T_H^2} \sin \phi]$$

W	=	gross weight of the container and its cargo, in M. Tons or L- Tons
A _V	=	vertical static gravitational component and acceleration from ship motion, non-dimensional (in "g's")
A _T	=	transverse static gravitational component and acceleration from ship motion, non-dimensional (in "g's")
A _L	=	longitudinal static gravitation component and acceleration from ship motion, non-dimensional (in "g's")
θ _R	=	maximum roll single amplitude, in degrees
φ	=	maximum pitch single amplitude, in degrees
T _R	=	roll period of the vessel (complete cycle), in seconds
T _H	=	heave period of the vessel (complete cycle), in seconds
T _P	=	pitch period of the vessel (complete cycle), in seconds
L	=	length between perpendiculars of the vessel, in m or ft
x	=	longitudinal distance from the vessel's pitch center to the center of gravity of the container, in m or ft
y	=	transverse distance from the vessel's centerline to the center of gravity of the container, in m or ft
z	=	vertical distance from the vessel's roll center to the center of gravity of the container, in m or ft

Considering the expression for A_V in Condition A, the middle term within the brackets represents the roll acceleration and has a ± sign. In one case, the roll acceleration is to be taken as positive and the container stack analyzed for A_V maximum and A_T. In a second case, the roll acceleration is to be taken as negative and the container stack analyzed for A_V minimum and A_T.

Considering the expression for A_V in Condition B, the middle term within the brackets represents the pitch acceleration and has a ± sign. In a similar manner as described for Condition A, the container stack is to be analyzed for A_L and A_V maximum and minimum.

For Conditions A and B, A_V minimum is not to be taken greater than 1.0g.

The ship motion equations in Section 3.2.2 and the loading cases above are to be used for the design of container securing systems only. They represent a simplified approach to ship motion forces.

C9 CLASS

GM = 2.0 feet	Transverse Accelerations (Roll and Heave) A	Longitudinal Accelerations (Pitch and Heave) (Pitch and Heave) B
HATCH	All Hatches	All Hatches
Tier 5	0.492	0.276
Tier 4	0.487	0.258
Tier 3	0.483	0.240
Tier 2	0.479	0.222
Tier 1	0.475	0.204
Below deck		
Top of Hold	0.461	0.141
Bottom Of Hold	0.470	0.068

Vertical Accelerations, Roll and Heave (Cond. A)

1.133

Vertical Accelerations, Pitch and Heave (Cond. B)

Hatch Number	1	3	5	7	10	12
	1.788	1.558	1.366	1.226	1.514	1.712

GM = 4.0 feet	Transverse Accelerations (Roll and Heave) A	Longitudinal Accelerations (Pitch and Heave) (Pitch and Heave) B
HATCH	All Hatches	All Hatches
Tier 5	0.604	0.276
Tier 4	0.594	0.258
Tier 3	0.585	0.240
Tier 2	0.575	0.222
Tier 1	0.565	0.204
Below deck		
Top of Hold	0.532	0.141
Bottom Of Hold	0.555	0.068

Vertical Accelerations, Roll and Heave (Cond. A)

1.081

Vertical Accelerations, Pitch and Heave (Cond. B)

Hatch Number	1	3	5	7	10	12
	1.788	1.558	1.366	1.226	1.514	1.712

Accelerations calculated at Load line Draft

C9 CLASS

GM = 6.0 feet	Transverse Accelerations (Roll and Heave) A	Longitudinal Accelerations (Pitch and Heave) (Pitch and Heave) B
HATCH	All Hatches	All Hatches
Tier 5	0.666	0.276
Tier 4	0.651	0.258
Tier 3	0.636	0.240
Tier 2	0.621	0.222
Tier 1	0.606	0.204
Below deck		
Top of Hold	0.555	0.141
Bottom Of Hold	0.590	0.068

Vertical Accelerations, Roll and Heave (Cond. A)

1.072

Vertical Accelerations, Pitch and Heave (Cond. B)

Hatch Number	1	3	5	7	10	12
	1.788	1.558	1.366	1.226	1.514	1.712

GM = 10.0 feet	Transverse Accelerations (Roll and Heave) A	Longitudinal Accelerations (Pitch and Heave) (Pitch and Heave) B
HATCH	All Hatches	All Hatches
Tier 5	0.749	0.276
Tier 4	0.724	0.258
Tier 3	0.699	0.240
Tier 2	0.674	0.222
Tier 1	0.649	0.204
Below deck		
Top of Hold	0.563	0.141
Bottom Of Hold	0.622	0.068

Vertical Accelerations, Roll and Heave (Cond. A)

1.072

Vertical Accelerations, Pitch and Heave (Cond. B)

Hatch Number	1	3	5	7	10	12
	1.788	1.558	1.366	1.226	1.514	1.712

Accelerations calculated at Load line Draft

3.6.3 Design Calculations

Internal Container Cargo Load Distribution

For design purposes the transverse and longitudinal forces from ship motion are to be divided equally between container end or side walls. Unless otherwise specified, the center of gravity of the container is to be taken at 50% of the container height above the base and the "self-racking" force of the container is to be as indicated in Figure 3.2. When the location of the center of gravity of the container is different, the actual container center of gravity is to be submitted.

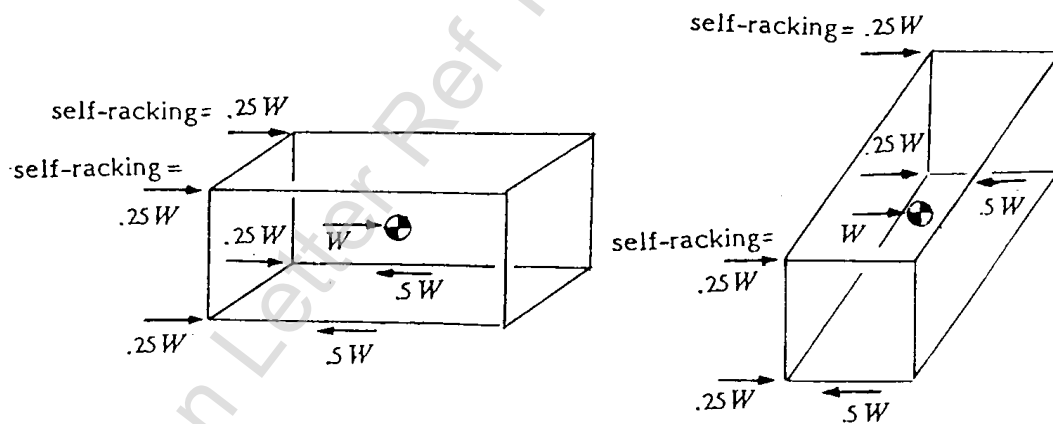
When a container with an extra high center of gravity or a center of gravity significantly shifted towards one end is to be stowed, an engineering analysis of the stack is to be performed which takes the unusual weight distribution into account.

Securing Systems Employing Only Lock Fittings

As there is only one load path, the container stack, a conventional free body diagram of the stack may be employed to obtain the support reactions. They are above the lashing point and can be regarded as an unlashed stack only lock fittings.

FIGURE 3.2

Cargo Load Distribution



$W = W [A']$
 W = Container weight
 A' = Transverse or Longitudinal
 Ship motion coefficient, A_T or A_L

Rigid Securing Systems

- a Cell Guides** The vertical load per container corner at deck, inner bottom, or hatch cover be obtained from the following equation.

$$P = [n W A_V]/4$$

The maximum corner post load may be obtained from the following equation.

$$P_C = [(n-1) W A_V]/4$$

For containers with center of gravity at 50% of the container's height, the horizontal reactions the container corners bearing on the guide are as follows.

$$R = [W A_T]/4$$

$$R = [W A_L]/4$$

See page 36 for expressions of A_V , A_T and A_L . Note that values of A_T and A_L vary for tier of containers due to location.

- b Bridge Strut and Shoring Systems** For rigid securing systems, the vertical load per contain corner at deck or inner bottom may be obtained from the following equation.

$$P = [n W A_V]/4$$

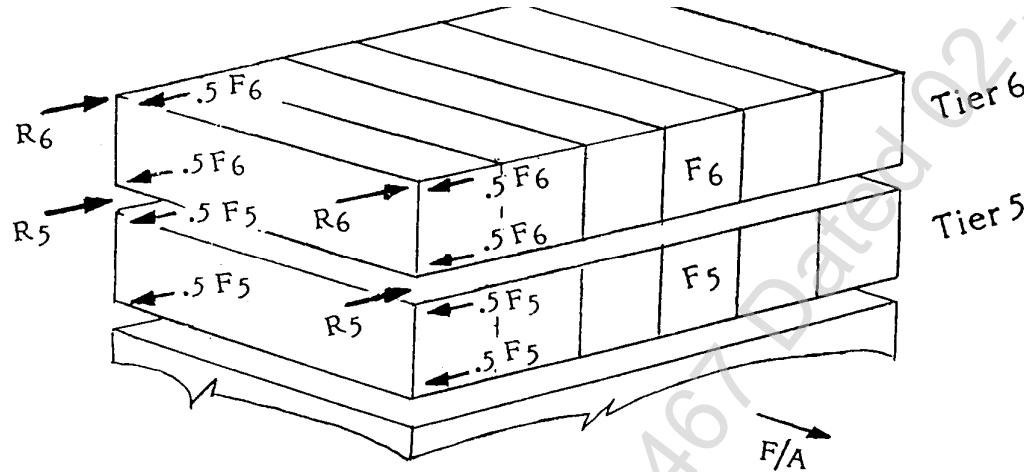
The maximum corner post load may be obtained from the following equation.

$$P_C = [(n-1) W A_V]/4$$

- P = vertical compression bottom corner fitting, in M. Ton or L. Ton
 P_C = corner post compression, in M. Ton or L. Ton
 R = horizontal reaction at cell guide, in M. Ton or L. Ton
 n = total number of containers in a stack
 W = gross weight of the container and its cargo, in M. Ton or L. Ton
 A_V = vertical static gravitational component and acceleration from ship motion, non-dimensional (in 'g's')
 A_T = transverse static gravitational component and acceleration from ship motion, non-dimensional (in "g's")
 A_L = longitudinal static gravitational component and acceleration from ship motion, non-dimensional (in "g's")

Where rigid side supports are provided at every tier of containers and the center of gravity of containers are at 50% of the container height, the side support reactions may be determined as indicated in Figure 3.3.

FIGURE 3.3
Side Supports Provided at Each Tier of Containers



F_i = tier load per end = $[m W_i A_{Ti}]/2$
 m = total number of containers in a tier
 W_i = gross weight of container at the i -th tier
 A_{Ti} = transverse static gravitational component and acceleration at the i -th tier

R_i = side support reaction at the i -th tier
 R_6 = $F_6 (.5)$
 R_5 = $F_5 (.5) + F_6 (.5)$

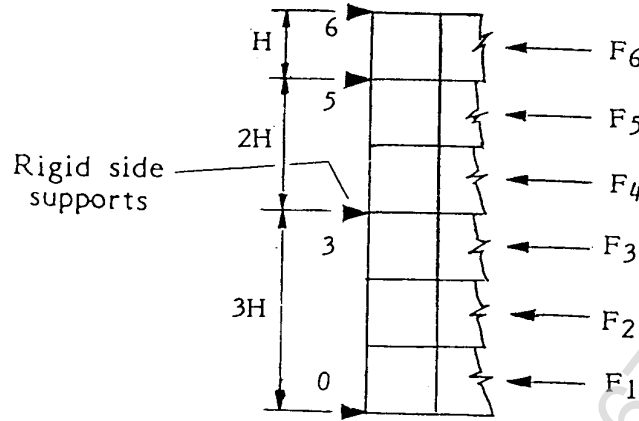
Remaining equations for side support reactions are similar.

Note For container center of gravity locations other than at 50% of the container height, these equations will change.

Where rigid side supports are not provided at every tier of containers and the center of gravity of the containers are at 50% of the container height, the side support reactions can generally be determined as indicated in the example shown in Figure 3.2.

FIGURE 3.2

Example for Six High Container Block with Four Rigid Side Supports



- F_i = tier load per end = $[m W_i A_{Ti}]/2$
 m = total number of containers in a tier
 w_i = gross weight of container at the i -th tier
 A_{Ti} = transverse static gravitational component and acceleration at the i -th tier
 R_i = side support reaction at the i -th tier
 R_0 = $F_1 (2.5H/3H) + F_2 (1.5H/3H) + F_3 (0.5H/3H)$
 R_3 = $F_1 (0.5H/3H) + F_2 (1.5H/3H) + F_3 (2.5H/3H) + F_4 (1.5H/2H) + F_5 (0.5H/2H)$
 R_5 = $F_4 (0.5H/2H) + F_5 (1.5H/2H) + F_6 (0.5H/H)$
 R_6 = $F_6 (0.5H/H)$
 H = height of the container

Note For other side support arrangements and for container center of gravity locations other than at 50% of the container height, these equations will change.

See page 4 of this section for expressions of A_v and A_T . Note that values of A_T vary for each tier of containers due to location.

Where bridge struts are fitted on the port and starboard sides of a container block at the same tier and the struts are designed to transmit both tension and compression loads and the clearances at the double stackers are negligible, the loads in the bridge struts may be obtained by dividing the side support reactions by a factor of two. Where there are significant clearances at each double stacker, the load distribution will be specially considered.

Double stacking cones are to be fitted throughout the tier of a container block in line with the hull support points.

Reactions for supports in the longitudinal direction depend upon support arrangements and will be specially considered.

Flexible Securing Systems

For flexible securing systems the relative stiffness of the securing elements and the container end walls must be considered in the determination of the loads acting on the container end wall structure and on the securing elements. Considering a lashed stack of containers, one method of finding the lashing tension is to equate the displacement of the container corner at the lashing point to the elongation of the lashing.

In general, separate calculations for both the container door end and closed end are required due to the difference in racking stiffness between the two. The container closed end generally produces the maximum loads on the container's structure, twistlocks, and deck sockets; the door end produces the maximum loads in the lashings. Containers which are stowed above the lashing or securing points and are secured by lock fittings only are to be checked in accordance with " on page 9.

a Container Spring Constants In the absence of container test data or container specifications, the values given in Table 3.1 will be used for 8 or 8.5 foot containers.

TABLE 3.1
Racking Spring Constant, K_C

	<u>M. Ton/mm</u>	<u>L. Ton/In</u>
Container Door End	0.38	9.50
Container Closed End	1.60	40.0
Container Side	0.59	4.75

b Lashing Spring Constants

The elongation or stretch of a lashing, Δ_L , can be expressed by the following equation.

$$\Delta_L = T l / AE_L = T / K_L$$

T = lashing tension

K_L = lashing spring constant = $(AE_L)/l$

A = area of lashing component, in mm^2 or in^2

E_L = equivalent elastic modulus of lashing, in kg/mm^2 or psi

l = length of lashing, in mm or in .

In general, manufacturers should supply the effective elastic moduli, E_L , or the spring constant values, K_L , for their lashing products. These values should be based on tests of the entire lashing assembly - rod or wire rope, tensioner, and lashing point - working as a unit.

In the absence of submitted lashing test data, the values given in Tables 3.2 and 3.3 will be used in the above expression for K_L .

TABLE 3.2
Equivalent Elastic Moduli, E_L

Lashing Element	E_L	
	Kg/mm2	psi
Steel Wire Rope	9000	12,797,400
Steel Rod - From Deck to Top of 1st Tier or Bottom of 2nd Tier*	9900	14,077,125
Steel Rod - From Deck to Top of 2nd Tier and Above	18,000	25,594,800
Steel Chain	10,000	14,219,325

*1st tier of containers is always the lowest tier

Note For consistency with E_L in the above table, the length of the lashing, l , is to be measured from the deck or securing point, to the container corner (no deduction for tensioning device).

Lashing made of material other than steel will be specially considered. Each wire rope lashing element is to be pre-stretched to remove its construction stretch by loading to 50% of its rated breaking strength before being placed in service.

TABLE 3.3
Area of Lashing Component, A

<u>Lashing Element</u>	<u>A</u>
Steel Wire Rope	nominal area
Steel Rod	actual area
Steel Chain	one side of

c Bridge Strut and Shoring Spring Constants For flexible securing systems, bridge strut and shoring spring constants depend on the nature of the element and will be specially considered. The displacement of the bridge strut or shoring, Δ_s , is expressed by the following equation.

$$\Delta_s = T'/K_s$$

K_s = bridge strut or shoring spring constant
 T' = bridge strut or shoring force

The effect of clearance between the container stack and the shoring should be taken into account when determining container racking loads.

Final Checks

For all three types of securing systems discussed in foregoing, both the loads in the container structure and the loads in the securing elements are to be checked against the safe working loads in following Section. 3.6.4.

3.6.4 Safe Working Loads, Permissible Stresses and Safety Factors

General

The calculated load in the container securing device or container structure is not to exceed the safe working load. The calculated load in the buttress, cell guide, shoring or attachment is not to result in stress exceeding the permissible stress.

In order to account for such unpredictable factors as deterioration of securing devices, deterioration of containers, manufacturing imperfections, extreme ship motions and variations in container and lashing spring constants, the following safety factors are to be used in the design of container securing systems.

Lashing and Securing Fittings

The safe working loads are obtained by dividing the design breaking load of the element by the specified safety factor.

TABLE 3.2
Factors of Safety

<i>Item</i>		<i>Safety Factor</i>
Steel Wire Rope		2.0
Steel Rod	MS	2.0
	HTS	1.67
Steel Chain		2.0
Other Steel Fittings and Securing Devices	MS	2.0
	HTS	1.67
Nodular Iron Fittings		2.0

MS = ordinary strength steel

HTS = higher strength Steel with $Y \geq 32 \text{ kg/mm}^2$ (45,500 psi)

Buttresses, Shoring and Cell Guides

Permissible stresses based on the minimum yield strength of the material, Y , are given in Table 3.5.

TABLE 3.5
Permissible Stresses

Buttresses Cell Guides $f_p = 0.8 Y$

Shoring Attachments $q = 0.53 Y$

f_p = normal stress, in kg/mm^2 or psi

q = shear stress, in kg/mm^2 or psi

Y = minimum yield strength of the material, in kg/mm^2 or psi

For higher strength steels, Y is not to be taken as greater than 72% of the specified minimum tensile strength.

Loads on Container Structure

The combined static, dynamic and securing loads imposed on the container structure are not to exceed those given in Table 3.6.

TABLE 3.6
Loads on Containers

Metric Units

Item	20 Ft. Container W= 20.32 M. Ton (W = 24.39 M. Ton - Bracketed Values)	40 Ft. Container W= 30.48 M. Ton
End Wall Racking	15.2 M. Ton	15.2 M. Ton
Side Wall Racking	12.7 M. Ton	10.2 M. Ton
Corner Post Compression	86.4 M. Ton	86.4 M. Ton
Vertical Compression on Bottom Corner Fitting	97.2 M Ton	97.2 M Ton
Transverse Securing ⁽¹⁾ Force on Corner Fitting:		
Top Corner	Tension – 25.4M. Ton Compression – 25.4 M. Ton	Tension - 25.4 M. Ton Compression - 25.4 M. T
Bottom Corner	Tension – 35.6 M. Ton Compression – 35.6 M- Ton	Tension - 35.6 M. Ton Compression - 35.6 M. Ton
Longitudinal Securing ⁽¹⁾ Force on Corner Fitting:		
Top Corner	Tension - 10.3 M. Ton Compression - 5.1 M. Ton	Tension - 10.2 M. Ton Compression - 5.1 M. Ton
Bottom Corner	Tension - 20.3 M. Ton Compression - 20.3 M- Ton	Tension - 30.5 M. Ton Compression - 30.5 M. Ton
Lashing Force ⁽²⁾ on Corner Fitting	Vertical - 30.5 M. Ton Horizontal - 15.2 M. Ton	Vertical - 30.5 M. Ton Horizontal - 15.2 M. Ton
Vertical Tension on Top Corner Fitting	20.3 M. Ton	20.3 M. Ton
Vertical Tension on Bottom Corner Fitting	25.4 M. Ton	25.4 M. Ton

(1) See Figure 3.5

(2) See Figure 3.6

For containers which cannot support the above loads due to the container construction standards, the loads are to be properly reduced.

TABLE 3.6
Loads on Containers

Inch/Pound Units

Item	20 Ft. Container W= 20 L. Ton (W = 24 L. Ton - Bracketed Values)	40 Ft. Container W= 30 L. Ton
End Wall Racking	15 L. Ton	15 L. Ton
Side Wall Racking	12.5 L. Ton	12.5 L. Ton
Corner Post Compression	85 L. Ton)	85 L. Ton
Vertical Compression on Bottom Corner Fitting	95.7 L. Ton	95.7 L. Ton
Transverse Securing ⁽¹⁾ Force on Corner Fitting:		
Top Corner	Tension - 25 L. Ton Compression - 25 L. Ton	Tension - 25 L. Ton Compression - 25 L. Ton
Bottom Corner	Tension - 35 L. Ton Compression - 35 L. Ton	Tension - 35 L. Ton Compression - 35 L. Ton
Longitudinal Securing ⁽¹⁾ Force on Corner Fitting:		
Top Corner	Tension - 10 L. Ton Compression - 5 L. Ton	Tension - 10 L. Ton Compression - 5 L. Ton
Bottom Corner	Tension - 20 L. Ton Compression - 20 M- Ton	Tension - 30 L. Ton Compression - 30 L. Ton
Lashing Force ⁽²⁾ on Corner Fitting	Vertical - 30 L. Ton Horizontal - 15 L. Ton	Vertical - 30 L. Ton Horizontal - 15 L. Ton
Vertical Tension on Top Corner Fitting	20 L. Ton	20 L. Ton
Vertical Tension on Bottom Corner Fitting	25 L. Ton	25 L. Ton

(1) See Figure 3.5

(2) See Figure 3.6

For containers which cannot support the above loads due to the container construction standards, the loads are to be properly reduced.

FIGURE 3.5
Securing Force on Corner Fitting

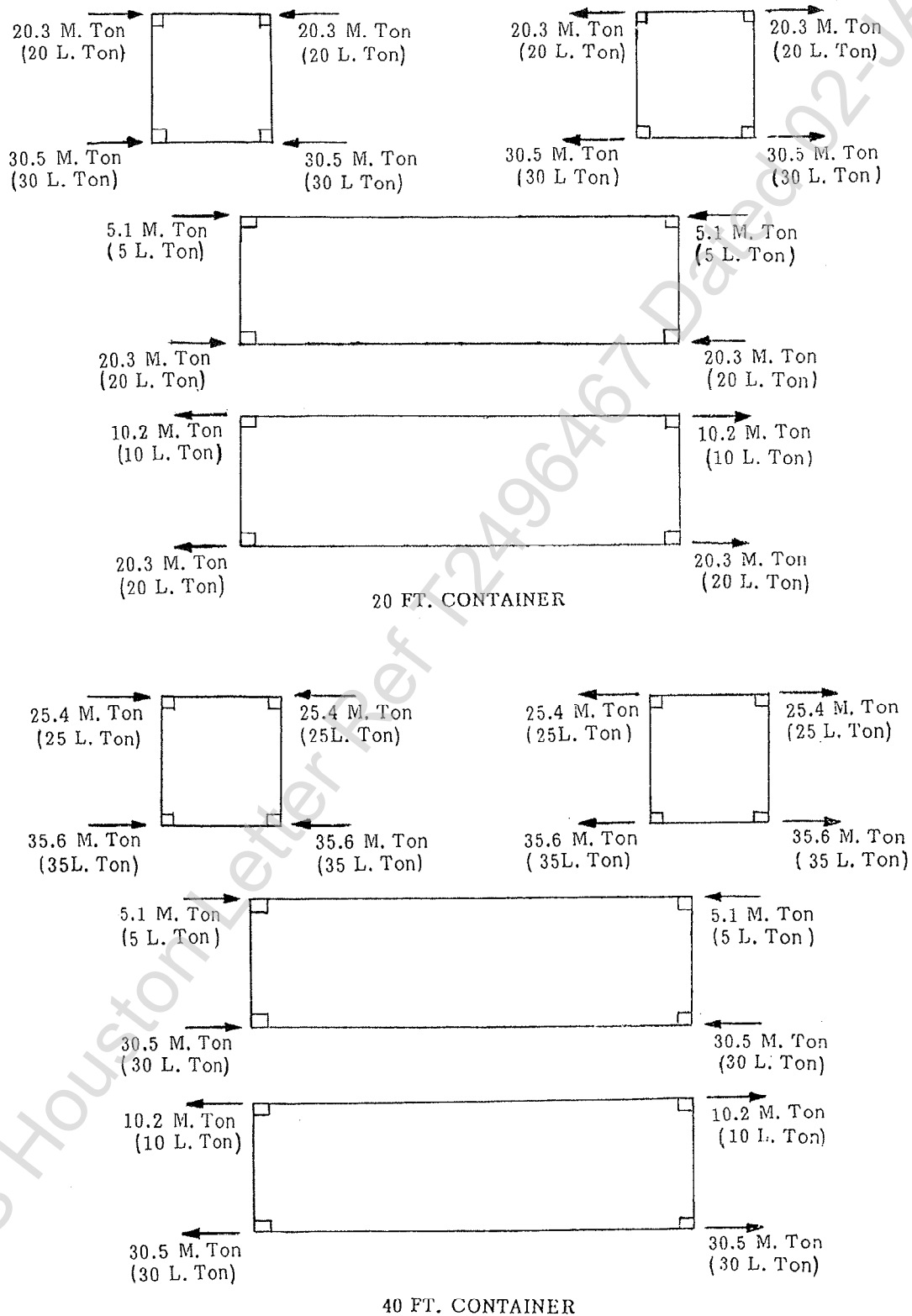
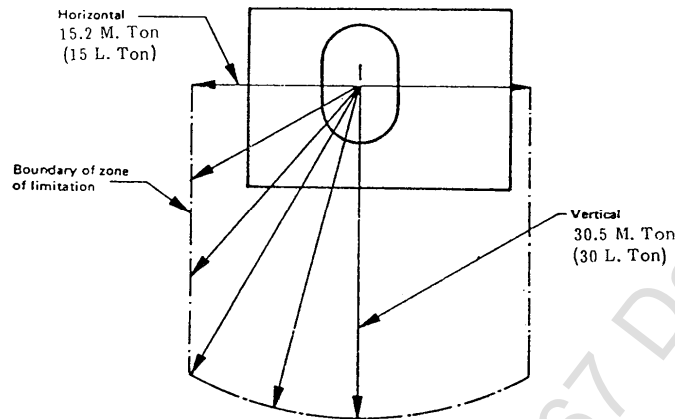


Figure 3.6

Lashing Force on Corner Fitting



In general the following types of imposed loadings on the container structure must be checked for the securing systems indicated.

- a** Lashing and Lock Fitting Systems
 - End wall racking
 - Side wall racking
 - Lashing force on corner fitting
 - Vertical tension on top and bottom corner fittings
- b** Hinged Frame and Cell Guide Systems
 - Transverse securing force on corner fitting
 - Longitudinal securing force on corner fitting
- c** Shoring and Bridge Strut Systems
 - End wall racking
 - Side wall racking
 - Transverse securing force on corner fitting
 - Longitudinal securing force on corner fitting
 - Vertical tension on top and bottom corner fittings
- d** All Securing Systems
 - Corner post compression
 - Vertical compression of bottom corner fitting

4.1 General principles of cargo securing

All cargoes should be stowed and secured in such a way that the ship and persons on board are not put at risk.

The safe stowage and securing of cargoes depend on proper planning, execution and supervision.

Personnel commissioned to tasks of cargo stowage and securing should be properly qualified and experienced.

Personnel planning and supervising the stowage and securing of cargo should have a sound practical knowledge of the application and content of this Cargo Securing Manual.

In all cases, improper stowage and securing of cargo will be potentially hazardous to the securing of other cargoes and to the ship itself.

Decisions taken for measures of stowage and securing cargo should be based on the most severe weather conditions which may be expected by experience for the intended voyage.

Ship-handling decisions taken by the master, especially in bad weather conditions, should take into account the type and stowage position of the cargo and the securing arrangements.

Behavior of cargoes

Some cargoes have a tendency to deform or to compact themselves during the voyage, which will result in a slackening of their securing gear.

Cargoes with low friction coefficients, when stowed without proper friction-increasing devices such as dunnage, soft boards, rubber mats, etc., are difficult to secure unless tightly stowed across the ship.

Equipment

The ship's cargo securing equipment should be:

- 1 available in sufficient quantity;
- 2 suitable for its intended purpose, taking into account the recommendations of this Cargo Securing Manual;
- 3 of adequate strength;
- 4 easy to use; and
- 5 well maintained
- 6 there should be a sufficient quantity of reserve cargo securing gear on board the ship.

Special cargo transport units

The shipowner and the ship operator should, when necessary, make use of relevant expertise when considering the shipment of a cargo with unusual characteristics which may require special attention to be given to its location on board vis-a-vis the structural strength of the ship, its stowage and securing, and the weather conditions which may be expected during the intended voyage.

Cargo information

Prior to shipment the shipper should provide all necessary information about the cargo to enable the shipowner or ship operator to ensure that:

- 1 the different commodities to be carried are compatible with each other or suitably separated;
- 2 the cargo is suitable for the ship;
- 3 the ship is suitable for the cargo; and
- 4 the cargo can be safely stowed and secured on board the ship and transported under all expected conditions during the intended voyage.

The master should be provided with adequate information regarding the cargo to be carried so that its stowage may be properly planned for handling and transport.

Suitability of cargo for transport

Cargo carried in containers, road vehicles, shipborne barges, railway wagons and other transport units should be packed and secured within these units so as to prevent, throughout the voyage, damage or hazard to the ship, to the persons on board and to the marine environment.

Cargo distribution

It is of utmost importance that the master takes great care in planning and supervising the stowage and securing of cargoes in order to prevent cargo sliding, tipping, racking, collapsing, etc.

The cargo should be distributed so as to ensure that the stability of the ship throughout the entire voyage remains within acceptable limits so that the hazards of excessive accelerations are reduced as far as practicable.

Cargo distribution should be such that the structural strength of the ship is not adversely affected.

Cargo securing arrangements

Particular care should be taken to distribute forces as evenly as practicable between the cargo securing devices. If this is not feasible, the arrangements should be upgraded accordingly.

If, due to the complex structure of a securing arrangement or other circumstances, the person in charge is unable to assess the suitability of the arrangement from experience and knowledge of good seamanship, the arrangement should be verified by using an acceptable calculation method.

The cargo securing gear should be adapted to the quantity and properties of the cargo to be carried and, when required, additional gear should be provided.

Lashings should be kept as short as possible. Long lashings are difficult to tighten and difficult to keep taut.

Residual strength after wear and tear

Cargo securing arrangements and equipment should have sufficient residual strength to allow for normal wear and tear during their lifetime.

Friction forces

Where friction between the cargo and the ship's deck or structure or between cargo transport units is insufficient to avoid the risk of sliding, suitable material such as soft boards or dunnage should be used to increase friction. (See paragraph 7.2.1 of the CSS Code).

Shipboard supervision

The principal means of preventing the improper stowage and securing of cargoes is through proper supervision of the loading operation and inspections of the stow.

As far as practicable, cargo spaces should be regularly inspected throughout the voyage to ensure that the cargo, vehicles and cargo transport units remain safely secured.

Entering enclosed spaces

The atmosphere in any enclosed space may be incapable of supporting human life through lack of oxygen or it may contain flammable or toxic gases. The master should ensure that it is safe to enter any enclosed space.

General elements to be considered by the master

Having evaluated the risk of cargo-shifting, taking into account the criteria for estimating the risk of cargo shifting the master should ensure, prior to loading of any cargo, cargo transport unit or vehicle that:

- 1 the deck area for their stowage is, as far as practicable, clean, dry and free from oil and grease;
- 2 the cargo, cargo transport unit or vehicle appears to be in suitable condition for transport, and can be effectively secured;
- 3 all necessary cargo securing equipment is on board and in good working condition; and
- 4 cargo in or on cargo transport units and vehicles is, to the extent practicable, properly stowed and secured on to the unit or vehicle.

Cargo stowage and securing declaration

Where there is reason to suspect that a container or vehicle into which dangerous goods have been packed or loaded is not in compliance with the provisions of sections 12 or 17, of regulation VII/5.2 or 5.3 of SOLAS 1974, as amended, or with the provisions as appropriate, of the General Introduction to the IMDG Code, or where a container packing certificate/vehicle packing declaration is not available, the unit should not be acceptable for shipment.

Where practicable and feasible, road vehicles should be provided with a cargo stowage and securing declaration, stating that the cargo on the road vehicle has been properly stowed and secured for the intended sea voyage, taking into account the IMO/ILO Guidelines for Packing Cargo in Freight Containers or Vehicles. The vehicle packing declaration, recommended by the IMDG Code may be acceptable for this purpose.

Causes of cargo loss

Some of the most common causes of cargo loss, which should be given careful consideration when securing cargoes, are as follows:

- 1 Severe adverse weather.
- 2 Insufficient or ineffective use of dunnage.
- 3 Lashings inadequate in number or strength.
- 4 Port and starboard or forward and aft lashings ill-balanced.
- 5 Wire attachment eyes or loops badly formed.
- 6 Incorrect use of bulldog grips.
- 7 Lack of strength continuity as between wire, attachment eyes, chain, turnbuckles, lashing webbing, shackles and fixed point terminations.
- 8 Lashings secured around sharp or unprotected edges.
- 9 Failure to appreciate the forces generated on a sea-going vessel.
- 10 Failure to provide sufficient personnel or time to effectively complete the work before sailing.

4.2 Method to assess securing arrangements for non-standardized cargo

MSC/Circ.1026

ANNEX

AMENDMENTS TO THE CODE OF SAFE PRACTICE FOR CARGO STOWAGE AND SECURING (CSS CODE)

ANNEX 13

Methods to assess the efficiency of securing arrangements for non-standardized cargo

1 SCOPE OF APPLICATION

1 In paragraph 1, after the second sentence a new sentence is added as follows:

“All lashing assemblies used in the application of the methods described in this annex must be attached to fixed securing points or strong supporting structures marked on the cargo unit or advised as being suitable, or taken as a loop around the unit with both ends secured to the same side as shown in Annex 5, Figure 2 of the Code. Lashings going over the top of the cargo unit, which have no defined securing direction but only act to increase friction by their pre-tension, cannot be credited in the evaluation of securing arrangements under this annex.”

4 STRENGTH OF SECURING EQUIPMENT

2 In paragraph 4.2, the second sentence in the first sub-paragraph is replaced by the following text:

“Safe Working Load (SWL) may be substituted for MSL for securing purposes, provided this is equal to or exceeds the strength defined by MSL.”

3 In Table 1 (as amended by MSC/Circ. 812), “70% of breaking strength” on the line regarding web lashing is replaced by “50% of breaking strength”.

5 SAFETY FACTOR

4 Existing paragraph 5 is replaced by the following text and re-numbered as paragraph 6:

“When using balance calculation methods for assessing the strength of the securing devices, a safety factor is used to take account of the possibility of uneven distribution of forces among the devices or reduced capability due to the improper assembly of the devices or other reasons. This safety factor is used in the formula to derive the calculated strength (CS) from the MSL and shown in the relevant method used.

$$CS = MSL/safety\ factor$$

Notwithstanding the introduction of such a safety factor, care should be taken to use securing elements of similar material and length in order to provide a uniform elastic behaviour within the arrangement.”

MSC/Circ.1026
 ANNEX
 Page 2

6 RULE-OF-THUMB METHOD

5 Existing paragraph 6 is re-numbered as paragraph 5. Existing sub-paragraphs 6.1, 6.2 and 6.3 are re-numbered as 5.1, 5.2 and 5.3 accordingly.

7 ADVANCED CALCULATION METHOD

6 After Table 3 the following text and formula are added:

“For length/speed combinations not directly tabulated, the following formula may be used to obtain the correction factor with v = speed in knots and L = length between perpendiculars in metres:

$$\text{correction factor} = (0.345 \cdot v / \sqrt{L}) + (58.62 \cdot L - 1034.5) / L^2$$

This formula shall not be used for ship lengths less than 50 m or more than 300 m.”

7 Under the existing paragraph 7.2, the following text and a new table are added:

“Friction contributes towards prevention of sliding. The following friction coefficients (μ) should be applied.

Table 5 – Friction coefficients

Materials in contact	Friction coefficient, (μ)
timber-timber, wet or dry	0.4
steel-timber or steel-rubber	0.3
steel-steel, dry	0.1
steel-steel, wet	0.0

8 In paragraph 7.2.1, the text from ($\mu = 0.3$ for steel-timber or steel-rubber) to ($\mu = 0.0$ for steel-steel, wet) is deleted; “table 5” in the definition of f is replaced by “table 6”; and a formula is added under the definition of CS as follows:

$$CS = \frac{MSL}{1.5}$$

9 Existing Table 5 is re-numbered as Table 6.

10 Under the re-numbered Table 6, the following text is added:

“As an alternative to using Table 6 to determine the forces in a securing arrangement, the method outlined in paragraph 7.3 can be used to take account of transverse and longitudinal components of lashing forces.”

11 In paragraph 7.2.3, under the definition of CS a formula is added:

$$“CS = \frac{MSL}{1.5}”$$

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- 12 A new paragraph 7.2.4 is added as follows:

“7.2.4 Calculated example

A calculated example for this method is shown in Appendix 1.”

- 13 A new paragraph 7.3 is added as follows:

“7.3 Balance of forces – alternative method

The balance of forces described in paragraph 7.2.1 and 7.2.3 will normally furnish a sufficiently accurate determination of the adequacy of the securing arrangement. However, this alternative method allows a more precise consideration of horizontal securing angles.

Securing devices usually do not have a pure longitudinal or transverse direction in practice but have an angle β in the horizontal plane. This horizontal securing angle β is defined in this annex as the angle of deviation from the transverse direction. The angle β is to be scaled in the quadrantal mode, i.e. between 0 and 90°.

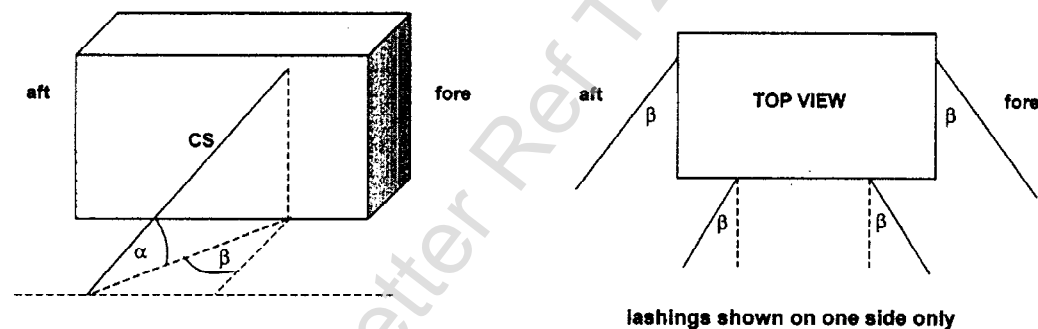


Figure 3 – Definition of the vertical and horizontal securing angles α and β

A securing device with an angle β develops securing effects both in longitudinal and transverse direction, which can be expressed by multiplying the calculated strength CS with the appropriate values of f_x or f_y . The values of f_x and f_y can be obtained from Table 7.

Table 7 consists of five sets of figures, one each for the friction coefficients $\mu = 0.4, 0.3, 0.2, 0.1$ and 0. Each set of figures is obtained by using the vertical angle α and horizontal angle β . The value of f_x is obtained when entering the table with β from the right while f_y is obtained when entering with β from the left, using the nearest tabular value for α and β . Interpolation is not required but may be used.

The balance calculations are made in accordance with the following formulae:

$$\text{Transverse sliding : } F_y \leq \mu \cdot m \cdot g + f_{y1} \cdot CS_1 + \dots + f_{yn} \cdot CS_n$$

$$\text{Longitudinal sliding : } F_x \leq \mu(m \cdot g - F_z) + f_{x1} \cdot CS_1 + \dots + f_{xn} \cdot CS_n$$

$$\text{Transverse tipping : } F_y \cdot a \leq b \cdot m \cdot g + 0.9(CS_1 \cdot c_1 + CS_2 \cdot c_2 + \dots + CS_n \cdot c_n)$$

MSC/Circ.1026
ANNEX
Page 4

Caution:

Securing devices, which have a vertical angle α of less than 45° in combination with horizontal angle β greater than 45°, should not be used in the balance of transverse tipping in the above formula.

All symbols used in these formulae have the same meaning as defined in paragraph 7.2 except f_y and f_x , obtained from Table 7, and CS is as follows:

$$CS = \frac{MSL}{1.35}$$

A calculated example for this method is shown in Appendix 1.

Table 7 – f_x -values and f_y -values as a function of α , β and μ

Table 7.1 for $\mu = 0.4$

β for f_y	α														β for f_x
	-30	-20	-10	0	10	20	30	40	45	50	60	70	80	90	
0	0.67	0.80	0.92	1.00	1.05	1.08	1.07	1.02	0.99	0.95	0.85	0.72	0.57	0.40	90
10	0.65	0.79	0.90	0.98	1.04	1.06	1.05	1.01	0.98	0.94	0.84	0.71	0.56	0.40	80
20	0.61	0.75	0.86	0.94	0.99	1.02	1.01	0.98	0.95	0.91	0.82	0.70	0.56	0.40	70
30	0.55	0.68	0.78	0.87	0.92	0.95	0.95	0.92	0.90	0.86	0.78	0.67	0.54	0.40	60
40	0.46	0.58	0.68	0.77	0.82	0.86	0.86	0.84	0.82	0.80	0.73	0.64	0.53	0.40	50
50	0.36	0.47	0.56	0.64	0.70	0.74	0.76	0.75	0.74	0.72	0.67	0.60	0.51	0.40	40
60	0.23	0.33	0.42	0.50	0.56	0.61	0.63	0.64	0.64	0.63	0.60	0.55	0.48	0.40	30
70	0.10	0.18	0.27	0.34	0.41	0.46	0.50	0.52	0.52	0.53	0.52	0.49	0.45	0.40	20
80	-0.05	0.03	0.10	0.17	0.24	0.30	0.35	0.39	0.41	0.42	0.43	0.44	0.42	0.40	10
90	-0.20	-0.14	-0.07	0.00	0.07	0.14	0.20	0.26	0.28	0.31	0.35	0.38	0.39	0.40	0

Table 7.2 for $\mu = 0.3$

β for f_y	α														β for f_x
	-30	-20	-10	0	10	20	30	40	45	50	60	70	80	90	
0	0.72	0.84	0.93	1.00	1.04	1.04	1.02	0.96	0.92	0.87	0.76	0.62	0.47	0.30	90
10	0.70	0.82	0.92	0.98	1.02	1.03	1.00	0.95	0.91	0.86	0.75	0.62	0.47	0.30	80
20	0.66	0.78	0.87	0.94	0.98	0.99	0.96	0.91	0.88	0.83	0.73	0.60	0.46	0.30	70
30	0.60	0.71	0.80	0.87	0.90	0.92	0.90	0.86	0.82	0.79	0.69	0.58	0.45	0.30	60
40	0.51	0.62	0.70	0.77	0.81	0.82	0.81	0.78	0.75	0.72	0.64	0.54	0.43	0.30	50
50	0.41	0.50	0.58	0.64	0.69	0.71	0.71	0.69	0.67	0.64	0.58	0.50	0.41	0.30	40
60	0.28	0.37	0.44	0.50	0.54	0.57	0.58	0.58	0.57	0.55	0.51	0.45	0.38	0.30	30
70	0.15	0.22	0.28	0.34	0.39	0.42	0.45	0.45	0.45	0.45	0.43	0.40	0.35	0.30	20
80	0.00	0.06	0.12	0.17	0.22	0.27	0.30	0.33	0.33	0.34	0.35	0.34	0.33	0.30	10
90	-0.15	-0.10	-0.05	0.00	0.05	0.10	0.15	0.19	0.21	0.23	0.26	0.28	0.30	0.30	0

MSC/Circ.1026
 ANNEX
 Page 5

Table 7.3 for $\mu = 0.2$

β for fy	α															β for fx
	-30	-20	-10	0	10	20	30	40	45	50	60	70	80	90		
0	0.77	0.87	0.95	1.00	1.02	1.01	0.97	0.89	0.85	0.80	0.67	0.53	0.37	0.20	90	
10	0.75	0.86	0.94	0.98	1.00	0.99	0.95	0.88	0.84	0.79	0.67	0.52	0.37	0.20	80	
20	0.71	0.81	0.89	0.94	0.96	0.95	0.91	0.85	0.81	0.76	0.64	0.51	0.36	0.20	70	
30	0.65	0.75	0.82	0.87	0.89	0.88	0.85	0.79	0.75	0.71	0.61	0.48	0.35	0.20	60	
40	0.56	0.65	0.72	0.77	0.79	0.79	0.76	0.72	0.68	0.65	0.56	0.45	0.33	0.20	50	
50	0.46	0.54	0.60	0.64	0.67	0.67	0.66	0.62	0.60	0.57	0.49	0.41	0.31	0.20	40	
60	0.33	0.40	0.46	0.50	0.53	0.54	0.53	0.51	0.49	0.47	0.42	0.36	0.28	0.20	30	
70	0.20	0.25	0.30	0.34	0.37	0.39	0.40	0.39	0.38	0.37	0.34	0.30	0.26	0.20	20	
80	0.05	0.09	0.14	0.17	0.21	0.23	0.25	0.26	0.26	0.26	0.26	0.25	0.23	0.20	10	
90	-0.10	-0.07	-0.03	0.00	0.03	0.07	0.10	0.13	0.14	0.15	0.17	0.19	0.20	0.20	0	

Table 7.4 for $\mu = 0.1$

β for fy	α															β for fx
	-30	-20	-10	0	10	20	30	40	45	50	60	70	80	90		
0	0.82	0.91	0.97	1.00	1.00	0.97	0.92	0.83	0.78	0.72	0.59	0.44	0.27	0.10	90	
10	0.80	0.89	0.95	0.98	0.99	0.96	0.90	0.82	0.77	0.71	0.58	0.43	0.27	0.10	80	
20	0.76	0.85	0.91	0.94	0.94	0.92	0.86	0.78	0.74	0.68	0.56	0.42	0.26	0.10	70	
30	0.70	0.78	0.84	0.87	0.87	0.85	0.80	0.73	0.68	0.63	0.52	0.39	0.25	0.10	60	
40	0.61	0.69	0.74	0.77	0.77	0.75	0.71	0.65	0.61	0.57	0.47	0.36	0.23	0.10	50	
50	0.51	0.57	0.62	0.64	0.65	0.64	0.61	0.56	0.53	0.49	0.41	0.31	0.21	0.10	40	
60	0.38	0.44	0.48	0.50	0.51	0.50	0.48	0.45	0.42	0.40	0.34	0.26	0.19	0.10	30	
70	0.25	0.29	0.32	0.34	0.35	0.36	0.35	0.33	0.31	0.30	0.26	0.21	0.16	0.10	20	
80	0.10	0.13	0.15	0.17	0.19	0.20	0.20	0.20	0.19	0.19	0.17	0.15	0.13	0.10	10	
90	-0.05	-0.03	-0.02	0.00	0.02	0.03	0.05	0.06	0.07	0.08	0.09	0.09	0.10	0.10	0	

Table 7.5 for $\mu = 0.0$

β for fy	α															β for fx
	-30	-20	-10	0	10	20	30	40	45	50	60	70	80	90		
0	0.87	0.94	0.98	1.00	0.98	0.94	0.87	0.77	0.71	0.64	0.50	0.34	0.17	0.00	90	
10	0.85	0.93	0.97	0.98	0.97	0.93	0.85	0.75	0.70	0.63	0.49	0.34	0.17	0.00	80	
20	0.81	0.88	0.93	0.94	0.93	0.88	0.81	0.72	0.66	0.60	0.47	0.32	0.16	0.00	70	
30	0.75	0.81	0.85	0.87	0.85	0.81	0.75	0.66	0.61	0.56	0.43	0.30	0.15	0.00	60	
40	0.66	0.72	0.75	0.77	0.75	0.72	0.66	0.59	0.54	0.49	0.38	0.26	0.13	0.00	50	
50	0.56	0.60	0.63	0.64	0.63	0.60	0.56	0.49	0.45	0.41	0.32	0.22	0.11	0.00	40	
60	0.43	0.47	0.49	0.50	0.49	0.47	0.43	0.38	0.35	0.32	0.25	0.17	0.09	0.00	30	
70	0.30	0.32	0.34	0.34	0.34	0.32	0.30	0.26	0.24	0.22	0.17	0.12	0.06	0.00	20	
80	0.15	0.16	0.17	0.17	0.17	0.16	0.15	0.13	0.12	0.11	0.09	0.06	0.03	0.00	10	
90	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0	

Remark: $f_x = \cos \alpha \cdot \sin \beta + \mu \cdot \sin \alpha$ $f_y = \cos \alpha \cdot \cos \beta + \mu \cdot \sin \alpha$

14 The existing text under the heading “Advanced calculation method: calculated example” with the heading are deleted from section 7 and added in as new Appendix 1 to the Annex with modifications as following paragraphs 15 and 16.

MSC/Circ.1026
ANNEX
Page 6

15 In new Appendix 1, the words “Advanced calculation method: calculated example” are replaced by the follows:

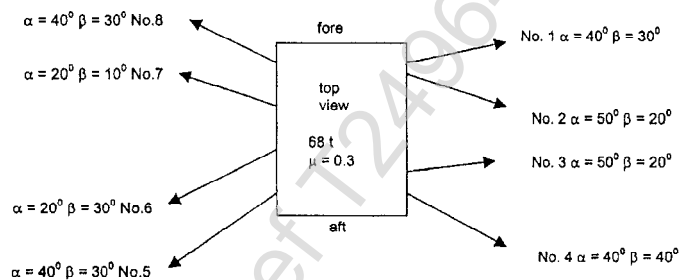
“Calculated example 1
(refer to paragraph 7.2, Balance of forces and moments)”

16 In new Appendix 1, calculated example 2 is added after calculated example 1 as follows:

“Calculated example 2
(refer to paragraph 7.3, Balance of forces – alternative method)

A cargo unit of 68 t mass is stowed on timber ($\mu = 0.3$) in the tween deck at 0.7 L of a vessel. $L = 160\text{m}$, $B = 24\text{ m}$, $v = 18\text{ kn}$ and $GM = 1.5\text{ m}$. Dimensions of the cargo unit are height = 2.4 m and width = 1.8 m. The external forces are: $F_x = 112\text{ kN}$, $F_y = 312\text{ kN}$, $F_z = 346\text{ kN}$.

The top view shows the overall securing arrangement with eight lashings.



Calculation of balance of forces:

No.	MSL (kN)	CS (kN)	α	β	f_y	$C_s * f_y$	f_x	$C_s * f_x$
1	108	80	40° stbd	30° fwd	0.86	68.8 stbd	0.58	46.4 fwd
2	90	67	50° stbd	20° aft	0.83	55.6 stdb	0.45	30.2 aft
3	90	67	50° stbd	20° fwd	0.83	55.6 stdb	0.45	30.2 fwd
4	108	80	40° stbd	40° aft	0.78	62.4 stdb	0.69	55.2 aft
5	108	80	40° port	30° aft	0.86	68.8 port	0.58	46.4 aft
6	90	67	20° port	30° aft	0.99	66.3 port	0.57	38.2 aft
7	90	67	20° port	10° fwd	1.03	69.0 port	0.27	18.1 fwd
8	108	80	40° port	30° fwd	0.86	68.8 port	0.58	46.4 fwd

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Transverse balance of forces (STBD arrangement) Nos. 1, 2, 3 and 4:

$$312 < 0.3 \cdot 68 \cdot 9.81 + 68.8 + 55.6 + 55.6 + 62.4$$

$$312 < 443 \quad \text{this is OK !}$$

Transverse balance of forces (PORT arrangement) Nos. 5, 6, 7 and 8:

$$312 < 0.3 \cdot 68 \cdot 9.81 + 68.8 + 66.3 + 69.0 + 68.8$$

$$312 < 473 \quad \text{this is OK !}$$

Longitudinal balance of forces (FWD arrangement) Nos. 1, 3, 7, 8:

$$112 < 0.3 (68 \cdot 9.81 - 346) + 46.4 + 30.2 + 18.1 + 46.4$$

$$112 < 237 \quad \text{this is OK !}$$

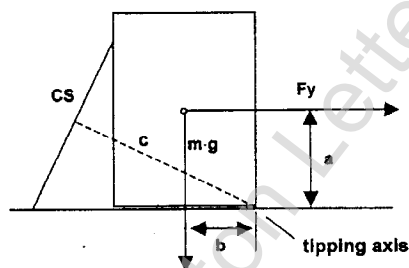
Longitudinal balance of forces (AFT arrangement) Nos. 2, 4, 5, 6:

$$112 < 0.3 (68 \cdot 9.81 - 346) + 30.2 + 55.2 + 46.4 + 38.2$$

$$112 < 266 \quad \text{this is OK !}$$

Transverse Tipping

Unless specific information is provided, the vertical center of gravity of the cargo unit can be assumed to be at one half the height and the transverse center of gravity at one half the width. Also, if the lashing is connected as shown in the sketch, instead of measuring c, the length of the lever from the tipping axis to the lashing CS, it is conservative to assume that it is equal to the width of the cargo unit.



$$F_y \cdot a \leq b \cdot m \cdot g + 0.9 \cdot (CS_1 \cdot c_1 + CS_2 \cdot c_2 + CS_3 \cdot c_3 + CS_4 \cdot c_4)$$

$$312 \cdot 2.4/2 < 1.8/2 \cdot 68 \cdot 9.81 + 0.9 \cdot 1.8 \cdot (80 + 67 + 67 + 80)$$

$$374 < 600 + 476$$

$$374 < 1076 \quad \text{this is OK !}$$

17 The existing text under the heading “Explanation and interpretation of “Methods of assess the efficiency of securing arrangements for non-standardized cargo” with the heading are deleted from section 7 and added in as new Appendix 2 to the Annex.

HAND CALCULATION SHEET FOR ADVANCED CALCULATION

In accordance with Annex 13 (Section 4.2) to the Code of Safe Practice for Cargo Stowage and Securing (*WITH EXAMPLE DATA*)

Procedure to Calculate Acceleration

1. Determine vertical position (on deck - high, on deck - low, 'tween deck or lower hold)
2. Determine the longitudinal position with respect to A.P. (within the range 0 to 1.0 L, where L is LBP)
3. Determine the basic accelerations in transverse, longitudinal and vertical directions using the Table 2 in Section 4.2
4. Apply correction to the basic acceleration for the length and speed of the specific vessel using the Tables in Section 4.2
5. If the B/GM ratio is less than 13, the transverse accelerations are to be corrected using the Tables in Section 4.2

Abbreviations:

CS : Calculated Strength in kN
 ϕ : angle of lashing device to horizontal plane
c : lever-arm of securing force
 $f = \mu \cdot \sin \phi + \cos \phi$

Friction Coefficient	μ
steel to timber or rubber	0.3
steel to steel (dry)	0.1
steel to steel (wet)	0.0

Ship	Length		m	120.00
	Breadth		m	20.0
	GM		m	1.4
	Speed		knots	15
	B/GM			14.286
Cargo Location	Vertical Location (DH/DL/TD/LH)			DL
	Longitudinal Location (* L from aft)			0.7L
Acceler'n	Longitudinal (x)		m/s ²	2.9
	Transverse (y)		m/s ²	6.3
	Vertical (z)		m/s ²	6.2
	Length Correction Factor			0.89
	GM Correct'n for Transv. Acceler'n			1.0
Cargo Details	Mass		tons	62
	Dimensions	Longitudinal	m	6
		Transverse	m	4
		Vertical	m	4
	Center Gravity Height		m	1.8
	Half Base	Longitudinal	m	2
		Transverse	m	2
	Friction Coefficient		μ	0.3
Wind Affected?				1
Sea Sloshing Affected?				1

Securing Material Details

Securing Device	Breaking Strength (KN)	MSL (KN)	CS (KN)
Wire Rope	125	$0.8 \times 125 = 100$	$100/1.5 = 66.7$
Shackles, turnbuckles, deck rings	180	$0.5 \times 180 = 90$	$90/1.5 = 60$

Lashing Details and Securing Forces

	No.	CS	Angle(°)	Lever Arm (c)	f	CSn.fn	CSn.cn
Long'l Lashing (Fore)	1						
	2						
	3						
	4		NOT	APPLI	CABLE		
	5						
	6						
	7						
Total							

	No.	CS	Angle(°)	Lever Arm (c)	f	CSn.fn	CSn.cn
Long'l Lashing (Aft)	1						
	2						
	3						
	4		NOT	APPLI	CABLE		
	5						
	6						
	7						
Total							

	No.	CS	Angle(°)	Lever Arm (c)	f	CSn.fn	CSn.cn
Transv. Lashing (Port)	1	60	40	2.3	0.96	57.53	138.0
	2	60	40	2.3	0.96	57.53	138.0
	3	60	10	0.8	1.04	62.21	48.0
	4	60	10	0.8	1.04	62.21	48.0
	5						
	6						
	7						
Total						239.48	372.0

	No.	CS	Angle(°)	Lever Arm (c)	f	CSn.fn	CSn.cn
Transv. Lashing (Starboard)	1	60	40	2.3	0.96	57.53	138.0
	2	60	40	2.3	0.96	57.53	138.0
	3	60	40	2.3	0.96	57.53	138.0
	4	60	40	2.3	0.96	57.53	138.0
	5						
	6						
	7						
Total						230.12	552.0

Corrections to Basic Accelerations

Basic Acceleration (m/s ²)	Length Correction Factor	GM Correct'n for Transv. Acceler'n	Corrected Acceleration (m/s ²)
Longitudinal (x)	0.89	N.A.	2.9 x 0.89 = 2.581
Transverse (y)	0.89	1.0	6.3 x 0.89 x 1 = 5.607
Vertical (z)	0.89	N.A.	6.2 x 0.89 = 5.518

Calculation of External Forces

Wind and Sea Sloshing		Horizontal Dimension (m)	Vertical Dimension (m)	Project Area (m ²)	Force (KN)
Wind	Longl Force	4	4	4 x 4 = 16	16 x 1 = 16
	Transv. Force	6	4	6 x 4 = 24	24 x 1 = 24
Sea Sloshing	Longl Force	4	2	4 x 2 = 8	8 x 1 = 8
	Transv. Force	6	2	6 x 2 = 12	12 x 1 = 12

Remarks: Sea Sloshing apply only to deck cargo up to 2m above deck/hatch top

External Force	Corrected Acceleration	Acceleration Force	Wind Force	Sea Sloshing	total force	Vertical Lever	moment
Longl (x)	2.581	62 x 2.581	16	8	184.02	1.8	184.02x1.8
Trans (y)	5.607	62 x 5.607	24	12	383.63	1.8	383.63x1.8
Vert (z)	5.518	62 x 5.518	N.A.	N.A.	342.12	N.A.	N.A.

Calculation of Friction Forces

Friction Force	Longl (x)	(62 x 9.81 - 342.12) x 0.3 = 79.83
	Trans (y)	62 x 9.81 x 0.3 = 182.47
	Vert (z)	N.A.

Results

Sliding Forces		Securing Force			External Force	OK or NG?
		Friction	Lashing	Total		
Long'l	Fore	79.83	0.	79.83	184.02	Not OK
	Aft	79.83	0.	79.83	184.02	Not OK
Transv.	Port	182.47	239.48	421.95	383.63	OK
	Starboard	182.47	230.12	412.59	383.63	OK

Tipping Moment		Securing Moment					External Moment	OK or NG?
		Self-Weight	Lever	Moment	Lash mt	Total		
Long'l	Port	266.10	2.0	532.2	0.	532.2	331.24	OK
	Starboard	266.10	2.0	532.2	0.	532.2	331.24	OK
Transv.	Port	608.22	2.0	1216.44	372.0	1588.44	690.53	OK
	Starboard	608.22	2.0	1216.44	552.0	1768.44	690.53	OK

HAND CALCULATION SHEET FOR ADVANCED CALCULATION

In accordance with Annex 13 (Section 4.2) to the Code of Safe Practice for Cargo Stowage and Securing (BLANK FORM)

Procedure to Calculate Acceleration

1. Determine vertical position (on deck - high, on deck - low, 'tween deck or lower hold)
2. Determine the longitudinal position with respect to A.P. (within the range 0 to 1.0 L, where L is LBP)
3. Determine the basic accelerations in transverse, longitudinal and vertical directions using the Table 2 in Section 4.2
4. Apply correction to the basic acceleration for the length and speed of the specific vessel using the Tables in Section 4.2
5. If the B/GM ratio is less than 13, the transverse accelerations are to be corrected using the Tables in Section 4.2

Abbreviations:

CS : Calculated Strength in kN
 θ : angle of lashing device to horizontal plane
c : lever-arm of securing force
 $f = \mu \cdot \sin \theta + \cos \theta$

Friction Coefficient	μ
steel to timber or rubber	0.3
steel to steel (dry)	0.1
steel to steel (wet)	0.0

Ship	Length	m			
	Breadth	m			
	GM	m			
	Speed	knots			
	B/GM				
Cargo Location	Vertical Location (DH/DL/TD/LH)				
	Longitudinal Location (* L from aft)				
Acceler'n	Longitudinal (x)	m/s ²			
	Transverse (y)	m/s ²			
	Vertical (z)	m/s ²			
	Length Correction Factor				
	GM Correct'n for Transv. Acceler'n				
Cargo Details	Mass	tons			
	Dimensions	Longitudinal	m		
		Transverse	m		
		Vertical	m		
	Center Gravity Height		m		
	Half Base	Longitudinal	m		
		Transverse	m		
	Friction Coefficient		μ		
Wind Affected?					
Sea Sloshing Affected?					

Securing Material Details

Securing Device	Breaking Strength (KN)	MSL (KN)	CS (KN)

Lashing Details and Securing Forces

	No.	CS	Angle(θ)	Lever Arm (c)	f	CSn.fn	CSn.cn
Long'l Lashing (Fore)	1						
	2						
	3						
	4						
	5						
	6						
	7						
Total							

	No.	CS	Angle(θ)	Lever Arm (c)	f	CSn.fn	CSn.cn
Long'l Lashing (Aft)	1						
	2						
	3						
	4						
	5						
	6						
	7						
Total							

	No.	CS	Angle(θ)	Lever Arm (c)	f	CSn.fn	CSn.cn
Transv. Lashing (Port)	1						
	2						
	3						
	4						
	5						
	6						
	7						
Total							

	No.	CS	Angle(θ)	Lever Arm (c)	f	CSn.fn	CSn.cn
Transv. Lashing (Starboard)	1						
	2						
	3						
	4						
	5						
	6						
	7						
Total							

Corrections to Basic Accelerations

Basic Acceleration (m/s ²)	Length Correction Factor	GM Correct'n for Transv. Acceler'n	Corrected Acceleration (m/s ²)
Longitudinal (x)		N.A.	
Transverse (y)			
Vertical (z)		N.A.	

Calculation of External Forces

Wind and Sea Sloshing		Horizontal Dimension (m)	Vertical Dimension (m)	Project Area (m ²)	Force (KN)
Wind	Longl Force				
	Transv. Force				
Sea Sloshing	Longl Force				
	Transv. Force				

Remarks: Sea Sloshing apply only to deck cargo up to 2m above deck/hatch top

External Force	Corrected Acceleration	Acceleration Force	Wind Force	Sea Sloshing	total force	Vertical Lever	moment
Longl (x)							
Trans (y)							
Vert (z)							

Results

Sliding Forces		Securing Force			External Force	OK or NG?
		Friction	Lashing	Total		
Long'l	Fore					
	Aft					
Transv.	Port					
	Starboard					

Tipping Moment		Securing Moment					OK or NG?
		Self- Weight	Lever	Moment	Lash mt	Total Moment	
Long'l	Port						
	Starboard						
Transv.	Port						
	Starboard						

4.3 ALLOWABLE STOWAGE PATTERNS (flatracks, oversized containers, etc.)

Flat racks are secured as any other standard container with twist locks and lashing rods. Typically, no containers are stowed above flatracks unless the flatrack is equipped with standard height or high cube corner posts and fittings. In all cases standard containers are not stowed above loaded flat racks on deck. If stowed on deck loaded flatrack are typically stowed in top tier.

Oversized containers are typically stowed above deck on the top tier. The oversized containers are secured with twistlocks and lashing rods as other containers. Oversized means a container or its contents (flatrack) more than eight feet wide or more than forty-five feet long.

Special securing instructions will be provided when oversized cargo is to be loaded.

MV GEORGE II

Inspection Form (containers in place)

[Click HERE for printable form](#)

DATE _____

BY _____

Hatch	Twistlocks Locked	Turbuckles Slack removed	Short rods in place	Extension hooks in place
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

Comments;

MV GEORGE II

Maintenance Form (gear not in use)

[Click HERE for printable form:](#)

DATE _____

BY _____

Hatch	Twistlocks Serviced Maintained Greased	Turbuckles Serviced Maintained Greased	Short rods Serviced Maintained Greased	Extension Serviced Maintained Greased
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

Comments;

n10-97

Page 1 of 17

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NAVIGATION AND VESSEL INSPECTION CIRCULAR NO. 10-97

Subj: GUIDELINES FOR CARGO SECURING MANUAL APPROVAL

Ref (a) Guidelines for the Preparation of the Cargo Securing Manual, International Maritime Organization, Maritime Safety Committee, MSC/Circ.745 of June 13, 1996

(b) 1994 Amendments to Chapters VI/5.6 and VII/6.6 of the International Convention for the Safety of Life at Sea, 1974 (SOLAS)

1. PURPOSE. To provide guidelines for Coast Guard inspectors and qualified organizations to use when inspecting U.S. vessels for compliance with references (a) and (b). Regulations are forthcoming that will set forth the criteria for organizations that may qualify to assist the Coast Guard in the approval of Cargo Securing Manuals (CSM). In the meantime, because of the rapidly approaching SOLAS deadline of December 31, 1997, this Navigation and Vessel Inspection Circular (NVIC) also provides interim CSM submittal, review, approval, and appeal procedures.

2. DIRECTIVES AFFECTED. None.

3. BACKGROUND. During the early 1990's several maritime incidents underscored the risk of serious injury or death, vessel loss, property damage, and maritime environmental damage caused by improperly secured cargoes aboard vessels in international trade. For example, in January 1992 the M/V SANTA CLARA I lost 21 freight containers, including four containers of toxic arsenic trioxide, off the U.S. Atlantic Coast due to cargo securing failures in connection with bad weather and human error (U.S. Coast Guard Marine Board of Inquiry Report of May 18, 1992; Serial Number: 16732/03 HQS 92). In another incident, cargo securing failures were believed to have contributed to the loss of a British vessel carrying bagged cement. Growing international concern led to the creation of a working group to develop cargo securing standards for adoption by the Maritime Safety Committee (MSC) of the International Maritime Organization (IMO). In June 1996, reference (a) was published, providing a uniform approach to the preparation of CSMs. Reference (b) requires that cargo be loaded, stowed, and secured throughout the voyage in accordance with "Administration Approved" CSMs.

4. DISCUSSION.

a. Beginning on December 31, 1997, cargo vessels engaged in international trade which are equipped with cargo securing systems or individual securing arrangements must have on board a CSM which has been approved by the vessel's flag state administration. Under Title 46, Code of Federal Regulations

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 2 of 17

Part 90.05-10 (46 CFR 90.05-10) and reference (b), CSM requirements apply to all U.S. flag cargo vessels 500 gross tons or more engaged in international trade, except for those engaged solely in the carriage of bulk solid or liquid cargoes. Ship types affected include general cargo vessels, cellular containerships, roll on- roll off vessels, passenger/cargo vessels, supply vessels, bulk vessels capable of carrying non-bulk cargo, heavy lift ships, freight ships carrying packaged or break-bulk cargoes, and other similar vessels. Guidelines for preparation, submittal, review, approval, and appeal of CSMs are provided as enclosures (1) and (2). U.S. and foreign government port state control boardings may include examination of cargo securing for compliance with the vessel's CSM.

b. Executive Order (E.O.) 12234 of September 3, 1980, authorizes the Coast Guard to issue certificates and approvals required under the International Convention for the Safety of Life at Sea (SOLAS), 1974 including CSMs. The Coast Guard has delegated this authority to the American Bureau of Shipping (ABS). After the Coast Guard issues regulations explaining the necessary criteria, other organizations may be able to assist the Coast Guard in reviewing and approving CSMs on U.S. vessels. Further, under the authority of 46 U.S.C. § 3103, because of the rapidly approaching deadline to have approved CSMs on board U.S. vessels by December 31, 1997, and based on the National Cargo Bureau's (NCB) experience and familiarity in the field of loading and stowage, NCB is deemed to have met the criteria in enclosure (3). Therefore, Coast Guard units may direct U.S. vessel owners and operators to either ABS or NCB for CSM approval.

c. Other organizations have expressed interest in U.S. CSM Approval Authority authorization procedures. Application, review, and authorization guidelines are provided as enclosure (3).

5. IMPLEMENTATION.

a. Beginning December 31, 1997, U.S. cargo vessels meeting the criteria of 46 CFR 90.05-10 must have on board an approved CSM to maintain compliance with their SOLAS "Cargo Ship Safety Equipment Certificate" (CG-3347). Owners of U.S. cargo vessels subject to these requirements are required to prepare and submit CSMs for approval, using the guidance in enclosures (1) and (2).

b. Owners of U.S. cargo vessels less than 500 gross tons engaged in international commerce seeking voluntary CSM compliance shall prepare and submit CSMs for approval, using the guidance in enclosures (1) and (2).

c. Commandant (G-MSO) will initiate a rulemaking to specify U.S. flag vessel CSM responsibilities, establish U.S. CSM Approval Authority responsibilities, and identify application and selection procedures for organizations seeking U.S. CSM Approval Authority delegation.

d. Until regulations are issued, U.S. CSM Approval Authorities (those organizations delegated authority by the United States for the approval of U.S. flag vessel CSMs) will follow the guidance in enclosures (2) and (3).

e. Until regulations are issued, other organizations which desire to become a U.S. CSM Approval Authority may submit an application to Commandant (G-MOC) in accordance with enclosure (3). If the applicant meets the criteria and becomes qualified to approve CSMs, Commandant (G-MOC) will maintain an up-to-date listing of U.S. CSM Approval Authorities and shall publish a Notice in the Federal Register to notify the public of such determinations.

f. Commandant (G-MOC) will conduct periodic oversight audits of all U.S. CSM Approval Authorities to ensure compliance with the guidance in enclosures (2) and (3).

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 3 of 17

g. Beginning on December 31, 1997, OCMI and representatives of authorized classification societies under the Alternate Compliance Program (ACP) will include CSM compliance checks as part of routine SOLAS inspections conducted aboard affected U.S. cargo vessels. As part of Cargo Ship Safety Equipment Certificate issuance, the OCMI and ACP classification society representatives will check for compliance with references (a) and (b) requirements to certify that "in all other respects the ship complied with the relevant requirements of the Convention" per Section. 2.6 of the certificate (enclosure (4)). Until regulations on this matter are issued, inspectors should explain to cargo vessel masters that the U.S. Coast Guard expects conformance with these newly implemented SOLAS requirements in order to ensure the vessel's compliance with 46 CFR 91.60-10. The public will be notified of this course of action in a Notice to be published in the Federal Register.

6. **ACTION.** Vessel designers, builders, owners, operators, or their representatives, are encouraged to take advantage of the procedures and guidelines detailed in this Circular. District Commanders, Commanders of Activities, Commanding Officers of Marine Safety Offices, and the Commanding Officer, Marine Safety Center are encouraged to bring this NVIC to the attention of appropriate individuals in the marine industry.

R. C. NORTH
Rear Admiral, U. S. Coast Guard
Assistant Commandant for Marine
Safety and Environmental Protection

- Encl: (1) Guidelines for the Preparation of the Cargo Securing Manuals for U.S. Vessels
(2) Cargo Securing Manual Submittal, Review, Approval, and Appeal Procedures
(3) Authorization of U.S. Cargo Securing Manual Approval Authority
(4) SOLAS Cargo Ship Safety Equipment Certificate (CG-3347)

GUIDELINES FOR THE PREPARATION OF CARGO SECURING MANUALS FOR U.S. VESSELS

I. **GENERAL.** These guidelines provide the minimum standards for the preparation of Cargo Securing Manuals (CSM) for U.S. vessels to comply with the 1994 Amendments to Safety of Life at Sea (SOLAS), 1974 Regulations VI/5.6 and VII/6.6. Approved CSMs are required to provide up-to-date information and guidance to assist the vessel's Master and crew regarding the proper use of the equipment available to adequately stow and secure the vessel's cargo. An owner or manager must carry out a good deal of research and fact finding before drafting a CSM, often in consultation with the vessel's personnel responsible for cargo stowage and securing. IMO's Maritime Safety Committee Circular 745 (MSC/Circ. 745), "Guidelines for the Preparation of the Cargo Securing Manual", establishes a uniform international standard for CSM preparation and is provided as Attachment A to this enclosure.

II. **APPLICABILITY.** Compliance is required for all U.S. vessels 500 gross tons or more engaged in international trade, except those used solely in the carriage of bulk solid or liquid cargoes. Voluntary compliance is encouraged for other U.S. cargo vessels less than 500 gross tons engaged in international trade. CSMs will therefore be required for ship types such as:

A. General cargo vessels,

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 4 of 17

- B. Cellular Containerships,
- C. Roll on-Roll off (RO-RO) vessels,
- D. Passenger/Cargo vessels,
- E. Supply vessels,
- F. Bulk carriers which may, on some voyages, carry non-bulk cargo requiring individual stowage and securing arrangements,
- G. Heavy lift ships,
- H. Freight ships carrying packaged or break-bulk cargoes, and
- I. Vessels which are equipped with cargo securing systems or individual securing arrangements.

III. CSM PREPARATION GUIDELINES. CSMs for U.S. vessels shall comply with the provisions of MSC/Circ. 745, "Guidelines for the Preparation of the Cargo Securing Manual" (see Attachment A of this enclosure) and:

- A. Be specific for a single ship or for sister-ships with identical cargo stowage arrangements.
- B. Consistently use the same terminology, dimensional system, abbreviations, symbols, and other elements that must remain uniform throughout the manual.
- C. Describe the types of cargo for which the manual is applicable using the general cargo categories of standardized, semi-standardized, and non-standardized as defined in Section 1.1, Definitions, of MSC/Circ. 745, "Guidelines for the Preparation of the Cargo Securing Manual."
- D. Specifically identify applicable cargo securing standards to which the vessel was designed:
 - 1. New construction may comply with:
 - a. Applicable classification society standards. For example, ABS classed vessels carrying intermodal containers built after 1988 should comply with ABS's "Guide for Certification of Container Securing Systems" for the design and construction of container securing systems for the carriage of standardized cargo, or
 - b. A level of safety equivalent to applicable classification society standards as formally approved by the vessel's classification society after careful review of securing system particulars.
 - 2. Existing vessels shall comply with applicable international, maritime industry, or vessel owner's cargo securing equipment design standards in place at the time of vessel construction. When necessary, the vessel's owner shall provide complete copies of these applicable standards to the U.S. CSM Approval Authority in order to facilitate the review of the CSM.
- E. With respect to non-standard cargo, comply with IMO Resolution A.714(17), "The Code of Safe Practice for Cargo Stowage and Securing" (IMO Publication 292E).

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 5 of 17

Attachment: (A):
IMO Maritime
Safety Committee
Circular 745 of 13 June 1996



MSC/Circ.74513
June 1996
Ref. T3/2.01

GUIDELINES FOR THE PREPARATION OF THE CARGO SECURING MANUAL

1 In accordance with regulations VI/5 and VII/6 of SOLAS 1974, as amended, cargo units and cargo transport units shall be loaded, stowed and secured throughout the voyage in accordance with the Cargo Securing Manual approved by the Administration, which shall be drawn up to a standard at least equivalent to the guidelines developed by the Organization.

2 The Maritime Safety Committee, at its sixty-sixth session (28 May to 6 June 1996), considered the draft Guidelines for the Preparation of the Cargo Securing Manual (DSC 1/27, annex 11 and DSC/Circ.1) prepared by the Sub-Committee on Dangerous Goods, Solid Cargoes and Containers (DSC), at its first session (5 to 9 February 1996), and approved the Guidelines as amended and set out in the annex to this circular.

3 These Guidelines are based on the provisions contained in the annex to MSC/Circ.385 but have been expanded to include the applications explicit to ships which are equipped or adapted for the carriage of freight containers, taking into account the provisions of the Code of Safe Practice for Cargo Stowage and Securing (CSS Code), as amended. They are of a general nature and intended to provide guidance on the preparation of such Cargo Securing Manuals which are required on all types of ships engaged in the carriage of all cargoes other than solid and liquid bulk cargoes.

4 Member Governments are invited to bring these Guidelines to the attention of all parties concerned, with the aim of having Cargo Securing Manuals carried on board ships prepared appropriately and in a consistent manner, and to implement them as soon as possible and, in any case, not later than 31 December 1997.

5 This Circular replaces MSC/Circ.385 dated January 1985.

ANNEX GUIDELINES FOR THE PREPARATION OF THE CARGO SECURING MANUAL PREAMBLE

In accordance with the International Convention for the Safety of Life at Sea, 1974 (SOLAS) chapters

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 6 of 17

VI, VII and the Code of Safe Practice for Cargo Stowage and Securing, cargo units, including containers shall be stowed and secured throughout the voyage in accordance with a Cargo Securing Manual, approved by the Administration.

The Cargo Securing Manual is required on all types of ships engaged in the carriage of all cargoes other than solid and liquid bulk cargoes.

The purpose of these guidelines is to ensure that Cargo Securing Manuals cover all relevant aspects of cargo stowage and securing and to provide a uniform approach to the preparation of Cargo Securing Manuals, their layout and content. Administrations may continue accepting Cargo Securing Manuals drafted in accordance with MSC/Circ.383 provided that they satisfy the requirements of these guidelines. If necessary, those manuals should be revised explicitly when the ship is intended to carry containers in a standardized system.

It is important that securing devices meet acceptable functional and strength criteria applicable to the ship and its cargo. It is also important that the officers on board are aware of the magnitude and direction of the forces involved and the correct application and limitations of the cargo securing devices. The crew and other persons employed for the securing of cargoes should be instructed in the correct application and use of the cargo securing devices on board the ship.

CHAPTER 1- GENERAL

1.1 Definitions

Cargo Securing Devices are all fixed and portable devices used to secure and support cargo units.

Maximum Securing Load (MSL) is a term used to define the allowable load capacity for a device used to secure cargo to a ship. Safe Working Load (SWL) may be substituted for MSL for securing purposes, provided this is equal to or exceeds the strength definition by, MSL.

Standardized Cargo means cargo for which the ship is provided with an approved securing system based upon cargo units of specific types.

Semi-standardized Cargo means cargo for which the ship is provided with a securing system capable of accommodating a limited variety of cargo units, such as vehicles, trailers, etc.

Non-standardized Cargo means cargo which requires individual stowage and securing arrangements.

1.2 General Information

This chapter should contain the following general statements:

1. "The guidance given herein should by no means rule out the principles of good seamanship, neither can it replace experience in stowage and securing practice."
2. "The information and requirements set forth in this Manual are consistent with the requirements of the vessel's trim and stability booklet, International Load Line Certificate (1996), the hull strength loading manual (if provided) and with the requirements of the International Maritime Dangerous Goods (IMDG) Code (if applicable)."

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 7 of 17

3. "This Cargo Securing Manual specifies arrangements and cargo securing devices provided on board the ship for the correct application to and the securing of cargo units, containers, vehicles and other entities, based on transverse, longitudinal and vertical forces which may arise during adverse weather and sea conditions."

4. "It is imperative to the safety of the ship and the protection of the cargo and personnel that the securing of the cargo is carried out properly and that only appropriate securing points or fittings should be used for cargo securing."

5. "The cargo securing devices mentioned in this manual should be applied so as to be suitable and adapted to the quantity, type of packaging, and physical properties of the cargo to be carried. When new or alternative types of cargo securing devices are introduced, the Cargo Securing Manual should be revised accordingly. Alternative cargo securing devices introduced should not have less strength than the devices being replaced."

6. "There should be a sufficient quantity of reserve cargo securing devices on board the ship."

"Information on the strength and instructions for the use and maintenance of each specific type of cargo securing device, where applicable, is provided in this manual. The cargo securing devices should be maintained in a satisfactory condition. Items worn or damaged to such an extent that their quality is impaired should be replaced."

CHAPTER 2- SECURING DEVICES AND ARRANGEMENTS

2.1 Specification for fixed cargo securing devices

This sub-chapter should indicate and where necessary illustrate the number, locations, type and MSL of the fixed devices used to secure cargo and should as a minimum contain the following information:

1. a list and/or plan of the fixed cargo securing devices, which should be supplemented with appropriate documentation for each type of device as far as practicable. The appropriate documentation should include information as applicable regarding:

- Name of manufacturer
- Type designation of item with simple sketch for ease of identification
- Material(s)
- Identification marking
- Strength test result or ultimate tensile strength test result
- Result of non destructive testing
- Maximum Securing Load (MSL);

2. fixed securing devices on bulkheads, web frames, stanchions, etc. and their types (e.g. pad eyes, eyebolts, etc.), where provided, including their MSL;

3. fixed securing devices on decks and their types (e.g. elephant foot fittings, container fittings apertures, etc.) where provided, including their MSL;

4. fixed securing devices on deckhands, where provided, listing their types and MSL; and

5. for existing ships with non-standardized fixed securing devices, the information on MSL.

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 8 of 17

and location of securing points is deemed sufficient.

2.2 Specification for portable cargo securing devices

This sub-chapter should describe the number of and the functional and design characteristics of the portable cargo securing devices carried on board the ship, and should be supplemented by suitable drawings or sketches if deemed necessary. It should contain the following information as applicable:

1. a list for the portable securing devices, which should be supplemented with appropriate documentation for each type of devices as far as practicable. The appropriate documentation should include information as applicable regarding:

- Name of manufacturer
- Type designation of item with simple sketch for ease of identification
- Material(s), including minimum safe operational temperature
- Identification marking
- Strength test result or ultimate tensile strength test result
- Result of non destructive testing
- Maximum Securing Load (MSL);

2. container stacking fittings, container deck securing fittings, fittings for interlocking of containers, bridge-fittings, etc., their MSL and use;

3. chains, wire lashings, rods, etc., their MSL and use;

4. tensioners (e.g. turnbuckles, chain tensioners, etc.), their MSL and use;

5. securing gear for cars, if appropriate, and other vehicles, their MSL and use;

6. trestles and jacks, etc., for vehicles (trailers) where provided, including their MSL and use; and

7. anti-skid material (e.g. soft boards) for use with cargo units having low frictional characteristics.

2.3 Inspection and maintenance schemes

This sub-chapter should describe inspection and maintenance schemes of the cargo securing devices on board the ship.

2.3.1 Regular inspections and maintenance should be carried out under the responsibility of the master. Cargo securing devices inspections as a minimum should include:

1. routine visual examinations of components being utilized; and
2. periodic examinations/re-testing as required by the Administration. When required, the cargo securing devices concerned should be subjected to inspections by the Administration.

2.3.2 This sub-chapter should document actions to inspect and maintain the ship's cargo securing devices. Entries should be made in a recordbook, which should be kept with the Cargo Securing Manual. This recordbook should contain the following information:

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 9 of 17

1. procedures for accepting, maintaining and repairing or rejecting cargo securing devices; and
2. record of inspections.

2.3.3 This sub-chapter should contain information for the master regarding inspections and adjustment of securing arrangements during the voyage.

2.3.4 Computerized maintenance procedures may be referred to in this sub-chapter.

CHAPTER 3 -STOWAGE AND SECURING OF NON-STANDARDIZED AND SEMI-STANDARDIZED CARGO

3.1 Handling and safety instructions

This sub-chapter should contain:

1. instructions on the proper handling of the securing devices; and
2. safety instructions related to handling of securing devices and to securing and unsecuring of units by ship or shore personnel.

3.2 Evaluation of forces acting on cargo units

This sub-chapter should contain the following information:

1. tables or diagrams giving a broad outline of the accelerations which can be expected in various positions on board the ship in adverse sea conditions and with a range of applicable metacentric height (GM) values;
2. examples of the forces acting on typical cargo units when subjected to the accelerations referred to in paragraph 3.2.1 and angles of roll and metacentric height (GM) values above which the forces acting on the cargo units exceed the permissible limit for the specified securing arrangements as far as practicable;
3. examples of how to calculate number and strength of portable securing devices required to counteract the forces referred to in 3.2.2 as well as safety factors to be used for different types of portable cargo securing devices. Calculations may be carried out according to Annex 13 to the CSS Code or methods accepted by the Administration;
4. it is recommended that the designer of a Cargo Securing Manual converts the calculation method used into a form suiting the particular ship, its securing devices and the cargo carried. This form may consist of applicable diagrams, tables or calculated examples; and
5. other operational arrangements such as electronic data processing (EDP) or use of a loading computer may be accepted as alternatives to the requirements of the above paragraphs 3.2.1 to 3.2.4, providing that this system contains the same information.

3.3 Application of portable securing devices on various cargo units, vehicles and stowage blocks

3.3.1 This sub-chapter should draw the master's attention to the correct application of portable securing devices, taking into account the following factors:

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 10 of 17

1. duration of the voyage;
2. geographical area of the voyage with particular regard to the minimum safe operational temperature of the portable securing devices;
3. sea conditions which may be expected;
4. dimensions, design and characteristics of the ship;
5. expected static and dynamic forces during the voyage;
6. type and packaging of cargo units including vehicles;
7. intended stowage pattern of the cargo units including vehicles; and
8. mass and dimensions of the cargo units and vehicles.

3.3.2 This subchapter should describe the application of portable cargo securing devices as to number of lashings and allowable lashing angles. Where necessary, the text should be supplemented by suitable drawings or sketches to facilitate the correct understanding and proper application of the securing devices to various types of cargo and cargo units. It should be pointed out that for certain cargo units and other entities with low friction resistance, it is advisable to place soft boards or other anti-skid material under the cargo to increase friction between the deck and the cargo.

3.3.3. This sub-chapter should contain guidance as to the recommended location and method of stowing and securing of containers, trailers and other cargo carrying vehicles, palletized cargoes, unit loads and single cargo items (e.g. woodpulp, paper rolls, etc.), heavy weight cargoes, cars and other vehicles.

3.4 Supplementary requirements for ro-ro ships

3.4.1 The manual should contain sketches showing the layout of the fixed securing devices with identification of strength (MSL) as well as longitudinal and transverse distances between securing points. In preparing this sub-chapter further guidance should be utilized from IMO Assembly resolutions A.533(13) and A.581(14) as appropriate.

3.4.2 In designing securing arrangements for cargo units, including vehicles and containers, on ro-ro passenger ships and specifying minimum strength requirements for securing devices used, forces due to the motion of the ship, angle of heel after damage or flooding and other considerations relevant to the effectiveness of the cargo securing arrangement should be taken into account.

3.5 Bulk carriers

If bulk carriers carry cargo units falling within the scope of chapter VI/5 or chapter VII/6 of the SOLAS Convention, this cargo shall be stowed and secured in accordance with a Cargo Securing Manual, approved by the Administration.

CHAPTER 4- STOWAGE AND SECURING OF CONTAINERS AND OTHER STANDARDIZED CARGO

4.1 Handling and safety instructions

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 11 of 17

This sub-chapter should contain:

1. instructions on the proper handling of the securing devices; and
2. safety instructions related to handling of securing devices and to securing and unsecuring of containers or other standardized cargo by ship or shore personnel.

4.2 Stowage and securing instructions

This sub-chapter is applicable to any stowage and securing system (i. g. stowage within or without cellguides) for containers and other standardized cargo. On existing ships the relevant documents regarding safe stowage and securing may be integrated into the material used for the preparation of this chapter.

4.2.1 Stowage and securing plan

This sub-chapter should consist of a comprehensive and understandable plan or set of plans providing the necessary overview on:

1. longitudinal and athwartship views of under deck and on deck stowage locations of containers as appropriate;
2. alternative stowage patterns for containers of different dimensions;
3. maximum stack masses;
4. permissible vertical sequences of masses in stacks;
5. maximum stack heights with respect to approved sight lines; and
6. application of securing devices using suitable symbols with due regard to stowage position, stack mass, sequence of masses in stack and stack height. The symbols used should be consistent throughout the Cargo Securing Manual.

4.2.2 Stowage and securing principle on deck and under deck

This sub-chapter should support the interpretation of the stowage and securing plan with regard to container stowage, highlighting:

1. the use of the specified devices; and
2. any guiding or limiting parameters as dimension of containers, maximum stack masses, sequence of masses in stacks, stacks affected by wind load, height of stacks.

It should contain specific warnings of possible consequences from misuse of securing devices or misinterpretation of instructions given:

4.3 Other allowable stowage patterns

This sub-chapter should provide the necessary information for the master to deal with cargo stowage

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 12 of 17

situations deviating from the general instructions addressed to under sub-chapter 4.2, including appropriate warnings of possible consequences from misuse of securing devices or misinterpretation of instructions given. Information should be provided with regard to, inter alia:

1. alternative vertical sequences of masses in stacks;
2. stacks affected by wind load in the absence of outer stacks;
3. alternative stowage of containers with various dimensions; and
4. permissible reduction of securing effort with regard to lower stacks masses, lesser stack heights or other reasons.

4.4 Forces acting on cargo units

This sub-chapter should present the distribution of accelerations on which the stowage and securing system is based, and specify the underlying condition of stability. Information on forces induced by wind and sea on deck cargo should be provided.

It should further contain information on the nominal increase of forces or accelerations with an increase of initial stability. Recommendations should be given for reducing the risk of cargo losses from deck stowage by restrictions to stack masses or stack heights, where high initial stability cannot be avoided.

AUTHORIZATION OF U.S. CARGO SECURING MANUAL APPROVAL AUTHORITY

A. GENERAL. The Commandant (G-M) may authorize certain organizations to serve as designated U.S. CSM Approval Authorities of manuals for U.S. vessels. These guidelines summarize selection criteria, application procedures, and responsibilities of an authorized U.S. Cargo Securing Manual (CSM) Approval Authority.

B. SELECTION CRITERIA. Coast Guard authorization decisions will be based upon an assessment of each applicant's capability to fairly, professionally, and effectively review CSMs for approval. Authorization decisions by Commandant (G-M) are based upon the following criteria:

A. The organization's owners, managers, and employees are free from influence or control by vessel shipbuilders, owners, operators, lessors, or other related commercial interests as evidenced by past and present business practices.

B. The organization has demonstrated, through other related work, the capability to competently evaluate CSMs for completeness and sufficiency in accordance with SOLAS requirements.

C. The organization has an acceptable degree of financial security, as documented by recent audits from certified public accountants.

D. The organization has a documented ongoing training program and an established internal quality control process to ensure the consistent review of CSMs by its professional staff.

E. The organization maintains a corporate office in the United States that has adequate resources and staff to support all aspects of CSM review, approval, and record-keeping.

C. APPLICATION PROCEDURES.

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 13 of 17

A. Any organization seeking authorization to serve as a recognized U.S. CSM Approval Authority for U.S. vessels must submit a written application to: U.S. Coast Guard Commandant (G-MOC), 2100 Second Street, S.W., Washington, D.C. 20593-0001. Each application must be signed by an authorized officer of the organization.

B. Each application must include the following information:

1. A certified copy of the certificate of incorporation on file with the appropriate state, including the name and address of the organization,
2. A listing of the organization's principal executives, with their telephone and telefax numbers,
3. A description of the organization, including the ownership, managerial structure, and organizational components (including directly affiliated agencies and their functions) utilized for supporting technical services,
4. A list of the technical services offered,
5. A general description of the geographical area served,
6. A general description of the clients being served or intended to be served,
7. A description of the type of similar work performed by the organization in the past, noting the amount and extent of such work performed within the previous three years,
8. A listing by name of the full time professional staff available for technical review and approval of CSMs, including:
 - a) Naval architects and/or naval engineers, including copies of their professional credentials such as college degrees, specialized training, etc.,
 - b) Coast Guard licensed Masters of deep draft freight vessels, including a summary of the number of months experience aboard general cargo ships where cargo was secured and other credentials, and
 - c) Proof of staff competency to perform specific tasks, evidenced by detailed summaries of the person's experience (measured in months or years) during the past five years evaluating maritime cargo securing systems. Experience summaries and assessments shall be documented on company letterhead and signed by the company executive who had direct and daily observation of the individual and the quality of his work products,
9. A complete description of the organizations internal quality control processes which will be used to ensure consistent CSM review and approval procedures by qualified professionals,
10. A description of the organization's training program for assuring continued competence of the professionals identified above to ensure uniformity of review and approval,
11. Evidence of financial stability over the past five year period, such as financial reports completed independently by certified public accountants,

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 15 of 17

C. Grant the Coast Guard the right to inspect CSM records including approved CSMs, CSM changes, and supporting calculations or documents, and shall indicate the organization will cooperate in the conduct of such inspections, and

D. Comply with any other term or conditions stated in the Coast Guard's Letter of Authorization.

F. TERMINATION OF AUTHORIZATION.

A. A CSM Approval Authority may voluntarily terminate its authorization by giving written notice to the Commandant (G-MOC), U.S. Coast Guard, that it no longer wishes to review and approve Cargo Securing Manuals on behalf of the United States. This notice must contain the date on which the termination is to be effective.

B. Commandant (G-M), U.S. Coast Guard, may revoke an authorization when it is determined that the CSM Approval Authority has:

1. Made a material misrepresentation of itself in its application,
2. Failed to comply with a condition of the Coast Guard letter of designation or applicable provisions of this NVIC, and/or
3. Failed to fairly, professionally, and/or effectively execute its responsibilities.

C. When initiating a revocation of an U.S. CSM Approval Authority authorization, Commandant (G-M) shall provide the Approval Authority:

1. Written notice of the facts or conduct believed to warrant the revocation,
2. An opportunity to submit relevant information on its behalf about the discrepancy (ies), and
3. An opportunity to demonstrate or achieve compliance with the applicable requirement within a reasonable period of time.

D. In the case of termination for any cause:

1. Commandant (G-MOC) shall formally notify the Secretary - General of the IMO of the termination, including the organization's name, alphabetic Approval Authority identification code, termination date, and other particulars as necessary.

2. The affected organization within 30 days of the date of termination must:

a. For approved CSMs:

(1) Forward all records described in paragraph V.A. of this enclosure along with a complete inventory categorized by vessel owner, vessel name, and corresponding official number to: U.S. Coast Guard Commandant (G-MOC), 2100 Second Street, S.W., Washington, D.C. 20593-0001, and

(2) Notify each vessel owner of its termination and of the records forwarded to U.S. Coast Guard Commandant (G-MOC), categorized by the vessel name and corresponding official number.

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 15 of 17

C. Grant the Coast Guard the right to inspect CSM records including approved CSMs, CSM changes, and supporting calculations or documents, and shall indicate the organization will cooperate in the conduct of such inspections, and

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1. Made a material misrepresentation of itself in its application,
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(2) Notify each vessel owner of its termination and of the records forwarded to U.S. Coast Guard Commandant (G-MOC), categorized by the vessel name and corresponding official number.

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 16 of 17

b. For CSMs under review, return all records to the appropriate vessel owner along with a letter which catalogs the records being returned by vessel name, summarizes completed review actions, and individually addresses all areas pending resolution. A copy of each letter will be provided to U.S. Coast Guard Commandant (G-MOC), 2100 Second Street, S.W., Washington, D.C. 20593-0001.

CARGO SECURING MANUAL SUBMITTAL, REVIEW, APPROVAL, AND APPEAL PROCEDURES

A. GENERAL. These guidelines summarize the submittal and approval requirements for Cargo Securing Manuals (CSM) for U.S. vessels to comply with the 1994 Amendments to Safety of Life at Sea, 1974 Regulations VI/5.6 and VII/6.6. The Coast Guard has authorized certain organizations to approve CSMs for U.S. vessels.

B. SUBMITTAL PROCEDURES. Owners of U.S. vessels who must comply with SOLAS regulations shall:

A. Prepare their vessel's CSM in accordance with the guidelines provided by Enclosure (1) of this NVIC.

B. Submit their vessel's CSM for review and approval by an organization that has been designated by the U.S. Coast Guard. Authorized U.S. CSM Approval Authorities include:

1. The American Bureau of Shipping, ABS Plaza, 16855 Northchase Drive, Houston, TX 77060. [Telephone: (713) 873-0700/ Fax: (713) 874-8195],

2. The National Cargo Bureau, Inc., 30 Vesey Street, New York, NY 10007; [Telephone: (212) 571-5000/ Fax: (212) 571-5005], and

3. Any other organization authorized by the Coast Guard. In the future, other organizations may become authorized U.S. CSM Approval Authorities in addition to ABS and NCB. A complete up-to-date listing of U.S. CSM Approval Authorities is available from U.S. Coast Guard Commandant (G-MOC), 2100 Second Street, S.W. Washington, D.C. 20593; [Telephone: (202) 267-0495/ Fax: (202) 267-0506].

C. REVIEW PROCEDURES. U.S. CSM Approval Authorities designated by the Coast Guard shall:

A. Acknowledge receipt of each CSM and amendment submitted for approval.

B. Review submittals on a timely basis, requesting additional information whenever necessary.

C. Ensure only professional full time staff are used who are qualified to perform CSM review and approval functions, in accordance with the organization's internal quality control procedures. Qualifications considered to be satisfactory evidence of an individual's competency to perform specific CSM review and approval tasks are described in paragraph III.B.8 of Enclosure (3) to this NVIC.

D. Maintain a copy of each approved CSM and its amendments, together with supporting documents and calculations utilized in granting the approval, for review by the Coast Guard upon request.

D. APPROVALS. U.S. CSM Approval Authorities shall individually document CSM and amendment

http://www.uscg.mil/hq/g-m/nvic/10_97/n10-97.htm

12/17/2001

n10-97

Page 17 of 17

approvals on organization letterhead which:

A. Are dated,

B. Includes the following subject line: "APPROVAL OF CARGO SECURING MANUAL (AMENDMENT) FOR THE M/V _____, OFFICIAL NUMBER _____.",

C. Includes the following statement: "This is to certify that the Cargo Securing Manual (Amendment) dated _____ for the M/V _____, Official Number _____, has been approved on behalf of the United States. The Cargo Securing Manual (Amendment) was reviewed for compliance with Maritime Safety Committee Circular 745 for content, completeness, and correctness of the calculations on which the approval is based. This approval letter is to be kept with the Cargo Securing Manual, as proof of compliance with regulations VI/5.6 and VII/6.6 of the 1994 amendments to the International Convention for the Safety of Life at Sea (SOLAS) 1974. ",

D. Are signed by the official within the U.S. CSM Approval Authority responsible for Cargo Securing Manual review and approval, and

E. Includes the U.S. CSM Approval Authority's seal or stamp of authority.

E. APPEALS. Vessel owners shall use the following procedures to appeal a decision made by an U.S. CSM Approval Authority:

A. Submit to the CSM Approval Authority a written argument, including the rationale for reconsideration. Within 30 days of receiving the appeal, the CSM Approval Authority must formally provide the vessel owner a ruling on the appeal with a copy to the Commandant (G-MOC).

B. Vessel owners who are not satisfied with the CSM Approval Authority's ruling may appeal directly to the Commandant (G-MOC). The appeal must be made in writing, including any documentation and evidence that needs to be considered. Further, the appeal may ask the Commandant (G-MOC) to stay the effect of the appealed decision by the CSM Approval Authority while it is under review.

C. The Commandant (G-MOC) will make a decision on the appeal and send a formal response to the vessel owner and a copy to the U.S. CSM Approval Authority. The Coast Guard decision will constitute final agency action on the appeal request.

EQUIP TYPE	ISO TYPE	GENERAL DESCRIPTION	SPECIAL DESCRIPTION
BUL	8888	BULK SHIPMENT	SYSTEM GENERATED
BUN	9999	BUNDLED SHIPMENT	CHASSIS/FLAT BUNDLES-SYSTEM GE
OD	9999	LTL BKG EQP CODE	LESS THAN TRAILER LOAD
2B	2280	20 BULK CNTR	
2D	2210	20 STANDARD	GROSS WT LT 52000 LBS
2DC	2210	20 UPGRADE	24 TON CARGO CAPACITY BOX
2DG	2210	20 GARMENT	HANGING GARMENT EQUIPPED
2DQ	2010	20 DRY 8 FOOT HEIGHT	8 FT HIGH
2DS	2210	20 STEEL CONSTRUCTION	
2DU	2210	20 UPGRADE	30 TON NET CARGO CAPACITY
2DXX	2210	20 DRY	SALE BOX - TO BE SOLD
2F	2260	20 FLAT	FLAT FIXED END
2FC	2263	20 FLAT	COLLAPSIBLE FLATRACK
2FL	2263	20 FLAT	LOW FLAT
2H	2410	20 HIGH-CUBE DRY	GROSS WT LT 67200 LBS
2HH	2400	20 HIGH-CUBE	10.5 FT HIGH SPECIAL UNIT
2I	2220	20 INSUL	INSULATED AND VENTED CNTR
2IQ	2020	20 INSUL	8 FT HIGH
2O	2250	20 OPENTOP	OPENTOP CONTAINER
2OR	2251	20 OPENTOP	REMOVABLE END FRAMES
2R	2232	20 REEFER	
2RH	2432	20 REEFER	HEAT CAPABLE
2RTF	2232	24 REEFER	24' RFR CODE - INPUT TO ECR F1
2S	7500	20 SPECIAL	SPECIAL/MISCELLANEOUS USE
2SF	2160	20 SPECIAL	CONE FLATS (STEVEDORING)
2SFG	2160	20 SPECIAL	RAIL GENSET FRAMES
2SP	2309	20 SPECIAL	POWER BOX GENERATOR EQUIPPED
2T	2270	20 TANK-SHIPPER OWNED	
2TQ	2070	20 TANK	8 FT HIGH
2TR	2270	20 TANK REFRIGERATED	
2V	2215	20 VENT	
3D	8500	35 DV	
3FH	8560	35 FLAT	HIGH FLAT
3FL	8560	35 FLAT	LOW
3I	8520	35 INSUL	INSUL/VENTED
3IV	8520	35 INSUL	MODIFIED INSUL VAN COFFE/COCOA
3O	8550	35 OPENTOP	
3R	8532	35 REEFER	
3SP	8509	35 SPECIAL	POWER BOX
3T	8770	35 TANK	
3TW	8770	35 TANK	WHISKEY TANK
4D	4310	40 STANDARD	
4DBW	4300	40 STANDARD WIDE DOOR GARMENT	WIDE DOOR - GARMENT CAPABLE

EQUIP TYPE	ISO TYPE	GENERAL DESCRIPTION	SPECIAL DESCRIPTION
4DG	4300	40 GARMENT	HANGING GARMENT EQUIPPED
4DM	4310	40 ALUMINUM CONSTRUCTION	
4DQ	4000	40 DRY	8 FT HIGH
4DS	4310	40 STEEL CONSTRUCTION	
4DSW	4310	40 DRY STEEL CONSTRUCTION	STEEL WIDE DOOR OPENING
4DV	4300	40 DRY	SELF VENTED
4DW	4300	40 DRY	WIDE DOOR OPENING
4DWM	4310	40 DRY	ALUMINUM WIDE DOOR OPENING
4DXX	4310	40 DRY	SALE BOX - TO BE SOLD
4F	4361	40 FLAT	FLATRACK FIXED END
4FC	4363	40 FLAT	COLLAPSIBLE
4FCF	4363	40 FLAT	COLLAPSIBLE FLUSH FOLDING
4FCS	4363	40 FLAT	COLLAPSIBLE SPRING ASSISTED
4FH	4561	40 FLAT	MODIFIED 9' 6" FLAT ALASKA SVC
4FHM	4561	40 FLAT	MODIFIED 9' 6" FLAT ALASKA SVC
4FL	4960	40 FLAT	LOW PLATFORM
4FLM	4960	40 FLAT	MODIFIED PLATFORM FLAT
4FP	4960	40 FLAT	PLATFORM
4H	4500	40 HIGH CUBE	9' 6" HIGH
4HD	4510	40 HIGH CUBE	9' 6" HIGH
4HDG	4510	40 HIGH HANGING GARMENT	9' 6" HANGING GARMENT EQUIPPED
4HDM	4510	40 HIGH CUBE ALUMINUM	9' 6" HIGH ALUMINUM CONSTRUCTION
4HP	4510	40 HIGH CUBE	POP ON VENTS
4HS	4510	40 HIGH CUBE	STEEL CONSTRUCTION
4HSW	4510	40 HIGH STEEL WIDE DOOR	STEEL WIDE DOOR OPENING
4HWM	4510	40 HIGH CUBE	9' 6" HIGH WIDE DOOR OPENING
4HXX	4510	40 HIGH CUBE	SALE CONTAINER - TO BE SOLD
4I	4320	40 INSUL	CONVERTED REEFER - INSUL/VENT
4IH	4520	40 INSUL HIGH CUBE	INSUL/VENT
4IZ	4320	40 INSUL	CONVERTED FROM 35 FT
4O	4351	40 OPENTOP	
4OF	4750	40 OPENTOP	HALF HEIGHT
4OH	4551	40 OPENTOP	REMOVEABLE HARD TOP ROOF
4OI	4351	40 OPENTOP	ISO BULL RING EQUIPPED
4OR	4351	40 OPENTOP	REMOVABLE END FRAMES
4OXX	4351	40 OPENTOP	SALE BOX - TO BE SOLD
4R	4332	40 REEFER	
4RAE	4332	40 REEFER	ELECTRIFIED NON PROPANE
4RBE	4332	40 REEFER	ELECTRIFIED 8 CORNER POST
4RH	4532	40 REEFER HIGH CUBE	9 FT 6 IN HIGH
4RHA	4532	40 REEFER HIGH CUBE	CONTR ATMOS ONLY NOT USDA
4RHC	4532	40 REEFER HIGH CUBE	CNTL ATMOSPHERE
4RHD	4532	40 REEFER HIGH CUBE	INT SIDE RAILS - DOUBLE STACK

EQUIP TYPE	ISO TYPE	GENERAL DESCRIPTION	SPECIAL DESCRIPTION
4RHE	4532	40 REEFER HIGH CUBE	ELECTRIFIED STAINLESS STEEL INT
4RHF	4532	40 REEFER FRESHMIST HUMIDITY	FRESHMIST HUMIDITY CNTL
4RHH	4532	40 REEFER ATMOSPHERE, HUMIDITY	ATMOSPHERE, HUMIDITY, USDA COLD
4RHL	4532	40 REEFER LOW TEMP SMART RFR	LOW TEMP R404A SMART RFR
4RHM	4532	40 REEFER HIGH CUBE	MEATRAIL EQUIPPED
4RHN	4532	40 REEFER HIGH CUBE	EVERFRESH NITROGEN CNTL ATMOS
4RHS	4532	40 REEFER HIGH CUBE	STAINLESS STEEL
4RHT	4532	40 REEFER DUAL TEMP	DUAL TEMP CONTROL
4RHU	4532	40 REEFER HIGH CUBE	ATMOSPHERE, DE-HUMID, USDA COLD
4RI	4332	40 REEFER	
4RIE	4332	40 REEFER	ELECTRIFIED ISO
4RME	4332	40 REEFER	ELECTRIFIED MEATRAIL
4RNE	4532	40 REEFER	ELECTRIFIED 9 FT HIGH
4RS	4332	40 REEFER	STEEL EXTERIOR, STAINLESS STEEL
4RZ	4332	40 REEFER	STRETCHED UNIT NON-ISO
4RZE	4332	40 REEFER	ELECTRIFIED STRETCHED NON ISO
4S	4309	40 SPECIAL	CAR RAMPS INSTALLED IN 4W BOXE
4SF	4509	40 SPECIAL	SPECIAL USE FLAT
4SFG	4509	40 SPECIAL	SPECIAL USE GENSET FLAT
4SFT	4509	40 SPECIAL	FLANK 40' FLAT WITH 35' TANK
4SHP	4509	40 SPECIAL HIGH CUBE	9' 6" HIGH POWER BOX GENERATOR
4SP	4309	40 SPECIAL	POWER BOX GENERATOR EQUIPPED
4SRC	4361	40 SPECIAL	CHASSIS STACK RACKS
4SY	4509	40 SPECIAL	YARN RACKS - ALLIED / DUPONT
4T	4370	40 TANK	
4TS	4370	40 TANK	TEST TANK
4WC	4526	40 CARTAINER	STRETCHED 13' 6" HIGH
4WHV	4500	40 CARCARRIER HIGH CUBE	VTI UNIT EQPD W/ TEMP AUTO RAC
5H	9500	45 HIGH CUBE	9' 6" HIGH
5HD	9510	45 HIGH CUBE	
5HDM	9510	45 HIGH CUBE	ALUMINUM CONSTRUCTION
5HG	9500	45 HIGH CUBE	GARMENT EQUIPPED
5HS	9500	45 HIGH STEEL CONSTRUCTION	STEEL CONSTRUCTION
5IH	9520	45 INSULATED / HEATED	
5IHS	9520	45 INSULATED / HEATED	STEEL CONSTRUCTION
5PG	9909	45 RAIL PIG	RAIL PIG TRAILER
5RH	9532	45 REEFER HIGH CUBE	
8HD	9410	48 HIGH CUBE	
8HW	9410	48 HIGH CUBE	102 IN WIDE

Resolution A.748(18)

*Adopted on 4 November 1993
(Agenda item 11)*

**CODE FOR THE SAFE CARRIAGE OF IRRADIATED NUCLEAR FUEL,
PLUTONIUM AND HIGH-LEVEL RADIOACTIVE WASTES IN
FLASKS ON BOARD SHIPS**

A

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organization (IMO) concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety and the prevention and control of marine pollution from ships,

RECALLING FURTHER the provisions of chapter VII of the International Convention for the Safety of Life at Sea (SOLAS), 1974, as amended, and the provisions of the International Maritime Dangerous Goods Code (IMDG Code), which regulate, *inter alia*, the carriage of radioactive material by sea,

RECOGNIZING that the IMDG Code, which generally implements the International Atomic Energy Agency (IAEA) Regulations for the Safe Transport of Radioactive Material, contains no specific requirements for the design and equipment of ships engaged in the carriage of irradiated nuclear fuel, plutonium and high-level radioactive wastes,

NOTING the increase in maritime transport of irradiated nuclear fuel, plutonium and high-level radioactive wastes,

BELIEVING that adequate regulations for the construction, equipment and operation of ships engaged in the carriage of irradiated nuclear fuel, plutonium and high-level radioactive wastes would enhance maritime safety and protection of the marine environment,

NOTING ALSO paragraph 17.30(a)(ix) of Agenda 21 of the United Nations Conference on Environment and Development (UNCED), encouraging IMO and IAEA to work together to complete consideration of a code on the carriage of irradiated nuclear fuel in flasks on board ships,

HAVING NOTED with appreciation the work carried out by the joint IAEA/IMO/UNEP Working Group on the Safe Carriage of Irradiated Nuclear Fuel and other Nuclear Material by Sea,

HAVING CONSIDERED the recommendations made by the Maritime Safety Committee at its sixty-second session and by the Marine Environment Protection Committee at its thirty-fourth session,

1. ADOPTS the Code for the Safe Carriage of Irradiated Nuclear Fuel, Plutonium and High-Level Radioactive Wastes in Flasks on Board Ships, set out in the annex to the present resolution;
2. URGES Governments to implement this Code at the earliest possible opportunity.

Resolution A.748(18) – 249

THE ASSEMBLY,

HAVING ADOPTED the Code for the Safe Carriage of Irradiated Nuclear Fuel, Plutonium and High-Level Radioactive Wastes in Flasks on Board Ships (INF Code),

BEARING IN MIND the results of the United Nations Conference on Environment and Development, in particular chapter 17 of Agenda 21,

NOTING the reports of the Joint IAEA/IMO/UNEP Working Group on the Safe Carriage of Irradiated Nuclear Fuel and other Nuclear Material by Sea, in particular matters identified by the Working Group which require further consideration and which are complementary to the INF Code,

NOTING ALSO the importance of technical co-operation for the enhancement of international standards for maritime safety and marine environmental protection, as envisaged by the INF Code,

RECOGNIZING the need for further work to be carried out, with a view to augmenting or improving the INF Code,

REQUESTS the Maritime Safety Committee and the Marine Environment Protection Committee, in consultation with IAEA:

- (a) to keep the INF Code under regular review and to amend it, as necessary;
- (b) to consider, as a matter of high priority, relevant aspects of the transport of irradiated nuclear fuel and other nuclear material which are complementary to the INF Code, taking into account the recommendations of the Joint IAEA/IMO/UNEP Working Group and the objectives of paragraph 17.22 of Agenda 21 of UNCED;
- (c) to keep under review the need for international co-operation for the implementation of the INF Code and related matters; and
- (d) to report to the nineteenth session of the Assembly on the progress of work made in this field.

Annex

**CODE FOR THE SAFE CARRIAGE OF IRRADIATED NUCLEAR FUEL,
PLUTONIUM AND HIGH-LEVEL RADIOACTIVE WASTES
IN FLASKS ON BOARD SHIPS**

GENERAL

- 1 This Code applies to new and existing ships regardless of size, including cargo ships of less than 500 tons gross tonnage, engaged in the carriage of irradiated nuclear fuel, plutonium and high-level radioactive wastes in flasks approved in accordance with the applicable Regulations for the Safe Transport of Radioactive Material adopted by the International Atomic Energy Agency and carried in accordance with class 7 of the International Maritime Dangerous Goods (IMDG) Code, schedules 10, 11, 12 or 13.
- 2 For the purposes of this Code:
 - .1 *Irradiated nuclear fuel* means material containing uranium, thorium and/or plutonium isotopes which has been used to maintain a self-sustaining nuclear chain reaction.
 - .2 *Plutonium* means the resultant mixture of isotopes of that material extracted from irradiated nuclear fuel from reprocessing.
 - .3 *High-level radioactive wastes* means liquid wastes resulting from the operation of the first stage extraction system or the concentrated wastes from subsequent extraction stages, in a facility for reprocessing irradiated nuclear fuel, or solids into which such liquid wastes have been converted.
- 3 For the purposes of this Code, ships carrying materials covered by this Code in flasks have been assigned to three classes, depending on the total radioactive quantity which may be carried on board:
 - Class INF 1 - Ships carrying such materials with an aggregate radioactivity less than 4,000 TBq.
 - Class INF 2 - Ships carrying irradiated nuclear fuel or high-level radioactive wastes with an aggregate radioactivity less than 2×10^6 TBq and ships carrying plutonium with an aggregate radioactivity less than 2×10^5 TBq.
 - Class INF 3 - Ships carrying irradiated nuclear fuel or high-level radioactive wastes and ships carrying plutonium with no restriction on the aggregate radioactivity of the materials.
- 4 All ships, regardless of size, carrying materials covered by this Code should comply with the requirements of SOLAS 1974, as amended, and in addition with the requirements as prescribed in table 1 and paragraphs 7 to 25, concerning damage stability, fire protection, temperature control of cargo spaces, structural considerations, cargo securing arrangements, electrical supplies, radiological protection equipment and management, training and shipboard emergency plan.
- 5 In addition to the requirements of this Code, the provisions of the IMDG Code also apply.
- 6 The Administration should provide the ship with an appropriate document as evidence of compliance of construction and equipment with the requirements of this Code.

Table 1 – Requirements for ships carrying irradiated nuclear fuel, plutonium and high-level radioactive wastes in flasks on board ships

Ship's class	Damage stability		Fire protection		Temperature control of cargo spaces	Structural considerations	Cargo securing arrangements	Electrical supplies	Radiological protection equipment	Management, training and shipboard emergency plan
	Passenger ships*	Cargo ships	Passenger ships*	Cargo ships						
INF 1	7	7	11	11	14 + 15 + 16	17	18 + 19	20	24	25
INF 2	8	9	13	13	14 + 15 + 16	17	18 + 19	21 + 22	24	25
INF 3	N/A	10	N/A	12 + 13	14 + 15 + 16	17	18 + 19	21 + 22 + 23	24	25

N/A: INF 3 not allowed on passenger ships

* As defined in SOLAS 74, chapter 1, part A, regulation 2(f)

DAMAGE STABILITY

- 7 To the satisfaction of the Administration concerned.
- 8 Part B, chapter II-1, of the 1974 SOLAS Convention, as amended by resolution MSC.12(56), or resolution A.265(VIII).
- 9 Part B-1, chapter II-1 of the 1974 SOLAS Convention (resolution MSC.19(58)), regardless of ship length.
- 10 Type 1 ships survival capability and location of cargo spaces in chapter 2 of the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (IBC Code) or, regardless of ship length, requirements in part B-1, chapter II-1 of the 1974 SOLAS Convention (resolution MSC.19(58)) with subdivision index R_{INF} as given below:

$$R_{INF} = R + 0.2(1 - R)$$

FIRE PROTECTION

- 11 To the satisfaction of the Administration concerned.
- 12 Accommodation spaces, service spaces, control stations and machinery spaces of category A, as defined in regulation II-2/3.19 of SOLAS 74, as amended, should be fitted either forward or aft of the cargo spaces, due regard being paid to the overall safety of the ship.

13 The ship should be fitted with the following systems and equipment, irrespective of its gross tonnage, i.e. also for cargo ships of less than 500 tons gross tonnage:

- .1** a water fire-extinguishing system complying with the requirements of regulation II-2/4 of SOLAS 74, as amended;
- .2** a fixed fire-extinguishing system in machinery spaces (defined as above) complying with the requirements of regulation II-2/7 of SOLAS 74, as amended;
- .3** fixed cargo space cooling arrangements, complying with the requirements of regulation II-2/54.2.1.3 of SOLAS 74, as amended; and
- .4** a fixed fire-detection and fire-alarm system, protecting the machinery spaces, accommodation and service spaces, complying with the requirements of regulation II-2/13 of SOLAS 74, as amended.

TEMPERATURE CONTROL OF CARGO SPACES

- 14** Adequate ventilation or refrigeration of enclosed cargo spaces should be provided so that the average ambient temperature within such spaces does not exceed 55°C at any time.
- 15** Ventilation or refrigeration systems serving cargo spaces intended for the transport of materials covered by this Code should be independent of those serving other spaces.
- 16** Those items essential to operation, e.g. fans, compressors, heat exchangers, cooling water supply, etc., should be provided in duplicate for each cargo space and spare parts should be available, to the satisfaction of the Administration concerned.

STRUCTURAL CONSIDERATIONS

- 17** The structural strength of deck areas and support arrangements should be designed, taking into account the high loading which is to be sustained. In existing ships the structural strength should be evaluated with a view to limiting the locations where flasks may be stowed.

CARGO SECURING ARRANGEMENTS

- 18** Adequate permanent securing devices should be provided to prevent movement of the flasks within the cargo spaces. In designing permanent devices, due consideration should be given to the orientation of the flasks and the following ship acceleration levels should be taken into account:

- 1.5g longitudinally;
- 1.5g transversely;
- 1.0g vertically up;
- 2.0g vertically down;
- or alternatively,

where flasks are carried on the open deck or a vehicle deck they should be secured in accordance with the principles of safe stowage and securing of heavy, unitized and wheel-based (rolling) cargoes contained in Assembly resolution A.714(17) of November 1991, on the Code of Safe Practice for Cargo Stowage and Securing, taking into account Assembly resolution A.581(14) of 20 November 1985, on Guidelines for Securing Arrangements for the Transport of Road Vehicles on Ro-Ro Ships, and MSC/Circ.385 of 8 January 1985, on Provisions to be Included in the Cargo Securing Manual to be Carried on Board Ships.

19 Where collision chocks are used, these should be so arranged that they will not interfere or prevent cooling air flow which may be necessary (see 14, 15 and 16).

ELECTRICAL SUPPLIES

- 20 To the satisfaction of the Administration concerned.
- 21 An alternative source of electrical power, complying with the requirements of the International Electrotechnical Commission (IEC), should be provided so that damage involving the main supply would not also affect the alternative source.
- 22 The power available from the alternative source should be sufficient to supply the following services for at least 36 hours:
- .1 the equipment provided for the flooding and cooling arrangements referred to in 13.3, 14, 15 and 16; and
 - .2 all emergency services required by SOLAS 74, as amended.
- 23 The alternative source referred to in 21 should be located outside the extent of any damage envisaged in 7, 8, 9 and 10.

RADIOLOGICAL PROTECTION EQUIPMENT

- 24 Depending upon the degree of activity of the materials covered by this Code which are being carried, the ship's design may need to provide for additional arrangements or equipment for radiological protection to the satisfaction of the Administration concerned.

MANAGEMENT, TRAINING AND SHIPBOARD EMERGENCY PLAN

- 25 The management, training and shipboard emergency plan for a ship should take account of developments within the Organization to the satisfaction of the Administration concerned.